Informatica®
10.5.11

Catalog Administrator Guide

Informatica Catalog Administrator Guide
10.5.11
August 2026

© Copyright Informatica LLC 2015, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Subject to your opt-out rights, the software will automatically transmit to Informatica in the USA information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at https://www.informatica.com/in/privacy-policy.html. You may disable usage collection in Administrator tool.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved. Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright © University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved. Copyright © MapR Technologies Inc. All rights reserved. Copyright © Amazon Corporate LLC. All rights reserved. Copyright © Highsoft. All rights reserved. Copyright © Python Software Foundation. All rights reserved. Copyright © BeOpen.com. All rights reserved. Copyright © CNRI. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http:// www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http:// httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/ release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/ license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/

licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html; https://aws.amazon.com/asl/; https://github.com/twbs/bootstrap/blob/master/LICENSE; https://sourceforge.net/p/xmlunit/code/HEAD/tree/trunk/LICENSE.txt; https://github.com/documentcloud/underscore-contrib/blob/master/LICENSE, and https://github.com/apache/hbase/blob/master/LICENSE.txt.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer’s Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

See patents at https://www.informatica.com/legal/patents.html.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-08-18

Table of Contents

Preface 11

Informatica Resources 11

Informatica Network 11

Informatica Knowledge Base 11

Informatica Documentation 11

Informatica Product Availability Matrices 12

Informatica Marketplace 12

Informatica Global Customer Support 12

Chapter 1: Introduction to Catalog Administration 13

Enterprise Data Catalog Overview 13

Enterprise Unified Metadata Architecture 14

Catalog Administration Overview 14

Catalog Administration Process 15

Accessing Catalog Administrator 15

Prerequisites 16

Log In To Catalog Administrator 16

Changing the Password 16

Chapter 2: Enterprise Data Catalog Concepts 17

Enterprise Data Catalog Concepts Overview 17

Catalog 18

Resource Type 18

Resource 18

Scanner 18

Schedule 18

Business Example 19

Data Domains and Data Domain Groups 20

Data Domains and Data Domain Groups in Catalog Administrator 20

Predefined Data Domains 21

Rule-based Data Domains 21

Composite Data Domains 26

Composite Data Domain Discovery Workflow 28

Data Discovery 29

Supported Resource Types for Data Discovery 29

Running a Random N Sampling Option on a Scorecard 31

Domain Connection Settings 32

Basic Profile Settings 33

Similarity Profile Data Preparation and Value Frequency Settings 37

Unique Key Inference Settings 39

Column Similarity. 40
How Column Similarity Works. 40
Column Similarity Process. 41
Choosing Factors to Propagate Smart Data Domains. 42
Business Example. 43
Business Term Propagation. 45
Business Glossary Synchronization. 46
Data Preview and Provisioning. 47
Prerequisites. 48
Data Preview. 48
Data Provisioning 50

Chapter 3: Using Catalog Administrator. 54

Catalog Administrator Overview. 54
Overview Workspace. 56
Resource Workspace. 56
Monitoring Workspace. 56
Library Workspace. 57
Data Domains Workspace. 57

Chapter 4: Managing Resources. 58

Managing Resources Overview. 58
Resources and Scanners. 59
Resources and Schedules. 59
Resources and Attributes. 59
Reference Resources. 60
Benefits of Extracting Reference Assets. 60
Assign Connections to Reference Resources. 61
Sample Use Case. 61
Viewing Reference Resources. 62
Creating a Resource. 63
Resource Type. 64
Dataset or Schema Profile Filter. 65
Source Metadata and Data Profile Filter. 66
Case Insensitive Linking 67
Enable Data Discovery. 67
Composite Data Domain Discovery. 74
Enable Business Term Association 75
Enable Data Provisioning 77
Editing a Resource. 78
Cloning a Resource. 78
Running a Scan on a Resource. 79
System Resources. 79

Viewing a Resource. 80

Chapter 5: Managing Resource Security 81

Managing Resource Security Overview. 81
Configuring Default Permissions for Resources. 82
Configuring Permissions for Specific Users and User Groups. 83
Selecting Resources to Assign Permissions for Specific Users or User Groups. 84
Applying Restrictions on a Resource. 85
Assigning Configuration Permissions to Resources. 85

Chapter 6: Managing Schedules. 86

Managing Schedules Overview. 86
Schedule Types. 86
Reusable Schedules. 86
Custom Schedules. 87
Creating a Schedule. 87
Viewing the List of Schedules. 87

Chapter 7: Managing Attributes. 88

Managing Attributes Overview. 88
System Attributes. 88
Custom Attributes. 89
General Attribute Properties. 89
Search Configuration Properties. 89
Creating a Custom Attribute. 90
Editing a Custom Attribute and System Attribute. 90
Assigning Custom Attributes to Classes and Resources. 91

Chapter 8: Assigning Connections. 92

Assigning Connections Overview. 92
Auto-assigned Connections. 92
User-assigned Connections. 93
Managing Connections. 93
Assigning Connections to Reference Resources. 94
Missing Links Report 94
Missing Links Report Columns 95
Generating Missing Links Report 95

Chapter 9: Configuring Reusable Settings. 96

Reusable Configuration Overview. 96
General Configuration Properties. 96
Data Integration Service Connection Properties. 97
Setting Up a Reusable Data Integration Service Configuration. 97

Chapter 10: Monitoring Enterprise Data Catalog. 98

Monitoring Enterprise Data Catalog Overview. 98
Task Status. 99
Task Distribution. 99
Monitoring by Resource. 100
Monitoring by Task. 100
Managing Tasks 101
Applying Filters to Monitor Tasks. 101

Chapter 11: Managing Data Domains. 102

Managing Data Domains Overview. 102
Best Practices for Creating Data Domains Using Rules. 103
Creating a Rule-based Data Domain. 103
Creating a Data Domain Group. 105
Viewing Data Domains and Data Domain Groups. 106
Filtering Data Domains. 106
Filtering Data Domain Groups. 106
Modifying a Data Domain or Data Domain Group. 107
Permissions and Privileges. 107
Data Domain Discovery on the Spark Engine. 108
Data Domain Discovery on the Databricks Cluster. 109

Chapter 12: Managing Composite Data Domains. 111

Managing Composite Data Domains Overview. 111
Creating Composite Data Domains. 111
Viewing Existing Composite Data Domains. 112
Filtering Composite Data Domains. 113
Modifying Existing Composite Data Domains. 113
Deleting Existing Composite Data Domains. 113

Chapter 13: Manage Synonym Definitions. 115

Manage Synonym Definitions Overview. 115
Synonym Definition File Validation. 115
Uploading Synonym Definition Files. 116

Chapter 14: Managing Cloud Organization. 117

Managing Cloud Organization Overview 117
Cloud Organization Properties 117
Adding an Informatica Intelligent Cloud Services Organization. 118
Editing Informatica Intelligent Cloud Services Organization. 118

Chapter 15: Custom Metadata Integration Overview 119

Custom Metadata Integration Overview. 119
Custom Metadata Integration Workflow. 120
Downloading the Model Definition Files for Validation. 121
Creating Custom Models. 121
Updating Custom Models. 122
Exporting Models. 122
Deprecating Custom Models. 123
Custom Resource Type Overview. 123
    Custom ETL Resources. 123
Creating Custom Resource Types. 124
Creating Custom Resources. 124
Metadata Ingestion Overview. 126
    Exporting the Custom Resource Type Template. 126
    Entering Association Details. 126
    Entering Class Details. 127
    Entering Transformation Details for Custom ETL Resources. 128
Customize Icons and Configure Relationships Views for Custom Resources. 129
    Workflow. 130
    Exporting the Model Template. 130
    Creating the Custom Model XML File. 131
    Creating the Icons ZIP File. 131
    Creating the View Configuration XML File. 131
    Creating the Custom Model with Custom Icons. 140
Context Lineage for Custom Resources. 140
    Context Lineage Resource CSV File Structure. 142

Chapter 16: Catalog Backup and Log Collection Utilities..... 144

Overview. 144
Accessing Utilities. 145
Performing a Catalog Backup. 145
Log Collection Utilities. 146
    Collect Enterprise Data Catalog Service Logs. 146
    Collect Resource Logs. 146
Monitoring Utility Run Status. 147

Appendix A: Registering Unknown Data Types for Profile Run..... 148

Overview. 148
    Example. 148
Registering an Unknown Data Type as a Supported Data Type. 149

Appendix B: Managing Import from ServiceNow 151

Managing Import from ServiceNow Overview. 151
Importing Connections from ServiceNow 151

Appendix C: Custom Lineage Resource CSV File 153

Custom Lineage Resource CSV File 153

Appendix D: Enterprise Data Catalog Agent..... 155

Installing Enterprise Data Catalog Agent. 155
Installing Enterprise Data Catalog Agent as a Windows Service. 156
Configuring Enterprise Data Catalog Agent for SSL. 157

Appendix E: Lineage between PowerCenter Resources and Flat Files..... 158

Overview. 158
Configuring and Running a Flat File Resource on a Windows Machine. 159
Viewing Lineage between PowerCenter Resources and Flat Files. 160

Appendix F: Metadata Extraction from Offline and Inaccessible Resources.. 163

Overview. 163
Process. 165
Scanner Utility Components. 166
Prerequisites and Best Practices. 166
Running the Scanner Script 167
Script Command. 168
- generateResourceConfig. 169
- encryptPassword. 170
- validate. 170
- scan. 171
- cancel. 172
- pause. 173
- resume. 174
- describeArchive. 174
- publishArchive. 175

Structure of Metadata Archive File. 177

Appendix G: Enterprise Data Catalog Add-Ons..... 179

Enterprise Data Catalog Add-Ons Overview. 179
Deploy the Add-Ons. 180
Log in to the Enterprise Data Catalog Add-Ons. 180
View Resource Schedules and Duplicate Resources. 180
Accept Data Domains. 181
Moving Data Domains using the Data Domains Manager Utility. 182

Associate Business Terms from Synonym File. 183
Migrate Custom Attributes. 184

Index. 185

Preface

Use the *Informatica® Catalog Administrator Guide* to perform administration tasks in Enterprise Data Catalog using the Catalog Administrator application. The guide includes information about administration tasks, such as creating resources and custom resources, managing schedules and attributes, and configuring profiles.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the Support option.

CHAPTER 1

Introduction to Catalog Administration

This chapter includes the following topics:

- Enterprise Data Catalog Overview, 13
- Enterprise Unified Metadata Architecture, 14
- Catalog Administration Overview, 14
- Catalog Administration Process, 15
- Accessing Catalog Administrator, 15

Enterprise Data Catalog Overview

Enterprise Data Catalog brings together all data assets in an enterprise and presents a comprehensive view of the data assets and data asset relationships. A data asset is a type of data object, such as a physical data source, HDFS, or big-data repository. The data assets in the enterprise might exist in relational databases, purpose-built applications, reporting tools, HDFS, and other big-data repositories.

Enterprise Data Catalog captures the physical and operational metadata for a large number of data assets that you use to determine the effectiveness of enterprise data. Metadata is data about data. Metadata contains details about the structure of data sources. Metadata also includes information, such as data patterns, data types, relationships between columns, and relationships between multiple data sources.

Enterprise Data Catalog gathers information related to metadata across the enterprise. The metadata includes column data statistics, data domains, data object relationships, and data lineage information. A comprehensive view of enterprise metadata can help you make critical decisions on data integration, data quality, and data governance in the enterprise.

Enterprise Data Catalog addresses the following key questions related to metadata in the enterprise:

- What content does a data asset contain?
- What does the content in a data asset mean?
- Who is responsible for a specific data asset in the enterprise?
- Which is the source for data in a data asset?
- What sensitive data does the enterprise have?
- Where is the sensitive data located?

Enterprise Unified Metadata Architecture

The Enterprise Unified Metadata architecture consists of applications, services, and databases. The applications layer consists of client applications, such as Enterprise Data Catalog. The services layer has application services, such as the Catalog Service, Data Integration Service, and Model Repository Service. Enterprise Unified Metadata requires the Catalog Service to extract metadata from data sources and manage the administrative tasks. The databases layer consists of the Model repository and the Informatica cluster for metadata storage and analysis. Data sources and metadata sources include source data repositories such as Oracle, Microsoft SQL Server, PowerCenter repository, and SAP Business Objects.

The following table describes the architecture components:

Component	Description
External Application	An application, such as the Enterprise Data Catalog, that you use to discover, explore, and relate different types of metadata from disparate sources in the enterprise.
Scanner Framework	A framework that runs scanners and manages a registry of available scanners. A scanner is a pluggable component that extracts specific metadata from external data sources.
Model Repository Service	An application service that manages the Model repository.
Content Management Service	An application service that manages reference data. It provides reference data information to the Data Integration Service and Informatica Developer. You can use Informatica Developer to import data domains into Model repository.
Data Integration Service	An application service that performs data integration tasks for Enterprise Data Catalog and external applications.
Catalog Service	An application service that runs Enterprise Unified Metadata and manages connections between service components and external applications.
Model repository	A relational database that stores the resource configuration and data domain information.
Metadata Persistence Store	A staging database that stores extracted metadata for further analysis.
Search Index	Apache Solr-based search index information. The search index is based on the Model repository assets and assets in the metadata persistence store. Enterprise Unified Metadata uses the indexed information to display search results based on the appropriate asset metadata and relationships.
Data sources and metadata sources	The source databases or metadata sources that Enterprise Unified Metadata scans to extract relevant metadata for further use.

Catalog Administration Overview

Catalog Administrator is the administration tool that you can use to manage and monitor the resources, schedules, attributes, and connections.

You can use Catalog Administrator to perform the following tasks:

- Resource management. Create, edit, and remove resources.

- Schedule management. Create, edit, and remove schedules.
- Attribute management. View system-defined attributes for metadata object types. Create custom attributes and assign to metadata object types, such as tables, views, and columns.
- Connection management. View automatically assigned connections and schemas. Assign schemas and connections to resources. Unassign user-assigned connections.
- Profile configuration management. Create and edit reusable profile-definition settings.
- Resource monitoring. Monitor resources and tasks.
- Data domain management. Create and edit data domains and data domain groups. Assign logical data domains to data domain groups.

Catalog Administration Process

The administration tasks include configuring resources, assigning schedules, and custom attributes. You also need to monitor the tasks that extract metadata using the resources.

You can perform the following tasks as part of the administration process:

1. Create resources for each resource type based on the type of sources that you need to extract metadata from.
2. Choose whether you want to extract the source metadata, profiling metadata, or both.
3. Choose whether you want to run the resources one time or multiple times based on a common or custom schedule.
4. Optionally, assign a common schedule or custom schedule to the resources.
5. Monitor the tasks that extract metadata from different sources.
6. Define data domains based on predefined or user-defined Model Repository object based on the semantics of column data or a column name.
7. Troubleshoot tasks that do not perform as expected.

Accessing Catalog Administrator

Use Catalog Administrator to consolidate the administrative tasks for resources, attributes, and schedules. You launch Catalog Administrator from Informatica Administrator. You must know the host name of the gateway node and the Informatica Administrator port number to log in to Informatica Administrator.

Perform the following steps before you log in:

1. Launch Informatica Administrator using the gateway node and Informatica Administrator port number in the Informatica Administrator URL.
2. In Informatica Administrator, configure an Informatica domain, user accounts, database connections, and services if you haven't created them as part of the installation. The services include Data Integration Service, Model Repository Service, Content Management Service, Informatica Cluster Service and Catalog Service.
3. On the **Services and Nodes** tab of Informatica Administrator , select the Catalog Service, and then click the service URL to launch Catalog Administrator from Informatica Administrator.

4. Use the login credentials to log in to Catalog Administrator.

Prerequisites

The prerequisites to launch Catalog Administrator include an Informatica domain, domain connectivity information, and administrator user account.

You need to verify the following prerequisites before you log in to the Catalog Administrator:

1. The Informatica Domain is running.
2. The Informatica domain has Data integration Service, Model Repository Service, and Catalog Service enabled.
3. You have the domain connectivity information and administrator user account in Informatica Administrator.

Log In To Catalog Administrator

Use either Microsoft Edge or Google Chrome to log in to Catalog Administrator.

1. Start Microsoft Edge or Google Chrome.
2. In the Address field, enter the URL for the Catalog Administrator login page in the following format:

   ```
   http://<host>:<port>/ldmadmin
   ```

   The host is the gateway node host name. The port represents the port number configured for the Catalog Service.
3. In the Catalog Administrator login page, enter the user name and password.
4. Verify that the default domain option **Native** is selected. You can also select an LDAP domain.

   The Domain field appears when the Informatica domain contains an LDAP security domain.
5. Click **Log In**.

Changing the Password

Use the Administrator menu to change the password.

1. In the Catalog Administrator tool header area, click **Administrator** > **Change Password**.

   The **Change Password** page appears.
2. Enter the current password in the **Password** field and the new password in the **New Password** and **Confirm New Password** fields.
3. Click **Update**.

CHAPTER 2

Enterprise Data Catalog Concepts

This chapter includes the following topics:

- Enterprise Data Catalog Concepts Overview, 17
- Catalog, 18
- Resource Type, 18
- Resource, 18
- Scanner, 18
- Schedule, 18
- Business Example, 19
- Data Domains and Data Domain Groups, 20
- Composite Data Domains, 26
- Data Discovery, 29
- Column Similarity, 40
- Business Term Propagation, 45
- Business Glossary Synchronization, 46
- Data Preview and Provisioning, 47

Enterprise Data Catalog Concepts Overview

Enterprise Data Catalog helps you analyze and understand large volumes of metadata in the enterprise. You can extract physical and operational metadata for a large number of objects, organize the metadata based on business concepts, and view the data lineage and relationship information for each object.

The key concepts in Enterprise Data Catalog include catalog, resource, resource type, scanner, and schedule. Catalog stores all the metadata extracted from sources. Resource type represents different metadata source systems. Resource is a representation of a resource type, such as Oracle, SQL Server, or PowerCenter. Scanners fetch the metadata and save it in the catalog. Schedules determine the intervals at which scanners extract metadata from the source systems and save the metadata in the catalog.

Catalog

The catalog represents an indexed inventory of all the data assets in the enterprise that you configure in Catalog Administrator. Enterprise Data Catalog organizes all the enterprise metadata in the catalog and enables the users of external applications discover and understand the data.

You can find metadata and statistical information, such as profile results, data domains, and data asset relationships, from the catalog. Enterprise Data Catalog stores the metadata extracted from external sources in the MongoDB database.

Resource Type

A type of external data source or metadata repository from which scanners extract metadata. Examples include relational sources, Business Intelligence sources, and PowerCenter sources.

Resource

A resource is a catalog object that represents an external data source or metadata repository from where scanners extract metadata. A resource represents an instance of a specific resource type. The basic metadata operations, such as extraction and storage of metadata, are performed at the resource level.

A resource can be of resource types, such as relational databases, Business Glossary classification, and business intelligence sources. A resource might have an associated schedule. Each resource can extract both source metadata and profile metadata from the external data sources.

Scanner

A pluggable component of Enterprise Data Catalog that extracts specific metadata, such as source metadata or profile metadata, from external data sources and stores the metadata in the catalog. A scanner typically maps to a single resource type. However, there can be more than one scanner for a resource type. Examples are profiling scanner and lineage analyzer.

A scanner performs a scan job on the metadata sources to fetch metadata into the catalog. When you have scanners for newer resource types ready, you can plug in those scanners to Enterprise Data Catalog without having to upgrade Enterprise Data Catalog.

Schedule

A schedule determines when scanners extract metadata from sources. You can have recurring daily, weekly, and monthly schedules to extract metadata at regular intervals.

You can create the following types of schedules:

Global schedule

Reusable schedules that you can attach to more than one resource.

Custom schedule

A customized schedule assigned to a single resource.

Business Example

You are a catalog administrator in a multinational retail organization. The data analysts in your department need to view the metadata from different database schemas and database tables in multiple sources to perform an advanced data analysis. You also need to make sure that the data analysts understand and trust the data that they use. The organization might plan regular security audits to find sensitive data in the data sources and mask or protect them as required.

The retail organization that you work for has the following configured systems:

- Human resources management system set up on an Oracle database.
- Order management system set up on the same Oracle database.
- Data warehouse hosted on a Hadoop repository. The data warehouse has integrated information from multiple data sources.
- PowerCenter to perform data integration tasks across databases and schemas.
- Reporting system set up on an SAP Business Object source.

The administrator in the organization can perform the following tasks in Catalog Administrator to effectively meet the data governance needs in this example:

- Use Catalog Administrator to create an Oracle resource for the human resource management system and another Oracle resource for the order management system. You can configure the source metadata settings to extract the metadata into the catalog. You might not need to configure the profiling metadata settings for these resources. The resources provide the required database table and source column objects into catalog for analysis.
- Create a Hive resource for the Hadoop warehouse. The Hive resource fetches tables and columns to the catalog. In addition to the source metadata extraction, you can configure the profiling metadata settings so that you have information related to data quality for further analysis.
- Create a PowerCenter resource that maps to the data integration requirements. The resource configuration provides the links between the Oracle data objects and Hive objects.
- Create an SAP Business Objects resource and configure the resource to extract reporting metadata. The resource provides reporting metadata based on the links between the Business Objects and the Oracle and Hive objects.
- Set up a recurring schedule for each resource so that the scanners extract source and profiling metadata from the source systems at regular intervals.
- Periodically monitor the tasks and jobs in Catalog Administrator that extract metadata. Monitor the tasks and jobs to get a functional view of Enterprise Data Catalog. Monitoring also helps you to analyze and estimate the type of content the scanners fetch into the catalog.

Data Domains and Data Domain Groups

A data domain is a predefined or user-defined Model repository object that enables you to discover the functional meaning of column data or column names in a data source. Examples of data domains include Social Security number, account status, IP address, and UPC code. You can add one or more data domains into a data domain group. Data domain groups can contain data domains and not other data domain groups. Examples for data domain groups include PCI, PHI, and PII.

Data domain discovery is the process of discovering the functional meaning of data in the data sources based on the semantics of data. You can use data domains to identify and understand the meaning of critical source data or undiscovered source data so that you can take measures, such as data masking, to work effectively on it. For example, you might have legacy data systems that contain Social Security numbers in a Comments field. You need to find this information so that you can take appropriate measures before you move it to new data systems

Data domain groups help you categorize data domains into specific groups. For example, you can group the data domains first_name, last_name, and account_number into the Personal Health Information (PHI) data domain group. When you create or edit a data domain group, you can add one or more data domains to the group. You can delete a data domain from the data domain group. A data domain can be a part of multiple data domain groups. For example, the Social Security number can belong to both Payment Card Industry (PCI) and PII data domain groups.

Data Domains and Data Domain Groups in Catalog Administrator

In the Catalog Administrator, you can view data domains and data domain groups, create and edit rule-based data domains, edit smart data domains, create and edit data domain groups.

In Catalog Administrator, after you enable data domain discovery for a resource and run the resource, the profiling scanner uses the data domains to infer matching column data or column name patterns from the metadata extracted by the resources. In Enterprise Data Catalog, you can view a data domain as an asset. You can also view the data domains for tabular, column, and field assets. These data domains are inferred for the asset from the profile results or from similar columns. You can curate a inferred data domain based on your requirements.

In Catalog Administrator, you can view all the available data domains and data domain groups in the Library workspace. You can view or edit data domains, and view, edit, or delete data domain groups. You can use filters to view specific data domains or data domain groups.

To create or view data domains and data domain groups, you need appropriate permissions and privileges. Configure the **Domain Management: Admin - View Domain and Domaingroup** and **Domain Management: Admin - Edit Domain and Domaingroup** privileges in Informatica Administrator for a user to view, create, edit, or delete data domains or data domain groups in Catalog Administrator. For more information about privileges and permissions, see the *Informatica Administrator Reference for Enterprise Data Catalog Guide*.

In Enterprise Data Catalog, you can use the following types of data domains:

Predefined data domains

Predefined data domains are system-defined data domains. These data domains are imported automatically when you install Enterprise Data Catalog.

Rule-based data domains

Rule-based data domain are user-defined data domains. You can use one or more rules to create a rule-based data domain in Catalog Administrator.

You can also create data domains and data domain groups in Informatica Analyst or Informatica Developer and use it in Enterprise Data Catalog. You can create mapplet rules in the Developer tool and

use these rules to create data domains in Catalog Administrator. To use these data domains, Enterprise Data Catalog runs the DataDomain scanner which is auto-scheduled to run every 45 minutes. The scanner synchronizes the data domains between the Model repository and catalog.

Smart data domains

Smart data domains are user-defined data domains. You can create a smart data domain with an existing pattern of a column or field in the catalog. This data domain does not contain any rule. You can view smart data domains in Catalog Administrator. To convert a smart data domain to a rule-based data domain, add one or more rules, conformance values, and proximity data domains

Predefined Data Domains

Predefined data domains and predefined data domain groups are system-defined data domains and data domain groups that are available by default in Enterprise Data Catalog. The data domains are imported automatically when you install Enterprise Data Catalog. Predefined data domains use default conformance values. In Catalog Administrator, you can edit the predefined data domains to modify the default conformance values and add proximity data domains.

Predefined data domains use data rule, column name rule, or data and column name rules. Examples for predefined data domains include AccountNumber, India_NationalID, SSN_General, and ZipCode. After you install Enterprise Data Catalog, you can choose to import the predefined data domains either in the Developer tool or run the infacmd.sh oie importobjectscommand

Rule-based Data Domains

You can use one or more rules to define a rule-based data domain. In Catalog Administrator, you can create and edit rule-based data domains. When you enable data domain discovery on resources, Enterprise Data Catalog uses the data domains to discover matching column data or column name patterns from the metadata extracted by the resources. In Enterprise Data Catalog, you can view data domain assets and data domains in tabular, column, and field assets.

When you create or edit a data domain, you can add rules, choose conflict resolution options, configure conformance values, add proximity data domains, and data domain groups. You can curate the data domains in Enterprise Data Catalog.

Rules

You can use the rules that have a single output with Integer data type. If you do not use any rule in a data domain, Enterprise Data Catalog assigns the data domain to similar columns based on the data domain you assign to the column.

Choose one or both of the following rules to define a data pattern for a data domain:

- Data rule. Uses source data that match the metadata. The rule discovers columns with data that match a specific logic defined in the rule.
- Column name rule. Uses column-name patterns that match the metadata. The rule discovers columns that match the column name logic defined in the rule.

When you create or edit a data domain, choose one of the following rules as a data rule or column name rule:

Rule	Description
Reference table	Choose the reference table in the Model repository. When you choose a reference table, the rule uses the column data in the reference table to discover data domains.
Regular expression	Specify a regular expression to match string data. A regular expression is a specialized formula for matching text strings that follow a pattern. The syntax used for regular expression must conform to the grammar rules and tokens described here: https://docs.informatica.com/data-catalog/shared-content-for-data-catalog/10-5/transformation-language-reference/functions/reg_extract.html.
Existing rules	Choose existing rules in Model repository. Before you choose a rule, verify that the rules are available to you with appropriate permissions. You can choose mapplet based rules that were created using Informatica Analyst or Informatica Developer. For more information about using mapplet based rules, see: https://docs.informatica.com/data-engineering/shared-content-for-data-engineering/10-5/developer-mapping-guide/mapplets/mapplet-validation/mapplet-as-a-rule-validation.html.

Conflict Resolution

You can decide to use data rule or column name rule to configure auto-acceptance criteria. Select the **Conflict Resolution** option to auto-accept data domain on the basis of data or column name rule. You can choose one of the following options:

- If rules conflict, data rule overrides column name rule. Allows Enterprise Data Catalog to auto-accept data domains based on the data rule. If the data rule is not specified, column name rule result is used to auto accept the data domains
- If rules conflict, column name rule overrides data rule. Allows Enterprise Data Catalog to auto-accept data domains based on the column name rule. If the column name rule is not specified, data rule result is used to auto accept the data domains.
- Match data and column name rule. Enterprise Data Catalog auto-accepts data domains when data and column name rule results match. If the data rule is not specified, the data domains are not auto accepted.
- Match data or column name rule. Enterprise Data Catalog auto-accepts data domains when data or column name rule results match. If the data or column name rule are not specified, the data domains are not auto accepted.

If you do not select the **Conflict Resolution** option, Enterprise Data Catalog applies the data rule result to auto-accept data domains. Data domains are not auto accepted if you configure **Data Domain Match Criteria** as **Row** in the **Resource** section of the Catalog Administrator.

The data domains are auto accepted in the following scenarios:

Data rule overrides column name rule.

Column Name Rule Match	Data Rule Match	Data Domain State
Yes	No	No
Yes	Yes, reaches the configured minimum conformance value.	Inferred
Yes	Yes	Auto accepted

Column Name Rule Match	Data Rule Match	Data Domain State
No	No	No
No	Yes, reaches the configured minimum conformance value	Inferred
No	Yes	Auto accepted
Domain discovery on source metadata is not performed.	No	No
Domain discovery on source metadata is not performed.	Yes, reaches the configured minimum conformance value	Inferred
Domain discovery on source metadata is not performed.	Yes	Auto accepted
Yes	Domain discovery on source metadata is not performed.	Auto accepted
No	Domain discovery on source metadata is not performed.	No

Column name rule overrides data rule.

Column Name Rule Match	Data Rule Match	Data Domain State
Yes	No	Auto accepted
Yes	Yes, reaches the configured minimum conformance value	Auto accepted
Yes	Yes	Auto accepted
No	No	No
No	Yes, exceeds the configured auto accept threshold value.	No
No	Yes	No
No	Domain discovery on source data is not performed.	No
Yes	Domain discovery on source data is not performed.	Auto accepted
Domain discovery on source data is not performed.	Yes	Auto accepted

Column Name Rule Match	Data Rule Match	Data Domain State
Domain discovery on source data is not performed.	Yes, exceeds the configured auto accept threshold value.	Inferred
Domain discovery on source data is not performed.	No	No

Match data and column name rule.

Column Name Rule Match	Data Rule Match	Data Domain State
Yes	Yes, exceeds the configured auto accept threshold value.	Auto accepted
Yes	Yes, reaches the configured minimum conformance value	Inferred
Yes	No	No
No	No	No
No	Yes, reaches the configured minimum conformance value	No
No	Yes	No
No	Domain discovery on source metadata is not performed.	No
Yes	Domain discovery on source metadata is not performed.	Auto accepted
Domain discovery on source metadata is not performed.	Yes	Auto accepted
Domain discovery on source metadata is not performed.	No	No
Domain discovery on source metadata is not performed.	Yes, reaches the configured minimum conformance value	Inferred

Match data or column name rule.

Column Name Rule Match	Data Rule Match	Data Domain State
Yes	No	Auto accepted
Yes	Yes, reaches the configured minimum conformance value	Auto accepted
Yes	Yes	Auto accepted

Column Name Rule Match	Data Rule Match	Data Domain State
No	Yes, reaches the configured minimum conformance value	Inferred
No	Yes	Auto accepted
No	Domain discovery on source metadata is not performed.	No
Yes	Domain discovery on source metadata is not performed.	Auto accepted
Domain discovery on source metadata is not performed.	Yes	Auto accepted
Domain discovery on source metadata is not performed.	No	No
Domain discovery on source metadata is not performed.	Yes, reaches the configured minimum conformance value	Inferred

Proximity Data Domains

Enterprise Data Catalog uses proximity data domains to narrow down the inferred results to identify close-to-identical columns or fields for a data domain. Proximity data domains are one of the factors that help determine the type of data in a column. When one or more data domains are inferred with equal conformance or probability, proximity data domains can be used to identify the close-to-identical data domain. When you create or edit a data domain, add one or more data domains as proximity data domains.

Enterprise Data Catalog displays the results as a match score for the data domain. The match score is the ratio of number proximal data domains discovered in the data source to the number of configured proximal data domains for an inferred data domain.

When you enable data discovery, add data domains, and run the resource, the profiling scanner scans the data source for the data domain and the proximity data domains in the resource and displays a match score in Enterprise Data Catalog.

Example

You run a profile on a CSV file that has the following columns:

- ID. Contains seven-digit numbers
- Fname. Contains first names
- Lname. Contains last names
- Dname. Contains department names
- Addr1. Contains addresses
- Phone. Contains phone numbers

You want to determine whether the ID column refers to *EMPID*, *NODEID*, or *CUSTID* data domains that also contain seven digits.

When you analyze the file, you determine that the ID column has Employee IDs because the table has columns for first names, last names, and department names. The department names usually appear in employee-related tables. The Dname column is conclusive proof that the ID column is EMPID and not NODEID or CUSTID.

In Enterprise Data Catalog, you can add proximity data domains to determine the type of data in the ID column.

To identify the data in the ID column, perform the following tasks in Catalog Administrator:

1. Create the *DeptName* and *EMPID* data domains.
2. Add *FirstName*, *LastName*, and *DeptName* data domains as proximity data domains to the *EMPID* data domain.
3. Edit the resource with the CSV file, and add the *EMPID* data domain.
4. Run the resource.

During resource run, the ID column is identified as *EMPID*, *NODEID*, and *CUSTID* because of conformance scores, but the proximity data domains determine and infer the ID column as *EMPID*. The catalog displays the ID column as *EMPID*.

Conformance Criteria

When you choose a data rule for a data domain, you can configure the conformance criteria for the data domain. Enterprise Data Catalog identifies the data domains based on the conformance criteria.

You can configure the following conformance criteria for a data domain:

- Minimum conformance. Enter a minimum conformance percentage for the data domain. By default, this option displays 40.
- Auto accept if more than. Enter a percentage. Enterprise Data Catalog accepts the data domain automatically if the data domain match exceeds the configured percentage. By default, this option displays 80.00 percent.
- Row Count. Enter the number of minimum rows for data conformance. By default, this option displays 1.

The default values are called predefined conformance values. You can configure a custom value to override the predefined values. When you choose multiple data domains or data domain groups, Enterprise Data Catalog computes the conformance value based on the predefined values or custom values.

When you choose a column name rule for a data domain, you can configure the **Minimum conformance** value or **Auto accept if more than** value as 0 or 100. Column name rule does not use the **Row Count** value to compute the data domain match.

Note: You can specify a floating point value between 0 and 100 for the conformance criteria. For example, 40.5 and 85.7.

When you create or edit a resource, you can choose the data domains, its default or configured conformance values, or choose custom conformance values as conformance criteria. You can also choose to exclude the null values during data domain computation.

Composite Data Domains

A composite data domain is a collection of data domains or other composite data domains linked using rules. A composite data domain helps you search for the required details of an entity across multiple schemas defined for the database. An entity refers to a particular subject for which you need all the associated information. For example, for an entity such as customer, you might want to search for all the associated details such as the customer name, age, location, and contact number. The details might be spread across multiple tables and columns in the database. Composite data domains help you define a

search query that includes the entity details that you want to find from multiple schemas defined for the database.

Note: You must configure a primary key-foreign key relationship between tables if you want to search for entity details across multiple schemas.

A composite data domain helps you define search patterns based on a combination of existing data domains or composite data domains. Data domains in the example can be customer name, age, location and other details associated with the entity *customer*. In a composite data domain definition, you can define rules to link existing data domains or other composite data domains. Enterprise Data Catalog links multiple rules using OR operators and links the data domains or composite data domains in each rule using AND operators. Make sure that you create data domains before creating composite data domains.

After you create composite data domains and enable composite data domain discovery for resources, you can search for the entity using composite data domains in Enterprise Data Catalog. Enterprise Data Catalog finds the assets associated with the entity in the catalog.

Business Example

Sophie, a data steward at a retail chain, is asked to find specific details of customers from the customer details stored with the retail chain. The customer details are present across multiple tables in the retail chain database.

The tables in the database include the following details:

- Customer ID
- First name
- Last name
- Email
- Mobile number
- Street
- County
- Social media account information
- ZIP Code
- State

From the details, Sophie must find any of the following details that might help to identify the customer location:

- Street and County
- ZIP Code
- State

The following table lists the steps that Sophie must perform to collect the required customer details:

Scenario	Resolution
Prerequisite to creating composite data domains	Sophie must define data domains based on the existing assets in the catalog. Sophie creates the following data domains based on the customer details that she wants to collect: - Street - County - Zip_code - State
Create composite data domains	Sophie creates a composite data domain called `customer_location` and defines rules by linking data domains using the AND and OR operators as shown in the following list: - Street AND County OR - Zip_code OR - State
Identify the resources and enable composite data domain discovery	Sophie identifies the required resources by performing a search in Enterprise Data Catalog based on the composite data domains created. Sophie then enables composite data domain discovery for the identified resources.
Search for the required customer details	Sophie can search using the composite data domain `customer_location` in Enterprise Data Catalog. The search retrieves a list of assets and tables that include the customer details that the retail chain requires.

Alternatively, Sophie can create the following three composite data domains (`Street_county`, `Zip_code`, and `State`) and include the composite data domains in a composite data domain definition, called as `customer_location`:

- Street_county. Street AND County
- Zip_code.
- State.

The `customer_location` composite data domain definition is as follows:

- Street_county
  OR
- Zip_code
  OR
- State

Composite Data Domain Discovery Workflow

The composite data domain discovery workflow includes the following phases:

1. Synchronizing composite data domains between Model Repository and catalog.
2. Identifying the list of composite data domains configured for a resource.
3. Identifying the list of data domains required for each composite data domain.
4. Fetching data domain discovery results for the data domains for the current resource.
5. Preparing expressions for each composite data domain and the data domain discovery results for each composite domain.

6. Using internal machine logic to evaluate the expressions and results.
7. Publishing the results to the catalog for search and discovery.

Note: Enterprise Data Catalog performs the steps listed in the workflow during composite data domain discovery.

Data Discovery

When you enable data discovery for a resource and scan the resource, Enterprise Data Catalog identifies profiling-related metadata, such as null values, distinct values, inferred data types, unique keys, and data domains in the resource.

When you create a resource, choose the **Metadata Load Settings** > **Data Discovery** > **Enable Data Discovery** option to discover profiling-related metadata and unique keys. Based on your requirements, you can configure domain connection settings, basic profile settings, unique key inference settings, and column similarity settings.

Important: Profiling might fail with a licensing error if you run data domain discovery using a Data Integration Service on a grid.

Supported Resource Types for Data Discovery

You can enable data discovery for the following types of resources:

- Cloud
  - Amazon Redshift
  - Amazon S3
  - Microsoft Azure Data Lake Store
  - Azure Microsoft SQL Data Warehouse
  - Azure Microsoft SQL Server
  - Microsoft Azure Blob Storage
  - Salesforce
  - Google Big Query
  - Snowflake
- Data Engineering
  - HDFS
  - Hive
- File Management
  - File System. Supported protocols include Local File, SFTP, and SMB/CIFS protocol.
  - OneDrive
  - SharePoint
- Database Management
  - IBM DB2
  - IBM DB2 for z/OS

- IBM Netezza
- JDBC
- Microsoft SQL Server
- Oracle
- Sybase
- Teradata

- Application
  - SAP R/3
  - SAP S/4HANA
- No SQL
  - Apache Cassandra

You can also run data discovery on structured file types, unstructured file types, and extended unstructured formats.

Structured file types

Data discovery in Enterprise Data Catalog supports the following structured file types:

- Avro. Supported extension type is .avro.
  This file type is available for HDFS resource and File System resource. For the File System resource, you can choose only the Local File protocol.
- Delimited and Text
- JSON
- Parquet.
  This file type is available for HDFS, Amazon S3, Azure Data Lake Store Gen2, and File System resource. For the File System resource, you can choose only the Local File protocol.
- XML

Unstructured File Types and Extended Unstructured Formats

When you choose HDFS, Amazon S3, or File System as a resource, you can choose extended unstructured formats or unstructured file types. Extended unstructured formats include mp3, mp4, bmp, and jpg formats. Extended unstructured formats do not fall under structured or unstructured file types.

Data domain discovery in Enterprise Data Catalog supports the following unstructured file types:

- Apple Files. Supported extension types include .key, .pages, .numbers, .ibooks, and .ipa.
- Compressed Files. Supported extension types include gz, tgz, and emz.
- Email. Supported extension types include eml, emlx, and mime.
- Microsoft Excel
- Microsoft PowerPoint
- Microsoft Word
- Open Office Files. Supported extension types include .odt, .ott, .odm, .ods, .ots, .odp, .odg, .otp, .odg, .otg, and .odf.
- PDF
- Webpage Files. Supported extension types include chm, oth, and xhtml.
- CLOB. Supported on DB2 and DB2 z/OS datasources.

Running a Random N Sampling Option on a Scorecard

Run a random sampling option on a scorecard based on the number of rows you specify for a profile.

1. In the **Library** workspace, select the project or folder that contains the profile.
2. Click the profile to open the profile.

   The profile results appear in the summary view in the **Discovery** workspace.
3. Click **Actions** > **Add to scorecard**.

   The **Add to Scorecard** wizard appears.
4. In the **Add to Scorecard** screen, you can choose to create a new scorecard, or edit an existing scorecard to add the columns to a predefined scorecard. The **New Scorecard** option is selected by default. Click **Next**.
5. In the **Step 2 of 8** screen, enter a name for the scorecard. Optionally, you can enter a description for the scorecard. Select the project and folder where you want to save the scorecard. Click **Next**.

   By default, the scorecard wizard selects the columns and rules defined in the profile. You cannot add columns that are not included in the profile.
6. In the **Step 3 of 8** screen, select the columns and rules that you want to add to the scorecard as metrics. Optionally, click the check box in the left column header to select all columns. Optionally, select **Column Name** to sort column names. Click **Next**.
7. In the **Step 4 of 8** screen, you can add a filter to the metric.

   You can apply the filter that you created for the profile to the metrics, or create a new filter. Select a metric in the **Metric Filters** pane, and click the **Manage Filters** icon to open the **Edit Filter: column name** dialog box. In the **Edit Filter: column name** dialog box, you can choose to perform one of the following tasks:

   - Choose a filter that you created for the profile. Click **Next**.
   - Select an existing filter. Click the edit icon to edit the filter in the **Edit Filter** dialog box. Click **Next**.
   - Click the plus (+) icon to create filters in the **New Filter** dialog box. Click **Next**.

   Optionally, you can choose to apply the selected filters to all the metrics in the scorecard.

   The filter appears in the **Metric Filters** pane.
8. In the **Step 4 of 8** screen, click **Next**.
9. In the **Step 5 of 8** screen, select each metric in the **Metrics** pane to perform the following tasks:

   - Configure valid values. In the **Score using: Values** pane, select one or more values in the **Available Values** pane, and click the right arrow button to move them to the **Valid Values** pane. The total number of valid values for a metric appears at the top of the **Available Values** pane.
   - Configure metric thresholds. In the **Metric Thresholds** pane, set the thresholds for **Good**, **Acceptable**, and **Unacceptable** scores.
   - Configure the cost of invalid data. To assign a constant value to the cost for the metric, select **Fixed Cost**. To attach a numeric column as a variable cost to the metric, select **Variable Cost**, and click **Select Column** to select a numeric column. Optionally, click **Change Cost Unit** to change the unit of cost. If you do not want to configure the cost of invalid data for the metric, choose **None**.
10. Click **Next**.
11. In the **Step 6 of 8** screen, you can select a metric group to which you can add the metrics, or create a new metric group, To create a new metric group, click the group icon. Click **Next**.
12. In the **Step 7 of 8** screen, specify the weights for the metrics in the group and thresholds for the group.
13. In the **Step 8 of 8** screen, select **Random Sample** and enter the number of rows you want to run the scorecard. Then, select **Native** as a run-time environment to run the scorecard.

14. Click **Save** to save the scorecard, or click **Save & Run** to save and run the scorecard.

    The scorecard with the specified rows appears under the Total Rows column in**Scorecard** workspace.

Domain Connection Settings

In the **Domain Connection Settings** section, you can configure the properties for the Data Integration Service. After you configure the properties, the Data Integration Service runs the profile, performs data domain discovery, and infers column similarity for the resource. You can choose a different Data Integration Service to infer column similarity.

Choose one of the following options to configure Data Integration Service properties:

Custom

Configure the following Data Integration Service parameters:

Domain Name

Name of the Informatica domain.

Data Integration Service

Name of Data Integration Service to run profiles.

Username

User name that the Data Integration Service uses to access the Model Repository Service.

Password

Password for the Model repository user.

Security Domain

Name of the security domain to which the domain user belongs.

Host

Host name of the master gateway node.

Port

Port number of the master gateway node.

Operating System Profile

Choose an operating system profile if you do not have a default operating system profile. You need to assign an operating system profile to Enterprise Data Catalog users if the Data Integration Service uses operating system profiles. If you do not assign an operating system profile to a user and the user tries to run a profile in Catalog Administrator, the profile run fails. The Data Integration Service uses the operating system profile user credentials to run data discovery. Data discovery includes column profiles and data domain discovery profiles.

If you have multiple operating system profiles and you do not have a default operating system profile, choose an operating system profile in the **Domain Connection Settings** section.

In the **Informatica Administrator** > **Security** > **Operating System Profiles** tab, you can create or assign an operating system profile. To configure a default operating system profile for a user, click the **Assign or Change the Default Operating System Profile** option in the **Security** > **Users** > **Permissions** > **Operating System Profiles** section.

Note: You can choose an operating system profile only when you choose the **Custom** option for the **Specify the configuration settings for Data Integration Service** property. If you choose the **Global** option for the **Specify the configuration settings for Data Integration Service** property, then you must have a default operating system profile to run the profile.

For information about configuring the Data Integration Service to use operating system profiles, see *Informatica Application Service Guide*. For information about creating and assigning operating system profiles, see *Informatica Security Guide*.

Global

Choose this option to enable a reusable configuration.

A reusable configuration has Data Integration Service settings that you can use for a resource to extract profile metadata. You can configure one or more reusable configurations. Navigate to **Manage** > **Reusable Configuration** to view, create, or delete a reusable configuration.

Basic Profile Settings

In the **Basic Profile Settings** section, you can configure the following options for a resource:

Profile Run Option

Choose one of the following profile run options for the profiling scanner to run the profile job on the resource:

Column Profile

Identifies the number of null values, distinct values, non-distinct values, and infers data patterns and data types of the columns in the resource.

Data Domain Discovery

Discovers all the data domains associated with a column based on the column value or column name.

Column Profile and Data Domain Discovery

Identifies the number of null values, distinct values, non-distinct values, and infers data patterns, data types, and data domains in the resource.

When you run a scan on a resource multiple times, the last scan results include all the scans. For example, you choose column profile when you scan a resource. Then, before you run the scan again, you choose to perform data domain discovery. The results for the second scan includes both the column profile results and data domain discovery results.

Data domain discovery results display all the inferred data domains from all the runs. For example, if data domain D1 is inferred during the first resource scan and data domain D4 is inferred during the next scan, the scan results for the second time display both D1 and D4.

When you run a scan on a resource for the second time or for subsequent runs, you can optionally run only data discovery on the source. To run only data discovery on the resource, disable **Metadata Load Settings** > **Source Metadata** option.

Data Domain Discovery Type

Choose one of the following options for the profiling scanner to infer data domains based on column name, column data, or both:

Run Discovery on Source Data

Runs data domain discovery on source data.

Run Discovery on Source Metadata

Runs data domain discovery on column names.

Run Discovery on both Source Metadata and Data

Runs data domain discovery on source data and source metadata.

Run Discovery on Source Data Where Metadata Matches

Runs data domain discovery on the source metadata to identify the column names that match the data domains. The scanner then runs data domain discovery on the source data of the identified columns.

Note: You can choose only the **Run Discovery on Source Data** option when you run data domain discovery on unstructured data sources.

Sampling Option

Choose one of the following sampling options to determine the number of rows to run the profile job on:

All rows

Runs the profile on all the rows in the data source.

Auto Random rows

Runs the profile on a random sample of rows. Enterprise Data Catalog computes the number of random rows based on the number of source rows.

Random N rows

Runs the profile on the configured number of random rows

In the **Random Sampling Rows** field, enter the number of rows that you want to run the profile on.

First N rows

Runs the profile on the first N number of rows in the resource.

In the **Number of First N Sampling Rows** field, enter the number of rows to run the profile on.

Limit N rows

Runs the profile based on the number of rows in the data object.

In the **Number of Rows to Limit** field, enter the number of rows to run the profile on.

Random Percentage

Runs the profile on a percentage of rows in the data object.

In the **Random Percentage** field, enter the number of rows to run the profile on.

Table Sample

Runs the profile on a percentage of rows that the source database returns through a SQL `TABLESAMPLE` clause.

In the **Table Sample Percentage** field, enter the percentage of rows to sample. Use the option to profile JDBC resources on the Spark Databricks engine with pushdown enabled.

Priority

Choose one of the following values for the profiling scanner to prioritize the resource run:

- High
- Low

The profiling scanner runs the resources with High priority and then runs the resources with Low priority.

For example, you have three resources R1, R2, and R3. The priority value for R1 and R3 is set to high and the priority value for R2 is set to low. When you run the resources, the scanner first runs R1 and R3 followed by R2.

Exclude Views

Choose the Exclude Views option if you do not want the profiling scanner to scan the views in the relational data sources.

Incremental Profiling

Choose the option to run the profile only for the changes made to the data source. If you do not select this option, the profile runs on the entire data source.

When you enable incremental profiling for a resource that has one table and you run the profile multiple times on the resource, the profiling scanner validates and runs the profile on the same table every single time.

Note: Incremental profiling is available for all resource types except the relational database sources.

Data Profile Filter

You can include or exclude tables and views from the profile run. Use semicolons (;) to separate the table names and view names.

For more information about the Data Profile Filter field, see the "Source Metadata and Data Profile Filter" on page 66 topic.

Dataset or Schema Profile Filter

You can include or exclude data sets from the profile run for a Amazon Redshift, Google BigQuery, Google BigQuery V2, Hive, MySQL, Oracle, Snowflake, and Snowflake V2 resource.

For more information about the Dataset or Schema Profile Filter field, see the "Dataset or Schema Profile Filter" on page 65 topic.

Note: You can select the data sets associated with the resource that you specify in the Schema field.

Cumulative

Enterprise Data Catalog does not retain the previous scan results. It displays only the latest scan results. To retain the profile results from the previous run in the latest scan results, choose the **Cumulative** option. If you do not choose this option, the column profile and column similarity results from the previous run are deleted and only the latest results appear in Enterprise Data Catalog.

The following use cases explain how the **Cumulative** option with the **Data Profile Filter** field and **Incremental Profiling** option impacts the profiling results:

- **Cumulative** option with **Data Profile Filter** field
  - You run a resource after you enter the table names and view names in the **Data Profile Filter** field and then you choose the **Cumulative** option.
    In this scenario, the scanner retains the previous results, appends the latest results, and displays the consolidated profile results in Enterprise Data Catalog.
  - You run the resource after you enter the tables and views in the **Data Profile Filter** field, but you do not choose the **Cumulative** option.
    In this scenario, the previous profile results excluding the data domain discovery results are purged and the latest profile results appear in Enterprise Data Catalog.
- **Cumulative** option with **Incremental Profiling** option
  - You run a resource after you chose the **Incremental Profiling** option
    In this scenario, the scanner retains the previous profile results irrespective of whether you choose the **Cumulative** option or not. Enterprise Data Catalog displays the consolidated profile results.

- You do not choose the **Cumulative** option and **Incremental Profiling** option
  - You run a resource without choosing the **Cumulative** option and **Incremental Profiling** option.
    In this scenario, the previous results excluding the data domain discovery results are purged during the next profile run. Enterprise Data Catalog displays the latest profile results.

Data Integration Service

You can configure multiple Data Integration Service (DIS) nodes in the Administrator tool under the **Services and Nodes** tab to extract profiling information from data sources.

You can create reusable configurations and select multiple DIS instances. If you select the DIS pool for profile or similarity execution, the system automatically assigns a DIS node based on capacity and availability.

If a user selects a specific DIS instance in the DISPool, all profiling and mapping tasks run on the selected DIS instance.

Source Connection Name

Choose the source connection to run data discovery. You can create the connections in Informatica Administrator.

Note: This parameter is optional for a File System resource.

Run On

Choose one of the following run-time environments to run the profile:

Blaze

Runs the profile in the Hadoop environment on the Blaze engine.

Spark

Runs the profile in the Hadoop environment on the Spark engine.

Native

Runs the profile on the same machine where the Data Integration Service runs.

Databricks

Runs the profile in the Hadoop environment on the Spark engine in the Databricks cluster. The Databricks run-time environment supports JDBC and Azure Data Lake Store resources.

Note: Choose Blaze or Native run-time environment to run the profile job for all the resource except Hive resource. When you choose the Blaze engine or Spark engine, select a Hadoop connection to run the profiles.

Select Data Domain

Choose one of the following data domain options:

All Data Domains

Discovers all the data domains in the resource.

Specific Data Domain Groups

Discovers the data domains in the selected data domain groups.

In the Data Domain Groups field, choose one or more data domain groups.

Specific Data Domains

Discovers the selected data domains.

In the Data Domains field, choose one or more data domains.

In the **Library** workspace, you can view all the data domains and data domain groups that are available in Enterprise Data Catalog. To create a data domain or data domain group, navigate to **New** > **Data Domain**, or **New** > **Data Domain Group** page. In the **Library** workspace, you can view or delete data domains or data domain groups.

Use Conformance from

Choose one of the following conformance values for the data domain:

Data Domain

Uses the predefined conformance values that you configured for the data domains.

When you create a data domain, you configure the minimum percentage of source rows and minimum number of source rows as the conformance criteria for data domain match. These values are predefined conformance values.

Custom

Uses the conformance value that you enter in the **Custom Conformance Value** field for the data domains. The custom value overrides the predefined conformance values.

Data Domain Match Criteria

Choose one of the following conformance criteria for data domain match:

Percentage

Ratio of the number of matching rows divided by the total number of rows.

Rows

Total number of rows.

Enterprise Data Catalog uses the data conformance properties that you configured for the data domains. Navigate to **Library** > **Assets** > **Data Domains** to view the data domains. Open each data domain to view its configured properties.

Exclude Null Values from Data Domain Discovery

Choose the option to exclude the null values from the data source when you run data domain discovery. When you use this option, the data domain inference is more accurate and reliable. For example, you have a table with 100 rows and 30 rows contain null values . The conformance row count is 40. If you do not choose this option, data domain discovery runs on all the 100 rows to discover data domains which might result in an inaccurate inference. If you choose this option, data domain discovery runs on 70 rows and the results are more accurate.

When you select the minimum percentage of rows with the exclude null values option, the conformance percentage is the ratio of number of matching rows in a column to the number of rows that do not contain null values. For example, if the total number of rows in a column is T, the number of matching rows is M, the number of rows with null values is N, then the conformance percentage is M/(T-N)%.

Similarity Profile Data Preparation and Value Frequency Settings

Configure the column similarity properties to identify similar columns and value frequency in the resource.

Run Similarity Profile

Choose one of the following options:

- Yes. The profiling scanner scans the data source and prepares data to discover similar columns based on column names, column patterns, and unique values.
- No

Save Source Data

Choose one of the following options:

- Yes. The profiling scanner prepares data to discover similar columns based on column names, column patterns, and unique values. It also computes value frequencies. The scanner then persists the computed information in PostgreSQL. The computed information persists in PostgreSQL until you choose to delete or purge the resource.
- No. The profiling scanner prepares data to discover similar columns based on column names, column patterns, and unique values. The scanner then persists the computed information in PostgreSQL. The computed information persists in PostgreSQL until you choose to delete or purge the resource.

Sampling Option

Choose one of the following sampling options to determine the number of rows that Enterprise Data Catalog can run the profile on:

- Reuse Basic Profile Settings. Use the sampling option in the **Basic Profile Settings** section.
- All rows. Runs the profile on all the rows in the data source.
- Auto Random rows. Runs the profile on a random sample of rows. Enterprise Data Catalog computes the number of random rows based on the number of source rows.
- Random N rows. Runs the profile on the configured number of random rows.
  In the **Random Sampling Rows** field, enter the number of rows that you want to run the profile on.
- First N rows. Runs the profile on the first N number of rows in the resource.
  In the **Number of First N Sampling Rows** field, enter the number of rows to run the profile on.

Domain Connection Settings

Choose one of the following domain connection settings options:

- Use Profile Configuration Settings. Enterprise Data Catalog uses the Data Integration Service specified in the Domain Connection Settings section to identify similar columns in the data sources.
- Specify Domain Connection Settings. To use a different Data Integration Service to identify similar columns in the data sources, enter the domain connection settings for the Data Integration Service.

Permissions and Privileges

You can view the value frequency section in Enterprise Data Catalog if you have the following permissions and privileges:

- In the Administrator tool, the administrator must assign the **Data Privileges: View Data** privilege to the user.
- In Catalog Administrator, navigate to the **Manage** > **Security** > **Resources** page, and assign **Metadata and Data Read** or **All Permissions** permission to the resource.
- In Catalog Administrator, assign the **Read** permission to the DataDomain resource in the **Manage** > **Security** > **Resources** page.

If the data asset has sensitive data, then you can view the sensitive data in the data asset after the administrator assigns the **Data Privileges: View Sensitive Data** privilege. For more information about privileges and permissions, see the *Informatica Administrator Reference for Catalog* guide.

Unique Key Inference Settings

A unique key is a column or combinations of columns that uniquely identifies a row in a data source. The profiling service identifies columns in the data object to generate unique keys. Enterprise Data Catalog displays unique key inferences for the tabular assets.

The unique key cannot have duplicate values. If a column has a duplicate values, that column is not identified as the unique key. The unique key inference is supported on the native run-time environment.

In the **Unique Key Inference Settings** section, you can configure the following options for a resource to generate unique key:

Run Unique Key Inference

The profile scanner scans and infers unique keys from the data source.

Null Threshold % in Unique key Inference

Sets the threshold for null values in the unique key inference. You can enter a value between 0 and 1.

Skip Unique Key Inference When Accepted or Documented Unique Key Exists

Skips the table with the documented or accepted unique keys.

Unique Key Sampling Options

You can choose the following sampling options:

- All Rows. Runs the unique key inference on all the rows in the data object.
- First <number> Rows. Runs the unique key inference on the selected rows in the data object.

The following resources support the unique key inference:

Relational Resource Type

- Amazon Redshift
- Hive
- IBM Netezza
- Microsoft SQL Server
- Oracle
- Snowflake Advanced Scanner
- Teradata

File based Resources

- Amazon S3
- ADLS Gen1
- ADLS Gen 2 V2
- Azure Blob
- File System
- HDFS

File Types

CSV Files

When you configure a non-supported resource to infer unique keys, the following error message appears:

```
Unique Key Inference is not Supported for the Resource Type:Resource Type
```

Column Similarity

As a data analyst or data architect, you can scan your enterprise data to find similar columns. When you run the Data Domain Propagation scanner or when the Data Domain Propagation scanner runs based on a schedule, then the scanner uses the inferred similar columns to propagate smart data domains to other similar columns. This process allows you to search and discover assets of interest faster in the catalog.

In the Catalog Administrator, when you run a resource after you choose the **Run Similarity Profile** option, and then create and run the **SimilarityDiscovery** resource, the scanner discovers similar columns based on column name, column data pattern, and unique values factors in the resource.

By default, Enterprise Data Catalog uses all the factors to propagate the smart data domains to other similar columns. You can choose one or more factors to propagate smart data domains to other similar columns. For example, you identify an employee ID column as a potential data domain. You create a smart data domain for the column. To propagate this to other similar columns, you can choose only the name match and pattern match factors.

You can identify the frequency of values after you enable column similarity for a resource and choose the Save Source Data option. Based on your business requirement, you can use the value frequency to analyze data in a resource. You can compute value frequency in the view column, table column, CSV field, XML file field, and JSON file field for relational sources, semi-structured sources, and for sources that you can run the column profile on.

How Column Similarity Works

In an organization, a column name such as customer ID might exist across multiple data sources. To identify the data sources that contain such columns, you can use column similarity in Enterprise Data Catalog. It uses unsupervised clustering which is a machine learning technique to identify the similar columns. Enterprise Data Catalog performs unsupervised clustering across multiple data sources based on several factors, such as distinct value match, pattern match, and name match. It then assigns an overall similarity score as well as the match likelihood for each factor.

To determine column similarity, the unsupervised clustering uses the following factors:

Distinct value match

Determines similar columns across data sources based on distinct values. The distinct values overlap is calculated as a percentage of distinct values that overlap in two columns.

The catalog displays this measurement as **Distinct Values** in the **Similar Columns** > **Confidence%** section.

Pattern match

Determines the column data pattern in the data sources and calculates the percentage of pattern overlap across column pairs for all data types except numeric data types. The pattern match uses profiling to identify the dominant data patterns for each column and field. The pattern match, then, checks for an overlap of these patterns across column pairs. Enterprise Data Catalog does not infer X, XX, and XXX patterns for all data types.

The catalog displays the percentage as **Pattern** in the **Similar Columns** > **Confidence%** section.

For example, pattern match identifies column pairs that might contain the credit card pattern.

Name match

Determines similar columns across data sources based on the column names. The name match uses fuzzy string match to identify the columns that have similar names.

The catalog displays the percentage as **Name** in the **Similar Columns** > **Confidence%** section.

For example, name match can identify all the columns that has the column name, customerID.

Column Similarity Process

The column similarity process includes staging of data, discover, and ingestion of data in the catalog.

Column similarity process includes the following jobs:

Prepare Data

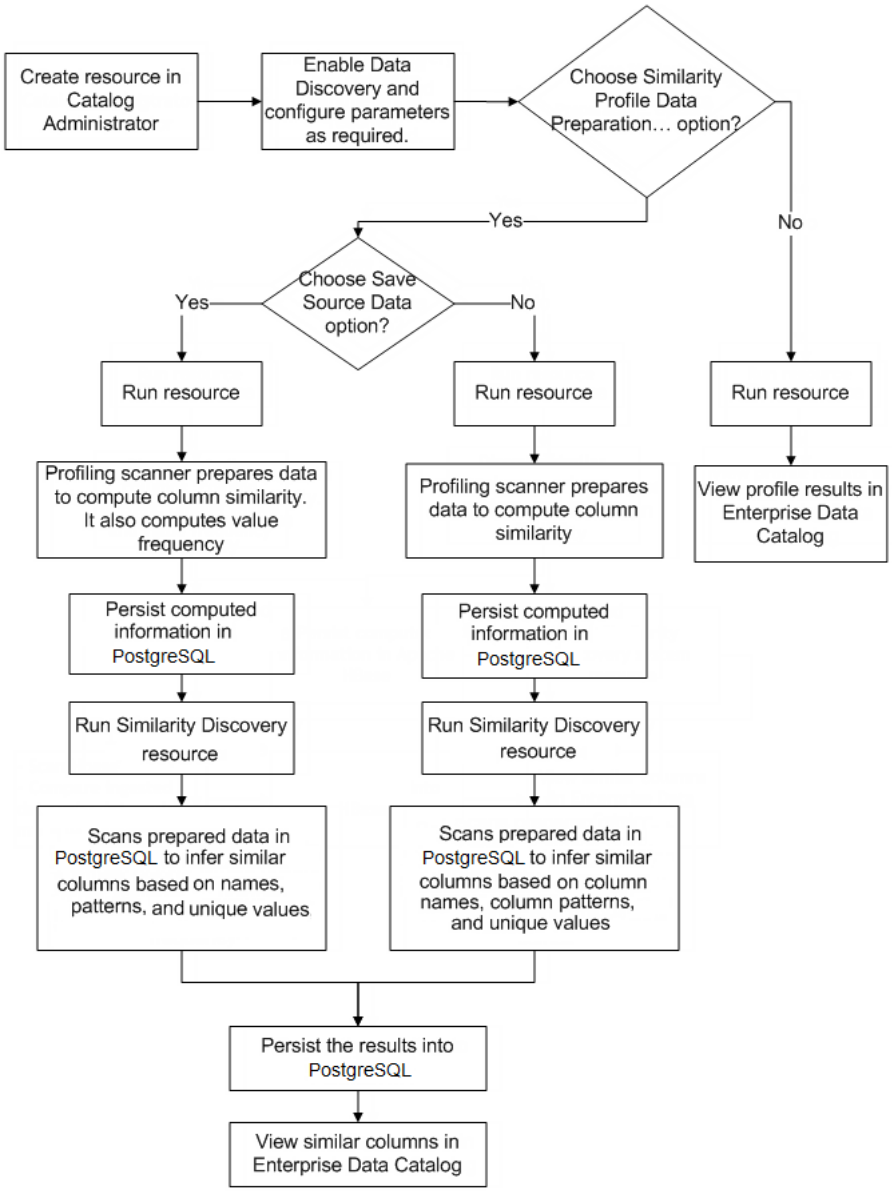
After you configure the parameters in the **Similarity Profile Data Preparation and Value Frequency Settings** section and run the resource, the profiling scanner prepares the data in the resource to infer similar columns and then persists the computed information in PostgreSQL. Similarly, you can prepare and persist data for multiple resources.

Discover Similar Columns

To identify similar columns, create and run the Informatica Similarity Discovery resource. The Informatica Similarity Discovery resource runs on multiple resources to discover similar columns in the catalog. When you run the Informatica Similarity Discovery resource, it scans PostgreSQL, compares the prepared data to infer similar columns, and persists the results into PostgreSQL.

Informatica administrators can either install the PostgreSQL database server bundled with the Enterprise Data Catalog installer or configure an external PostgreSQL database after installation. Informatica administrators can configure an external PostgreSQL database if you run the Informatica Similarity Discovery resource. For more information on how Informatica administrators can configure an external PostgreSQL database, see the KB article https://knowledge.informatica.com/s/article/You-can-t-create-logical-partitions-or-configure-high-availability-and-failover-options-when-you-use-the-PostgreSQL-database-bundled-with-the-Enterprise-Data-Catalog-installer?language=en_US.

The following image shows the column similarity process:



In Catalog Administrator, you can enable data discovery and configure the resource properties to discover column similarity for multiple resources. After you run the resources and Informatica Similarity Discovery resource in Catalog Administrator, you can view similar columns based on column names, column data patterns, and unique values for the data assets in Enterprise Data Catalog.

Choosing Factors to Propagate Smart Data Domains

You can choose one or more factors to propagate the smart data domains to other similar columns.

1. In Catalog Administrator, create or open a resource.
2. Run the resource after you choose the **Run Similarity Profile** option.
3. Click **Library** > **Resource**.
4. Create and run the **Informatica Similarity Discovery** resource.

   The catalog shows the assets related to the resource.

5. Create and save a text file on your local machine. Name the file as necessary.
6. Edit the file. Enter a smart data domain and the chosen factor abbreviations in the following format:

   <data domain name>=<factor1>,<factor2>

   The factor abbreviations are N for name match, P for pattern match, and U for distinct value match.
7. Continue to enter other data domains and their chosen factors in the file. Ensure that each data domain and its factors are in a new line. Data domain names are case-insensitive. Do not enter a whitespace in the file.

   The following snippet shows a sample text file:

   ```
   SDD_Oracle_DD1=P
   SDD_PatternCheck=P
   DD_NameCheck=N,VF
   DD_Check_Pattern=P,U
   ```
8. Save the file.
9. In the Catalog Administrator, edit the **DataDomainPropagation** system resource.
10. On the **Metadata Load Settings** tab, choose the text file for the **Enabled Similarity Types** option.
11. You can configure the following optional fields as required:
    - Choose to enable or disable the confidence score threshold.
      - Choose yes to enable the confidence score threshold and accept the smart data domains if the data domain match exceeds the configured auto accept threshold value. Specify the inference threshold and auto accept threshold for the **DataDomainPropagation** system resource.
      - Choose no if you want to curate the inferred smart data domains separately.
    - **Inference Threshold**. Enter a minimum conformance percentage required to consider the smart data domains for inference.
    - **Auto Accept Threshold**. Enter a percentage to accept the smart data domains if the data domain match exceeds the configured percentage.
12. Save and run the system resource.

    The **DataDomainPropagation** scanner propagates the smart data domains in the text file based on the chosen factors. To propagate the rest of the data domains, the scanner uses name match, pattern match, and distinct value match factors.

Business Example

Alex is a data analyst at a financial institution that has branches and franchises across North America. The institution has recently acquired another financial institution equal in size. There are customers who hold accounts in both the financial institutions. Alex and his team are asked to integrate all the customer details in a single database. Alex also wants to search for the customers based on the regions such as Northeast, South, Midwest, and West.

The following are the challenges that Alex and his team face:

- Browse multiple sources to identify similar customer data.
- Identify the lineage and impact analysis for data before removing duplicate data.
- Identify data assets that can be joined.
- Tag similar columns with additional attributes so that Alex and his team can search for required data faster.

The following table lists the scenarios that Alex and his team need to manage, and how Alex uses the Informatica Similarity Discovery resource and data domains to extract the required information:

Scenario	Resolution
Different database systems used by the financial institution and the acquired institution.	Identify the data sources that need to be scanned to find the required customers that match the eligibility criteria. Add these data sources as resources in Catalog Administrator to extract metadata from these resources. Alex identifies the databases in the enterprise that include the customer details.
Lack of consistency and context in the column names that makes it difficult to find and analyze source columns with similar data.	In Catalog Administrator, enable Data Discovery and Run Similarity Profile options for the selected resources. The profiling scanner and Informatica Similarity Discovery resource identifies the similar columns in the resources. The similar columns are identified based on column names, column data patterns, and unique values. Enterprise Data Catalog displays the similar columns for the data assets. Alex uses similar columns to identify the columns that contain similar data across all the data sources, identify data that can be joined, and duplicate data that can be removed. For example, Alex finds out from an existing bank report that both the organizations store the Social Security Number on all records that have customer information. Based on this information, he infers that if the tables contain a column with SSN information then the customer details might be present in the tables. To verify, Alex searches for an SSN column in the Catalog, Enterprise Data Catalog lists the searched column along with other columns from all the data sources that are similar to the searched column. Based on the search results, Alex and his team identify data that can be joined and duplicate data that can be removed.

Scenario	Resolution
Identify the lineage for each data asset, the other assets that are related to a particular asset, and the impact that joining or deleting a specific data asset might cause for the other related data assets.	In Enterprise Data Catalog, view the lineage, impact summary, and relationship view for identified assets. Alex and his team can view the lineage, impact summary, and related asset details to identify the impact before updating or deleting a specific asset.
Classify customers based on the regions and make searches faster.	In Catalog Administrator, create data domains and data domain groups to find a list of columns with customer details specific to a region. Alex performs the following steps to define data domains and data domain groups to classify customers based on the regions: 1. Creates a data domain called `customer_details`. 2. Assigns the data domain to one of the columns that contain the SSN in Enterprise Data Catalog. 3. Creates `ZIP_code_<area>` data domain. In Informatica Analyst or Informatica Developer, Alex specifies a rule to replace the `<area>` parameter in the data domain with the branch locations of the financial institutions. Enterprise Data Catalog uses the rule to match a column pattern with the ZIP code for a specific branch. **Note:** A rule is business logic that defines conditions applied to data when you run a profile. You can add a rule to the profile to cleanse, modify, or validate the data in the profile. 4. Creates four data domain groups based on the regions called `Northeast`, `South`, `Midwest`, and `West`, and includes the data domains in the respective data domain group. For example, the data domain that corresponds to the `ZIP_code_LosAngeles` ZIP Code is included in the `West` data domain group. 5. Alex performs a search in Enterprise Data Catalog for customer_details. Enterprise Data Catalog lists all the columns that include SSN details of the customers and also shows the data domains (`ZIP_code_<area>`) and the data domain groups associated with the column. Alex can also search based on the defined data domain groups to find a list of columns with customer details specific to a region.

Business Term Propagation

Enterprise Data Catalog identifies potential business terms that can be associated with columns or fields in the catalog. Enterprise Data Catalog periodically runs the **DataDomainPropagation** system resource to automatically accept or recommend the business terms for similar columns. This process minimizes manual intervention to identify and associate business terms with tables, columns, views, or fields.

In Enterprise Data Catalog, a business term with the confidence score of more than 80 percent is automatically assigned as a business title to an asset. The confidence score is calculated by an internal algorithm based on accepted business terms on data domains, column similarity, and name match between a column and business term. If there are multiple business terms with a threshold of more than 80 percent, the term with the highest confidence score is automatically assigned as a business title. Enterprise Data Catalog displays a maximum of three recommended business terms if the confidence score is between 60 and 80 percent for a column.

Business term propagation process involves the following steps:

1. **You Configure and Run a Resource**

   You create a resource and configure the parameters in the **Metadata Load Settings** > **Similarity Profile Data Preparation and Value Frequency Settings** section, and run the resource. Alternatively, along with the **Run Similarity Profile** option, configure the parameters in the **Metadata Load Settings** > **Business Term Association** section, and run the resource.

   Note: You can choose to configure only the **Business Term Association** without configuring the **Run Similarity Profile** option for a resource to propagate business terms.

   For more information about the **Business Term Association** configuration, see the "Enable Business Term Association " on page 75 topic.

2. **You Run a Business Glossary Resource or Axon Resource**

   You create and run a Business Glossary resource or Axon resource to retrieve and display the business terms in the catalog.

3. **You create and run the Informatica Similarity Discovery Resource**

   You can create and runs the **Informatica Similarity Discovery** resource. The **SimilarityDiscovery** system resource discovers similar columns based on column name, column data pattern, and unique values in the resource.

   Note: You do not have to configure the **Run Similarity Profile** option if you choose to propagate business terms only with the **Business Term Association** option.

4. **Enterprise Data Catalog Runs DataDomainPropagation System Resource**

   Enterprise Data Catalog runs the **DataDomainPropagation** system resource on a predefined schedule of 120 minutes. The **DataDomainPropagation** system resource performs the following tasks:

   - The scanner uses the inferred similar columns to propagate the smart data domains to other similar columns.
   - The scanner automatically accepts or recommends the business terms for other similar columns based on the confidence score.

The scanner automatically accepts business terms or displays them as recommendations for columns when one of the following conditions is true:

- A data domain has an accepted business term
- A column has an accepted business term
- The names of the column and business term are similar

Business Glossary Synchronization

When you associate a business term with a technical asset in Enterprise Data Catalog, the relationship information is updated in the Analyst tool business glossary. When you update asset links for a business term in an Analyst tool business glossary, Enterprise Data Catalog can extract the updated link information if the glossary is assigned to resource BG_DEFAULT_RESOURCE.

Synchronization between Enterprise Data Catalog and the Analyst tool business glossary occurs in the following ways:

From the catalog to the Analyst tool business glossary.

When you associate a business term with a technical asset in the catalog, the relationship is displayed in the catalog and in the Analyst tool business glossary. When you open the term in the Analyst tool business glossary, the Analyst tool imports the asset link and displays it in the Related Assets section.

For example, you associate the business term "Annual Salary" with the SALARY column in an Oracle table. Enterprise Data Catalog displays the term Annual Salary next to the column name in the search results and Asset Details view. When you open the business term in the Analyst tool business glossary, the SALARY column is listed as a related data asset.

From the Analyst tool business glossary to the catalog.

When you update an asset link for a term in the Analyst tool business glossary, Enterprise Data Catalog can extract the updated link information. Enterprise Data Catalog can extract the link information if the glossary is assigned to resource BG_DEFAULT_RESOURCE.

When an Informatica administrator creates and enables an Analyst Service with Catalog Service properties, the Analyst Service creates a default business glossary resource named "BG_DEFAULT_RESOURCE." The catalog administrator assigns business glossaries to BG_DEFAULT_RESOURCE. The metadata load job that is associated with BG_DEFAULT_RESOURCE extracts information from the Analyst tool business glossary and imports it into the catalog. The metadata load job starts when an Analyst tool user creates or updates an asset link in a business glossary that is assigned to the resource.

For example, a catalog administrator assigns business glossary Finance_Glossary to BG_DEFAULT_RESOURCE. An Analyst tool user links business term "Planned Cost" in Finance_Glossary to a flat file data asset named "COSTS." This action starts the metadata load job for BG_DEFAULT_RESOURCE. When the metadata load job completes, you view the term "Planned Cost" in the catalog. The Asset Details view lists the data asset "COSTS" in the Other Assets section.

If you create a business glossary resource to extract assets from a glossary that is not assigned to BG_DEFAULT_RESOURCE, Enterprise Data Catalog does not automatically synchronize assets. To synchronize the assets, you must load the business glossary resource manually through Catalog Administrator. Alternatively, you can configure a schedule in Catalog Administrator to load the resource at regular intervals.

For more information about the Analyst tool business glossary, see the *Informatica Business Glossary Guide*. For more information about loading resources through Catalog Administrator, see the *Informatica Catalog Administrator Guide*.

Data Preview and Provisioning

An effective data catalog enables business intelligence users and data scientists to locate, preview, prepare, and provision data for ad-hoc analysis.

Data provisioning is the process of moving data from a source in the catalog to a target for further processing. You can preview data for table and file asset types, and create provisioning tasks in Enterprise Data Catalog. Preview a sample of the source data before you create a provisioning task.

Prerequisites

Data Preview and Provisioning Prerequisites

Before you enable the data preview and provisioning capability, make sure that the following prerequisites are met:

1. Ensure that you have access to the Informatica Intelligent Cloud Services account as an administrator.
2. Create an Informatica Intelligent Cloud Services organization.
3. Create the necessary database connections in Informatica Intelligent Cloud Services for the sources and targets.
4. Ensure that you have created the `EdcProvisionMapping` mapping using the **Copy data into an existing target** integration template in Informatica Intelligent Cloud Services.
5. Configure the **LdmCustomOptions.enableDataProvision** and **LdmCustomOptions.provision.ics.master.app.url** custom properties in the Catalog Service for data provisioning in Informatica Administrator console.
6. Add the Informatica Intelligent Cloud Services organizations to the Catalog Administrator tool.

Data Preview

You can preview a sample of the source data in Enterprise Data Catalog. You might want to preview data before you perform a provisioning task. Preview data to assess the source data before you transfer the data to the target. For example, consider an Oracle source that contains some information about customers, and you want to migrate the data to an Oracle target. Before you move data from source to target, you might want to verify the data in the table and know if the table contains relevant information. To verify the data, you can preview the table and ensure that the table contains relevant columns, such as `Address`, `Phone Number`, and other information about the customers.

You can perform data preview for the following sources:

- Amazon Redshift
- Amazon S3
- Azure Microsoft SQL Data Warehouse
- Azure Microsoft SQL Server
- Hive
- JDBC
- Microsoft SQL Server
- Oracle
- Salesforce
- Teradata

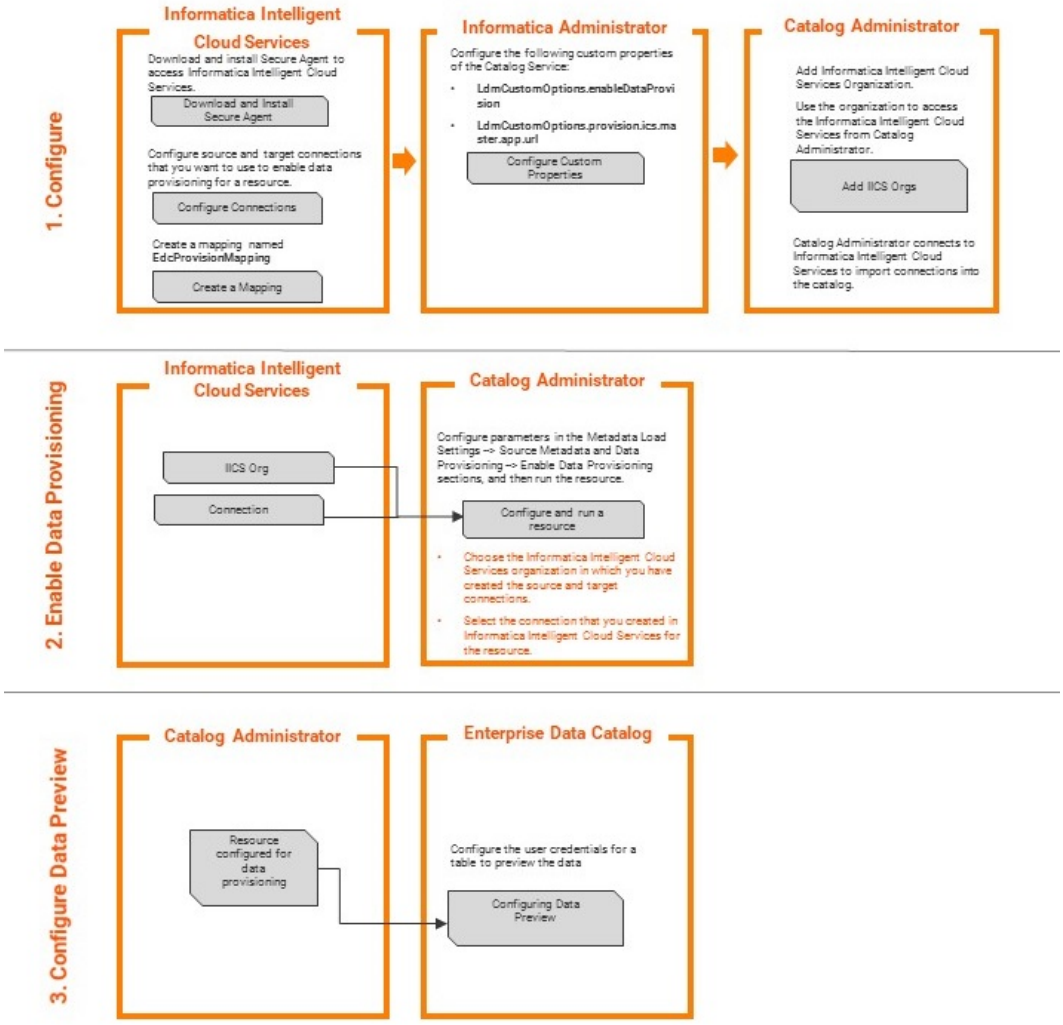
You can preview only table and file assets in the Catalog. Configure data provisioning in Catalog Administrator before you can preview data in the Catalog.

Alternatively, you can preview data for data object using the REST APIs. For more information, see the Preview Data for a Data Object topic in the *Enterprise Data Catalog REST API Reference* guide.

Data Preview Process

The data preview process involves various configuration steps in the Informatica Intelligent Cloud Services, Informatica Administrator, and Catalog Administrator. The data preview configuration steps are the same as the data provisioning configuration steps.

The following image shows the data preview process:



1. Configure. Set up Informatica Intelligent Cloud Services, configure custom properties for data provisioning in Informatica Administrator, and add Informatica Intelligent Cloud Services Organizations (Orgs) to Catalog Administrator.
2. Enable Data Provisioning. Configure a resource for data provisioning.
3. Configure Data Preview. Configure data preview for table and file asset types in Enterprise Data Catalog.

Data preview process involves the following steps:

1. **You configure custom properties for data provisioning in the Informatica Administrator console**

   For information about adding custom properties, see the Enable Data Preview, Provisioning and Preparation topic in the *Informatica Administrator for Enterprise Data Catalog* guide.

2. **You add Informatica Intelligent Cloud Services Organizations to Catalog Administrator**

   You add the organizations to get access to the Informatica Intelligent Cloud Services. Catalog Administrator connects to Informatica Intelligent Cloud Services to import database connections for the source and target. For more information about adding Informatica Intelligent Cloud Services Organizations, see the Adding an Informatica Intelligent Cloud Services Organization topic.

3. **You enable data provisioning for a resource**

   You create a resource and configure the parameters in the **Data Access** > **Enable Data Provisioning** section, and run the resource. When you enable data provisioning for the resource, Enterprise Data Catalog fetches source connection information for the data that you want to preview.

   Note: If you want to enable data provisioning for a resource, you must select the same schema name or directory name that you configured for a connection in Informatica Intelligent Cloud Services.

   For more information about enabling data provisioning for a resource, see the Enable Data Provisioning topic.

4. **You configure the connection properties for the data source in Enterprise Data Catalog**

   You specify the database user credentials in the connection properties for the data source, and then preview data.
   For more information about configuring connection properties for the source data, see the Data Preview topic in the *Informatica Enterprise Data Catalog User Guide*.

Data Provisioning

Data Provisioning is a process of transferring data from the source to target for further analysis and processing. You can provision tables and files to the target. Enterprise Data Catalog uses a mapping and database connections from Informatica Intelligent Cloud Services to perform provisioning of data. A mapping is a set of inputs and outputs that represent the data flow between sources and targets. A mapping contains components such as source objects, target objects, and transformations. Enable data provisioning in Catalog Administrator. After you enable data provisioning for a resource, you can create provisioning tasks in Enterprise Data Catalog. You can create provisioning tasks for table and file asset types.

You can perform data provisioning for the following sources:

- Amazon Redshift
- Amazon S3
- Azure Data Lake Store
- Azure Microsoft SQL Data Warehouse
- Azure Microsoft SQL Server
- Hive
- JDBC
- Microsoft SQL Server
- Microsoft Azure Blob Storage
- Oracle
- Salesforce
- Teradata

You can perform data provisioning for the following targets:

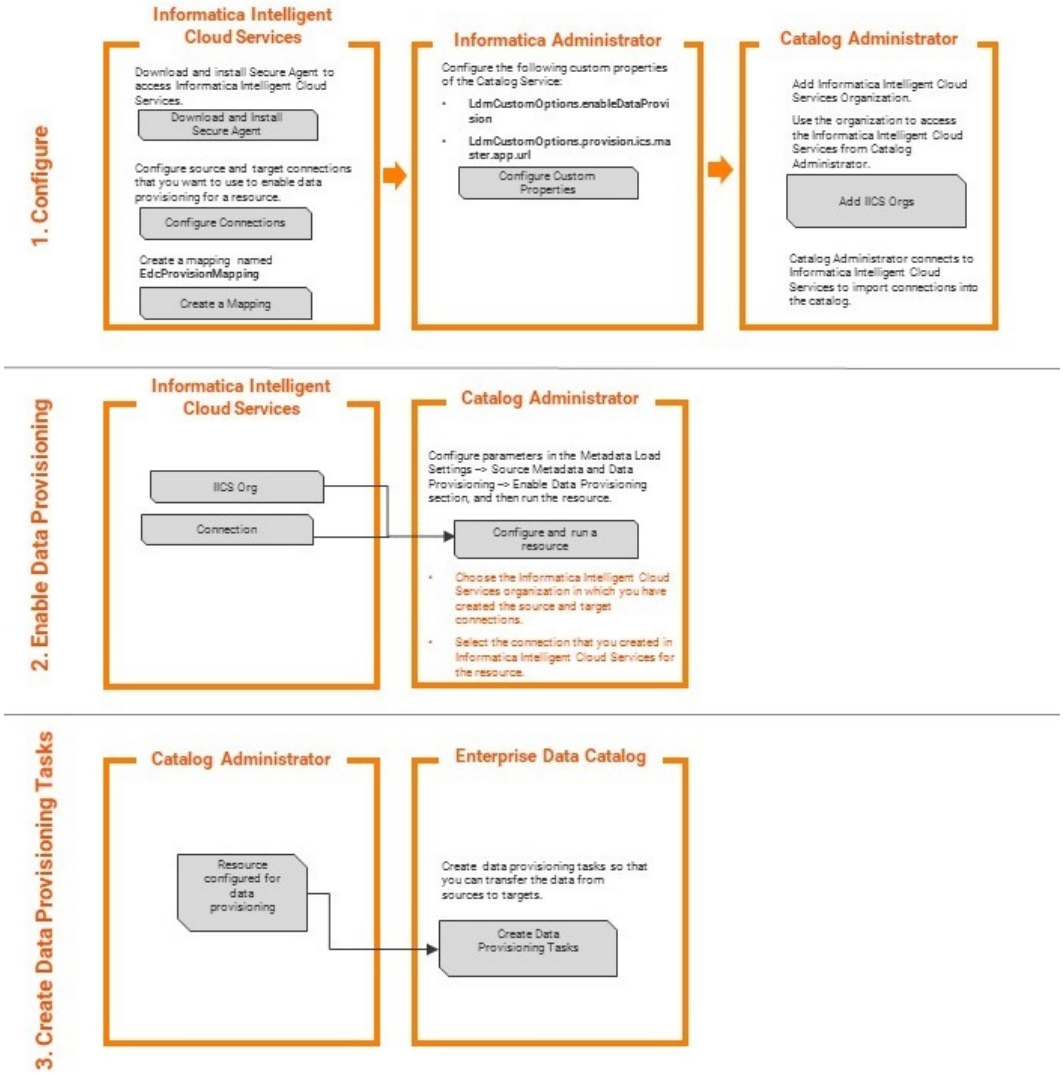
- Amazon Redshift

- Amazon S3
- Azure Data Lake Store
- Azure Microsoft SQL Data Warehouse
- Azure Microsoft SQL Server
- Google BigQuery
- Google Cloud Storage
- HDFS
- Hive
- JDBC
- Microsoft Azure Blob Storage
- Microsoft SQL Server
- Oracle
- QlikView
- Teradata
- Tableau Online
- Tableau Server

Data Provisioning Process

The data provisioning process involves various configuration steps in the Informatica Intelligent Cloud Services, Informatica Administrator, and Catalog Administrator.

The following image shows the data provisioning process:



1. Configure. Set up Informatica Intelligent Cloud Services, configure custom properties for data provisioning in Informatica Administrator, and add Informatica Intelligent Cloud Services Organizations (Orgs) to Catalog Administrator.
2. Enable Data Provisioning. Configure a resource for data provisioning.
3. Create Data Provisioning Tasks. Create data provisioning tasks for assets in the resource that is configured for data provisioning in Enterprise Data Catalog.

The Data Provisioning process involves the following steps:

1. **You configure custom properties for data provisioning in Informatica Administrator console**

   For information about adding custom properties, see the Enable Data Preview, Provisioning and Preparation topic in the *Informatica Administrator for Enterprise Data Catalog* guide.

2. **You add Informatica Intelligent Cloud Services Organizations to Catalog Administrator**

   You add the organizations to get access to the Informatica Intelligent Cloud Services. Catalog Administrator connects to Informatica Intelligent Cloud Services to import database connections for

the source and target. For more information about adding Informatica Intelligent Cloud Services Organizations, see the Adding an Informatica Intelligent Cloud Services Organization topic.

3. **You configure and run a resource in Catalog Administrator**

   You create a resource and configure parameters in the **Metadata Load Settings** > **Source Metadata** and **Data Access** > **Enable Data Provisioning** section, and then run the resource.

   Note: If you want to enable data provisioning for a resource, you must select the same schema name that you configured for a connection in Informatica Intelligent Cloud Services.

   For more information about enabling data provisioning for a resource, see the Enable Data Provisioning topic.

4. **You create data provisioning tasks in Enterprise Data Catalog**

   You configure the source and target connection information of the database to provision data. For more information about creating data provisioning tasks, see the Creating Provisioning Task topic in the *Informatica Enterprise Data Catalog User Guide.*

Alternatively, you can perform this task using the REST APIs. For more information, see the Data Provisioning REST APIs and List All the Provisioning Tasks topics in the *Enterprise Data Catalog REST API Reference* guide.

CHAPTER 3

Using Catalog Administrator

This chapter includes the following topics:

- Catalog Administrator Overview, 54
- Overview Workspace, 56
- Resource Workspace, 56
- Monitoring Workspace, 56
- Library Workspace, 57
- Data Domains Workspace, 57

Catalog Administrator Overview

Catalog Administrator is the administration tool that you can use to perform administrative tasks, such as the management of resources, schedules, and attributes.

Use the Catalog Administrator to complete the following types of tasks:

Manage Resources

Create, configure, edit, and remove resources. A resource is an object that represents an external data source or metadata repository from where scanners extract metadata. Enterprise Data Catalog performs all basic operations, such as extracting metadata, storing metadata in the Informatica cluster, and managing metadata, at the resource level.

Manage Schedules

Create schedules that you can attach to resources. You can create global, recurring schedules that you can assign to multiple resources.

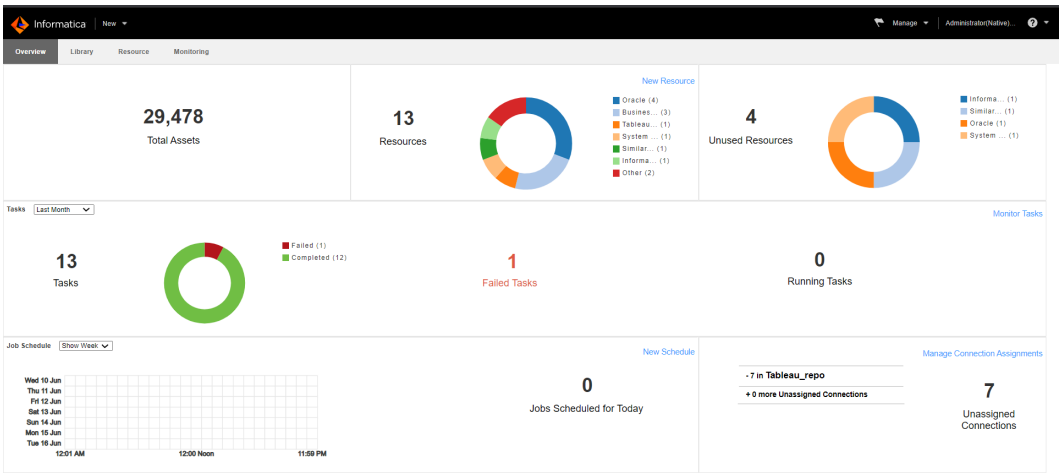
Manage Attributes

Assign predefined system attributes to specific metadata object types, such as table, column, report, and resource. You can create custom attributes and assign them to metadata object types based on the business requirements. Attributes assigned to resources can help Enterprise Data Catalog users to quickly find the data assets and related information. You can configure the system and custom attributes so that Enterprise Data Catalog displays the attributes as search filters.

Manage PowerCenter, SAP Business Objects, and Big Data Object Connections

You can view the details of connections that are automatically assigned to each resource. You can also view assigned and unassigned PowerCenter, SAP Business Objects, Cloudera, and Hive object connections that are user-defined and the schemas for each connection. You can assign specific schemas to the appropriate resources. Unassign the connections and schemas as required.

The following image displays the Catalog Administrator interface:



Catalog Administrator has the following tabs:

Overview

View the monitoring statistics for resources and tasks. You can view the task distribution by the task status and running time. You can also view the resource distribution and predictive job load for the current week.

Resource

Create resources. You can also open the recently configured objects.

Monitoring

View monitoring statistics by the task type and task status. Apply filters to shortlist tasks and resources that meet specific conditions.

Library

View the list of resources and schedules. Open a resource or schedule for further analysis.

Catalog Administrator has the following header items:

New

Create resources and schedules.

Manage

Manage system and custom attributes, connection assignments, and reusable configuration settings.

Administrator

Change the password for Catalog Administrator, and log out of Catalog Administrator.

Help

Access specific help topics for the current workspace or page, launch the online help, and view the Informatica version.

Overview Workspace

The Overview workspace displays visual charts that represent monitoring statistics of tasks, resources, and system load. Click the appropriate sections of the interactive charts to view the details.

You can view the following visual, monitoring statistics on the Overview workspace:

- Total number of assets in the catalog.
- Total number of resources and the number of resources by resource type. These numbers can help you monitor the resource load in Catalog Administrator.
- Total number of configured resources that you did not use yet to extract metadata. You can also view the number of such resources by resource type.
- Total number of tasks and their statuses. The task statuses include Running, Failed, Queued, and Paused. You can focus on any unexpected job discrepancy based on the task status details. You can view all the tasks in the last 24 hours, last week, or last month.
- Number of failed tasks and running tasks that need your attention. Click the task link to view the details in the Monitoring workspace.
- Number of running tasks based on the task run time. For example, you can view the details of the tasks that are running for more than a day or from 4 through 12 hours.
- Graphical representation of the predictive job load in terms of the number of jobs at different time slots in the current week or day.
- Total number of jobs scheduled for the day.
- Total number of unassigned connections.

Resource Workspace

You can create resources from the Resource workspace. You can also launch the recently opened data assets.

To create a resource, click the link under the **New Assets** panel. To open a recently opened data asset, click the asset under the **Recently Opened** panel.

Monitoring Workspace

You can monitor the status of Enterprise Data Catalog tasks on the Monitoring workspace. The workspace displays visual representation of tasks and their distribution. You can view the status of tasks, such as Running, Failed, and Paused.

You can view the following details of tasks in the bottom pane of the Monitoring workspace:

- Type. Indicates the type of task, such as metadata load, profile executor, and profile result fetcher.
- Resource name. Displays the name of the resource.
- Schedule. Displays the name of the schedule associated with the resource.
- Triggered by. Indicates whether the task was triggered manually or as part of a configured schedule.

- Status. Displays the task status, such as Running, Failed, Complete, and Paused.
- Start time of the task.
- End time of the task.
- Run time of the task.
- Next schedule. Displays the date and time when the scanner runs the job next.
- Log URL. Displays a link that you can click to open the log file for the task for completed tasks.

You can refresh the Monitoring workspace to view the latest tasks and task statuses. You can also apply filters on the task list based on conditions, such as resource type, job creation time, and job fail history.

Library Workspace

Use the Library workspace to browse, view, search, and apply filters on a collection of resources, composite data domains, data domains, data domain groups, reference resources, and schedules that you have the user privilege to access.

In Library workspace, specify the search criteria to find a specific resource or group of resources. You can sort the resources by resource name or resource type. You can also group the resources based on specific requirements.

You can open a resource or schedule from the assets list. When you click a resource, the resource opens in the Resource workspace. If you click a schedule, it opens in the Schedules workspace. You can filter the resource list based on multiple conditions, such as Resource Name, Created by, and Resource Type. You can filter a schedule based on conditions, such as schedule name and the time you created the schedule.

Data Domains Workspace

The Data Domains workspace appears when you view or create a data domain or a data domain group. You can also launch the recently opened data domains and data domain groups using the workspace.

To open the Data Domain workspace, perform any of the following steps:

- Click **New** > **Data Domain**
- Click **New** > **Data Domain Group**
- From the **Library** workspace, select **Data Domains** or **Data Domain Groups** and click an existing data domain or a data domain group to open the data domain or group details in the Data Domains workspace.

To open a recently opened data domain, point the mouse pointer on the **Data Domains** workspace tab and click the **Recently Opened** option that appears.

CHAPTER 4

Managing Resources

This chapter includes the following topics:

- Managing Resources Overview, 58
- Resources and Scanners, 59
- Resources and Schedules, 59
- Resources and Attributes, 59
- Reference Resources, 60
- Viewing Reference Resources, 62
- Creating a Resource, 63
- Resource Type, 64
- Dataset or Schema Profile Filter, 65
- Source Metadata and Data Profile Filter, 66
- Case Insensitive Linking , 67
- Enable Data Discovery, 67
- Composite Data Domain Discovery, 74
- Enable Business Term Association , 75
- Enable Data Provisioning , 77
- Editing a Resource, 78
- Cloning a Resource, 78
- Running a Scan on a Resource, 79
- System Resources, 79
- Viewing a Resource, 80

Managing Resources Overview

A resource is a repository object that represents an external data source or metadata repository from where scanners extracts metadata. The basic metadata operations, such as extraction, storage, and management of metadata, are performed at the resource level.

You can create, edit, and remove a resource. A resource has a resource type that determines the type of data source from where scanners extract metadata. You can also choose the specific types of metadata that you want scanners to extract from the data sources. For example, you can choose to extract basic source metadata or both basic source metadata and profile metadata. You can view classifications assigned to

resources. The classifications help you to quickly search and filter specific resources in Enterprise Data Catalog.

A resource has the following characteristics:

- A resource has a resource type that identifies the type of source system.
- A resource has a global, unique identity.
- A resource can have a one-time schedule or recurring schedule. You can attach a schedule to one resource or attach a reusable schedule to multiple resources.

Resources and Scanners

Scanners attached to resources identify the resource type including the name of the resource type, display name, description, and supported versions. You can configure multiple scanners for different types of metadata. You can configure the scanner properties for a resource when you create the resource. Each scanner has a unique ID that maps to the resource type.

A resource type can have multiple scanners. For example, one scanner might extract the metadata for an Oracle data source directly from the source. Another scanner might derive the metadata about the Oracle data source from other metadata repositories, such as a PowerCenter repository or Model repository.

Resources and Schedules

Resource schedules determine the frequency with which scanners extract metadata from the metadata sources. You can create schedules for specific resources or create reusable schedules that you want to assign to multiple resources.

You can create schedules that have an end date or schedules that run indefinitely. In addition to a start date, you can also set up the start time based on when you want scanners to start extracting metadata.

Resources and Attributes

Resources have attributes that represent specific properties of object types, such Alias Column, Category, Data Set, and Table, based on the data types of the attributes. You can use the resource attributes in Enterprise Data Catalog to search and find relevant information among a large number of data assets and data relationships.

Resource can have associated custom attributes. System attributes include predefined attributes that represent specific properties of object types, such as stored procedure, table, trigger, XML query, schema, and resource. For example, Comment is a system attribute assigned to a Hive table or Hive view. You can create custom attributes based on your business requirements that you want to assign to different resources. You can choose to create custom attributes based on the scanner types that are in use.

Reference Resources

A reference resource indicates that the resource you configured includes references from other data sources or resources. Assets from reference resources are known as reference assets.

Resources configured in Enterprise Data Catalog might include lineage information and connection data from other data sources or resources. Though the information about the data source might not exist in the catalog, you can view the data source as a reference resource when you view the lineage for assets in the resource.

For example, an Informatica PowerCenter resource that you want to configure might have mappings that point to tables or columns in an Oracle database. The PowerCenter resource is referred to as the primary resource. The Oracle database is a reference resource for the PowerCenter resource. The tables and columns in the Oracle database are reference assets from the reference resource.

After you extract reference resources and assets, you can view the lineage for an asset in the catalog with the reference assets. Reference assets include reference data sources and reference data sets that you can view in Enterprise Data Catalog. You can configure resources in Enterprise Data Catalog to include extraction of reference resources and assets. For more information about configuring resources to extract metadata about reference resources and assets, see the *Enterprise Data Catalog Scanner Configuration Guide*.

To view the lineage with the assets included in the reference resource data sources, you must configure resources for the reference resource data sources in Enterprise Data Catalog. You must then perform connection assignments between the reference resources and the resources you configured. For example, to view the column details from an Oracle table in an Oracle database referenced to by a PowerCenter resource, you must add the Oracle resource in Enterprise Data Catalog and perform connection assignments between the reference resource and the schema in the Oracle resource.

You can also assign connections from a reference resource to other reference resources. For example, if a Microsoft SQL Server database is referenced by a PowerCenter and a Tableau resource, you can assign connections from the database reference resource to PowerCenter and Tableau.

Benefits of Extracting Reference Assets

After you extract reference assets, you can use these assets similar to how you use other regular assets in the catalog.

Extracting reference assets in the catalog provides the following benefits:

- Search for reference resources and assets in Enterprise Data Catalog.
- Filter the search results based on reference resource and assets.
- View lineage for a primary resource that includes reference resources and assets.
- View Lineage, Impact, and Control Summary for reference assets.
- Filter lineage based on reference assets.
- Annotate and enrich reference assets with custom attributes and business terms.
- Perform a bulk export or import of the custom attributes, business term associations, and other enrichments on reference assets.
- View custom lineage between reference resources.

For more information about search, filters, lineage, and custom attributes and business terms, and bulk import and export, see the *Enterprise Data Catalog User Guide*.

Assign Connections to Reference Resources

Assigning connections to reference resources involves assigning connections from the reference resources to the resources that represent the data sources or other reference resources.

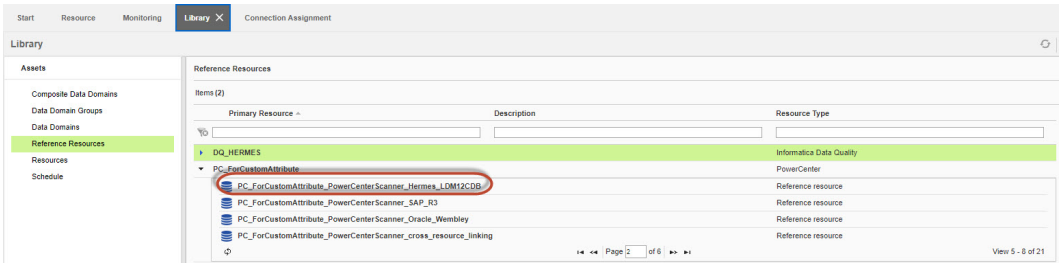
You must assign connections for reference resources to view the assets from the referenced data sources in the lineage. Before you assign connections, you must configure resources in Enterprise Data for the reference resource data sources. Refer to *Creating a Resource* section to create a resource. You can then perform the steps listed in the *Managing Connections* section to assign connections between reference resources and the configured resources. When you assign connections to reference resources, Enterprise Data Catalog converts the reference resources and assets to regular resources and assets in the catalog.

After you assign connections to reference resources, they cease to exist in the catalog as reference resources and assets. If you had enriched reference assets with multiple values for custom attributes and business terms, the values are merged and associated with the resource after connection assignment. If you had enriched reference assets with a single value for custom attributes and business terms, the most recent value is associated with the resource.

Sample Use Case

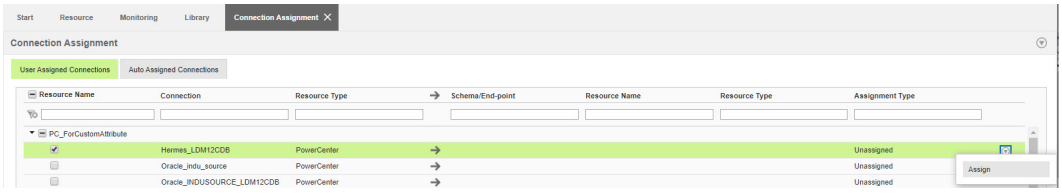
The sample use case describes the workflow involved in assigning connections between a sample reference resource and a schema in a resource in Enterprise Data Catalog.

A PowerCenter resource named `PC_ForCustomAttribute` includes the reference resource `PC_ForCustomAttribute_PowerCenterScanner_Hermes_LDM12CDB` on the **Reference Resources** page as shown in the following image:

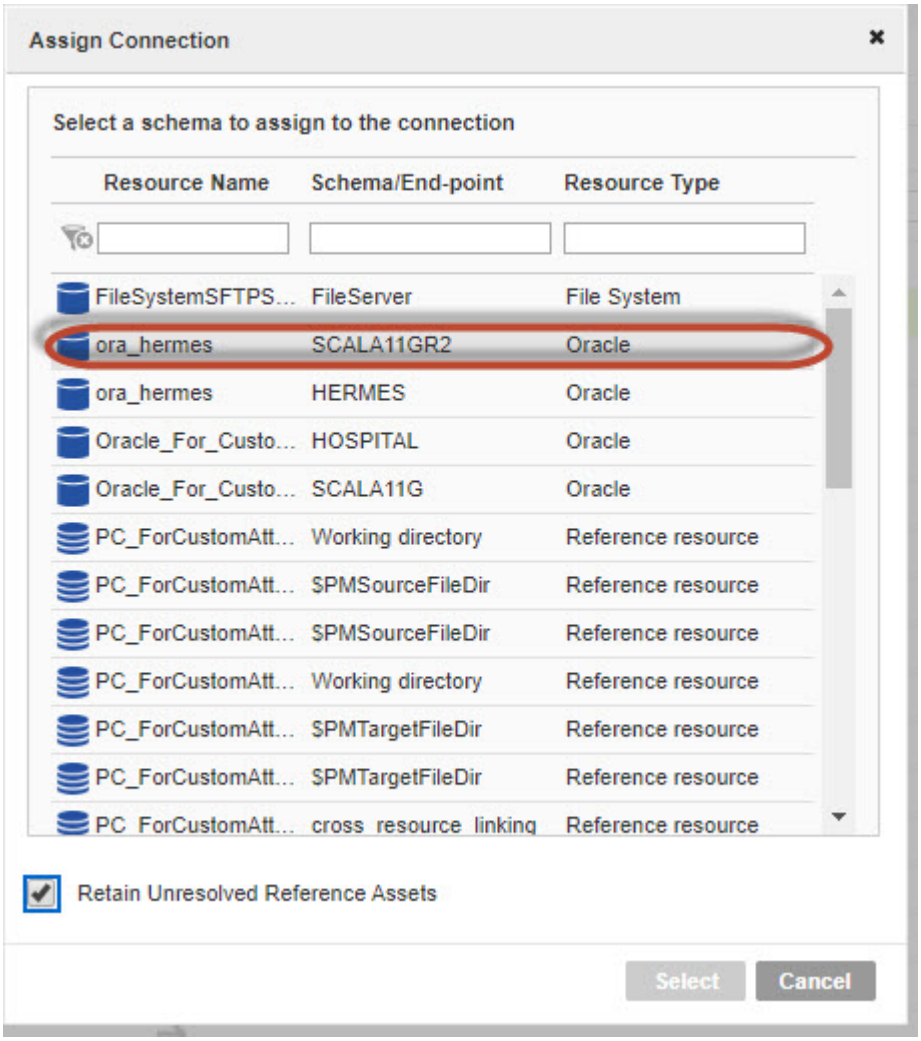


The reference resources are prefixed with the text `<resource name>_<Provider ID>_<Connection name>`.

From the **Connection Assignment** workspace, you can assign the connection in the reference resource to another resource as shown in the following image:



You can assign the schema **SCALA11GR2** in the Oracle resource **ora_hermes** to the connection `Hermes_LDM12CDB` as shown in the following image:



After you assign connections, the reference resource `Hermes_LDM12CDB` ceases to exist as a reference resource and the **ora_hermes** resource and assets included in the resource appear when you view the lineage for assets in the `PC_ForCustomAttribute` PowerCenter resource.

After connection assignment, the reference resource `PC_ForCustomAttribute_PowerCenterScanner_Hermes_LDM12CDB` ceases to exist in the catalog and the lineage for assets in the `PC_ForCustomAttribute` resource includes the assets in the **ora_hermes** resource.

Viewing Reference Resources

You can view the reference resources on the **Library** page.

To view the reference resources in the Catalog Administrator, perform the following steps:

1. In Catalog Administrator, click **Library**. The **Library** page opens.
2. From the **Library** pane on the left, click **Reference Resources**. The **Reference Resources** page appears.

You can view the primary resource, the description for the resource, and type of the resource on the **Reference Resources** page. You can click the drop-down list beside the primary resource to expand the list of reference resources.

Creating a Resource

When you create a resource, you can specify the resource type, type of metadata that Enterprise Data Catalog extracts, and an optional schedule for the resource. You can also assign custom attributes to the resource when you create it.

Note: You might want to validate the connection when you configure a resource. Click **Test Connection** to validate the connection parameters.

1. Click **New** > **Resource**.

   The **New Resource** wizard appears.
2. Enter a name and an optional description for the resource.
3. Optionally, if you want to assign custom attribute values to the resource, enter values for attributes, such as the Department, Data Owner, Data Steward, and Subject Matter Experts in the **Additional Properties** section.
   - Select **Propagate** under each custom attribute type if you want to apply custom attribute values to all the assets loaded through the resource.
4. Optionally, if you want users to receive scan failure notifications for the resource, select users as Failure Notification Subscribers in the **Additional Properties** section.
5. Click **Choose** to open the **Select Resource Type** dialog box.
6. Select a resource type, and click **Select**.

   More fields appear in the wizard based on the resource type you selected.
7. Based on the resource type, configure the connection properties.
8. Click **Next** to move to the **Metadata Load Settings** page.

   You can choose the type of metadata that you want Enterprise Data Catalog to extract from the source systems.
9. Configure the required source metadata and profiling metadata parameters.
10. Click **Next** to move to the **Custom Attributes** page, and configure the attribute settings. You can select the custom attributes that you want to associate with the resource.
11. Click **Next** to move to the **Data Access** page.
12. Optionally, enable data provisioning or preparation if you want to create provisioning or preparation tasks in Enterprise Data Catalog.
13. Click **Next** to move to the **Permissions** page, and restrict the permissions for the resource. You can also assign or modify the resource permissions from the security settings.
14. Click **Next** to go to the **Schedule** page.
15. Optionally, select schedules for source metadata load and profiling metadata. You can create a global schedule if required.

    Note: You can view the server time zone for both custom and reusable schedules. Select or create schedules based on the server time zone.
16. Click **Save**, or Click **Save and Run**.

Resource Type

A resource type represents the type of source system from where scanners extract metadata. A resource type identifies the required and optional metadata types that you need to configure for each resource. Examples of the resource type are Business Objects, Oracle, PowerCenter, and Teradata.

Each resource type has different properties that you need to configure when you create a resource. The properties include the connection properties and additional properties, such as resource owner and resource classification.

You can create the following types of resources:

- Amazon Redshift
- Amazon S3
- Apache Atlas
- Azure Microsoft SQL Server
- Azure Microsoft SQL Data Warehouse
- Azure Data Lake Store
- Microsoft Azure Blob Storage
- Business Glossary Classification
- Cloudera Navigator
- Custom Lineage
- Database Scripts
- Erwin
- File System
- Google BigQuery
- Hadoop File System (HDFS)
- Hadoop Multi-Part File Handler
- Hive
- IBM Cognos
- IBM DB2
- IBM DB2 for z/OS
- IBM Netezza
- Informatica Axon
- Informatica Cloud
- Informatica Platform
- JDBC
- MDM
- MicroStrategy
- OneDrive
- Oracle
- PowerCenter
- ProfileWarehouse

- Qlikview
- Salesforce
- SAP Business Objects
- SAP Business Warehouse (SAP BW)
- SAP BW/4HANA
- SAP S/4HANA
- SAP R/3
- SharePoint
- SQL Server Integration Services
- Sybase
- Tableau Server
- Teradata
- Workday

To run a scan job on resource types, such as Oracle, DB2, and SQL Server, use the standard resource types available in Enterprise Data Catalog. Informatica recommends that you do not configure the JDBC resource type to fetch data from the supported standard metadata sources.

However, you can use the JDBC resource type when there is no standard connector to fetch metadata and perform data discovery from the following sources:

- MySQL
- i5/OS
- Amazon Athena
- Vertica
- Postgres
- Databricks Delta Tables

For more information about the JDBC resource configuration, contact Informatica Global Customer Support.

For more information about configuring a resource, see the *Enterprise Data Catalog Scanner Configuration Guide*.

Dataset or Schema Profile Filter

In the Metadata Load Settings tab of the Resource wizard, the Dataset or Schema Profile filter specifies the data sets to include or exclude in a profile run.
You can select the data sets associated with the resources that you specify in the Schema field of the Source Metadata section. If the filter is empty, the profile runs on the selected schemas in the Schema field.

Source Metadata and Data Profile Filter

You can choose to extract source metadata for any object in the database and to extract profile metadata and data from tables and views that you identify.

You might configure the filter field for the following reasons:

- You want to extract metadata for one or more objects from a list of all objects in the database.
- You want to include or exclude one or more objects from the next resource run.

In a resource, you can use the following filters to include and exclude the database objects:

- Source Metadata Filter. In the **Metadata Load Settings** tab in the Resource wizard, enter the objects in the database that you want to include or exclude in a resource run. The scanner extracts the source metadata based on the objects in the Source Metadata Filter field.
- Data Profile Filter. In the **Metadata Load Settings** tab in the Resource wizard, enter the tables and views that you want to include or exclude in a profile run. The profile scanner extracts profile metadata based on the tables and views in the Data Profile Filter field.

To include tables and views, you can enter full names of tables and views, or use the % wildcard character with the name or pattern. Use semicolons (;) to separate the table names and view names. If the pattern contains a special character, such as a whitespace, enclose the pattern within single quotes. For example, HR_id; 'HR Emp%'; HR_job%.

Note: The Data Profile Filter supports Java regular expressions such as % , ^ , $, . , | , and (a | b), and the Source Metadata Filter supports the % expression. The filter constraint applies to Azure Microsoft SQL Server, Azure Microsoft SQL Data Warehouse, Hive, and all relational database sources.

You can choose one or more of the following patterns to identify the tables and views:

- <name>%. Includes tables and views that start with <name>.
  For example, A% includes all the tables and views that start with the character A.
- %<name>. Includes tables and views that end with <name>.
  For example, %id includes all the tables and views that end with the text 'id'.
- %<name>%. Includes tables and views that contain <name>.
  For example, %DB% includes all tables and views that contain the keyword "DB".
- <name>. Includes tables and views that match <name>.
  For example, HR_id filter includes any table or view in the resource that matches HR_id.

To exclude tables and views from the scan, prefix the term NOT before the name or pattern. For example, NOT %ID; NOT 'ID%'. Use semicolons (;) to separate the table names and view names.

Troubleshooting

What happens if I include a table in the Source Metadata Filter that depends on an external object?

When a filtered table or view depends on an external object such as a database, the resource scanner extracts the metadata of the external object even when the external object does not meet the filter conditions because the filter does not apply to the dependent external object.

How do I know which tables and views were excluded in the profile scan?

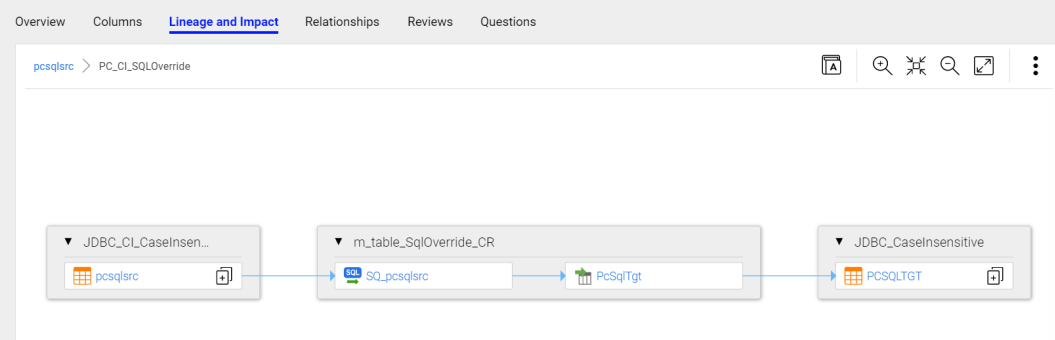
By default, the log files do not list all the tables and views that were excluded in a scan. To view the list of excluded tables and views, enable the *Debug* log level and run the scan. The log files list all the excluded tables and views in the scan.

Case Insensitive Linking

You can configure a resource as not case sensitive. If you set a resource as not case sensitive, Enterprise Data Catalog can create links between the source and target irrespective of the uppercase or lowercase names of the resources. The uppercase or lowercase names of the resources do not impact the lineage diagrams that you see in the Catalog. Case insensitive linking helps you view data asset lineage information for resources that have case sensitive source and target.

For example, consider that the Catalog Administrator contains Oracle and business intelligence resources between which you want to perform connection assignment. The source and target end points of the resources contain tables and columns that are both case sensitive and not case sensitive. In this scenario, traditionally, the Catalog would not create links between the source and target. To create links between the source and target, you configure the Oracle and business intelligence resources as case insensitive, and then perform connection assignment for the resources. Enterprise Data Catalog then creates links irrespective of the case sensitivity on the source and target, and displays lineage information.

The following image displays the sample lineage and impact view of a case insensitive table:



Enable Data Discovery

You can choose the **Enable Data Discovery** option for a resource to find the content, quality, and structure of the data source. You can run a column profile, perform data domain discovery, and prepare data to infer similar columns in multiple data sources. You can generate unique key from the column or combination of columns.

To perform data discovery on a resource, select the **Enable Data Discovery** option in the **Data Discovery** section in the **Metadata Load Settings** tab. After you enable data discovery for a resource, you can configure the Data Integration Service properties, profile-related settings, enable column similarity, and unique key inference settings. For more information about the options in the **Data Discovery** section, see the "Data Discovery" on page 29 topic.

Choose the following profiling options for unstructured file types and extended unstructured formats:

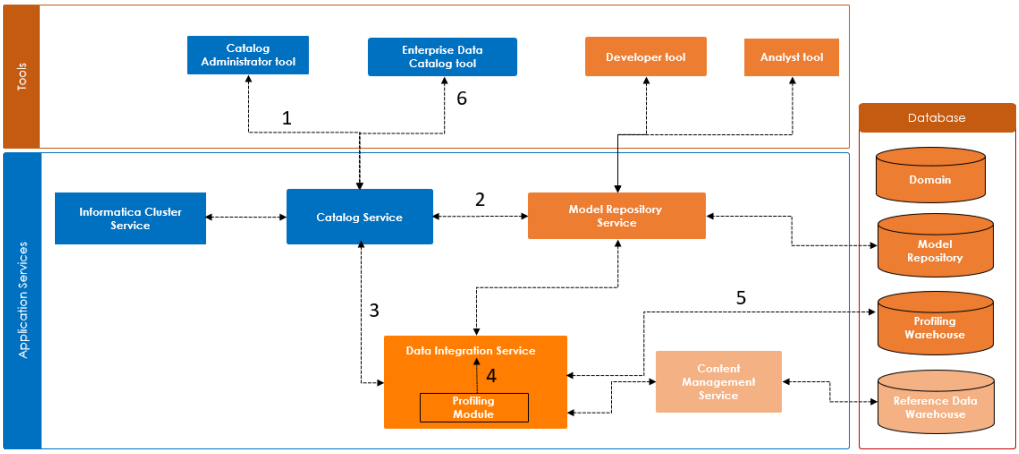
- **Data Domain Discovery** or **Column Profile and Data Domain Discovery** profile run option.
- **All Rows** sampling option.
- **Rows** as the data domain match criteria. Row count is the number of occurrences of a data domain in a data source.

You cannot run discovery on source metadata for unstructured file types and extended unstructured formats.

Profiling architecture

The profiling architecture consists of tools, services, and databases. The tools component consists of client applications. The services component has application services required to manage the tools, perform the data integration tasks, and manage the metadata of profile objects. The databases component consists of the Model Repository, the Profiling Warehouse, and the Reference Data Warehouse.

The following image shows the architecture components for profiling:



The following table describes the architecture components:

Component	Description
Catalog Administrator tool	A tool to configure the profiling metadata settings so that you have information related to data quality for further analysis.
Enterprise Data Catalog tool	A tool to discover, explore, and relate different types of metadata from disparate sources in the enterprise. You can view the profiling results in the Enterprise Data Catalog tool.
Developer tool	A client application that you use to design and implement data integration, data quality, data profiling, and data services solutions. You can use the Developer tool to create rules for profiling.
Analyst tool	A web-based client tool that is available to multiple Informatica products and is used by business users to collaborate on projects within an organization. You can use the Analyst tool to create rules for profiling.
Informatica Cluster Service	An application service that runs and manages all the Hadoop services, Apache Ambari server, and Apache Ambari agents on an embedded Hadoop cluster.
Catalog Service	An application service that runs and manages the connections between service components and the users that have access to the Catalog Administrator and Enterprise Data Catalog tools.
Model Repository Service	An application service that manages the Model repository.
Data Integration Service	An application service that performs data integration tasks for the Analyst tool, the Developer tool, and external clients.

Component	Description
Content Management Service	An application service that manages reference data for data domains that use reference tables. It uses the Data Integration Service to run mappings to transfer data between reference tables and external data sources.
Model repository	A relational database that stores the metadata for projects created in the Analyst tool or Developer tool.
Profiling warehouse	A database that stores profiling information such as profile results.
Reference Data warehouse	A database that stores data values for the reference table objects that you define in the Model repository.

As depicted in the profiling architecture diagram, the following processes occur when you enable profiling for a resource and scan the resource:

1. Enable data discovery for the resource in the Catalog Administrator tool to discover profiling-related metadata and unique keys.
2. The Catalog Service gets the profile definition from the Model repository.
3. The Catalog Service invokes the Profiling module in the Data Integration Service.
4. The Profiling module processes the profile job and submits the job to the Data Integration Service.
5. The Data Integration Service generates and writes the profiling results to the profiling warehouse.
6. Discover, explore, and view profiling results in the Enterprise Data Catalog tool.

Domain Connection Settings

Configure the properties for the Data Integration Service. After you configure the properties, the Data Integration Service runs the profile, performs data domain discovery, and infers column similarity. You can choose a different Data Integration Service to infer column similarity.

The following table describes the properties that you can configure in the **Metadata Load Settings** > **Domain Connection Settings** tab:

Property	Description
Specify the configuration settings for Data Integration Service.	Choose one of the following options: - Custom. Enter the domain connection settings for Data Integration Service - Global. Choose a reusable configuration.
Domain Name	Name of the Data Integration Service Domain.
Data Integration Service	Name of the Data Integration Service.
Username	Username to log in to the Data Integration Service.
Password	Password to log in to the Data Integration Service.
Security Domain	Name of the security domain.
Host	Host name for the Data Integration Service.

Property	Description
Port	Port number for the Data Integration Service.
Operating System Profile	Choose an operating system profile if you do not have a default operating system profile. The Data Integration Service uses the operating system profile to run the profile. **Note:** You can choose an operating system profile only when you choose the **Custom** option for the **Specify the configuration settings for Data Integration Service** property. If you choose the **Global** option for the **Specify the configuration settings for Data Integration Service** property, you need a default operating system profile to run the profile. For more information about using the operating system profiles in Enterprise Data Catalog, see the "Domain Connection Settings" on page 32 topic.

Basic Profile Settings

Configure the profile settings to run a column profile and perform data domain discovery for a resource.

The following table describes the properties that you can configure in the **Metadata Load Settings** > **Basic Profile Settings** tab:

Property	Description
Profile Run Option	Choose one of the following profile types: - Column Profile. - Data Domain Discovery. - Column Profile and Data Domain Discovery.
Priority	Choose one of the following priority values: - High - Low
Sampling Option	Choose one of the following sampling options: - All rows. - Auto Random rows. - Random N rows. - First N rows. - Limit N rows. - Random Percentage - Table Sample **Note:** For Hive resources, choose only All rows or First N rows sampling option. For XML, JSON, Avro, and Parquet resources, choose only All rows sampling option. **Note:** For Cassandra resource, choose All rows, First N rows, or Limit N rows sampling options. **Note:** For JDBC resources on the Spark Databricks engine with pushdown enabled, use Table Sample sampling option.
Exclude Views	Choose the option if you do not want the profiling scanner to scan the views in relational data sources. **Note:** For Cassandra resource, enable Exclude Views option as the data discovery is not supported on views.
Incremental Profiling	Choose the option to run the profile only on the changes made to the data source. If you do not select this option, the profile runs on the entire data source. For information about resources that support incremental profiling, see the "Basic Profile Settings" on page 33 topic.

Property	Description
Data Profile Filter	You can include or exclude tables and views from the profile run. Use semicolons (;) to separate the table names and view names. For more information about the filter field, see "Source Metadata and Data Profile Filter" on page 66.
Cumulative	Choose this option to retain the column profile and column similarity results from the previous scan in the next resource scan results. If you do not choose this option, the previous profile results are purged. For information about how to use the this option with the **Data Profile Filter** field and **Incremental Profiling** option, see the "Basic Profile Settings" on page 33 topic.
Source Connection Name	Choose a source connection to run data discovery. **Note:** This parameter is optional for a **File System** resource.
Run On	Choose one of the following run-time environments: - Blaze. Runs the profile in the Hadoop environment on the Blaze engine. Click **Select...**, and choose a Hadoop connection name in the **Select Hadoop Connection Name** dialog box. - Spark. Runs the profile in the Hadoop environment on the Spark engine. Click **Select...**, and choose a Hadoop connection name in the **Select Hadoop Connection Name** dialog box. - Databricks. Runs the profile in the Databricks environment on the Databricks Spark engine in the Databricks cluster. Click **Select...**, and choose a Hadoop connection name in the **Select Hadoop Connection Name** dialog box. **Note:** You can run JDBC and Azure Data Lake Store resources using the Databricks run-time environment. - Native. Runs the profile on the same machine where the Data Integration Service runs.

You can choose native, Hadoop, or Databricks as the run-time environment for a column profile. You can choose a Blaze or Spark engine in the Hadoop run-time environment. You can choose the Databricks Spark option in the Databricks run-time environment. Enterprise Data Catalog sets the run-time environment in the profile definition after you choose a run-time environment.

Note: Choose Blaze or Native run-time environment to run the profile job for all the resource except Hive resource. When you choose the Blaze engine or Spark engine, select a Hadoop connection to run the profiles.

You can run the profile in the following run-time environments:

Native environment

When you run a profile in the native run-time environment, the Data Integration Service runs the mappings and writes the profile results to the profiling warehouse. By default, all profiles run in the native run-time environment. You can use native data sources to create and run profiles in the native environment.

Hadoop environment

You can run profiles on the Spark or Blaze engine in the Hadoop environment or on the Databricks Spark engine in the Databricks environment. The Data Integration Service pushes the processing to the cluster. When processing is complete, the Data Integration Service writes the profile results to the profiling warehouse.

Databricks environment

You can choose the Databricks Spark option to run the profiles in the Databricks run-time environment. After you choose the Databricks Spark option, you can select a Databricks connection. The Data Integration Service pushes the profile logic to the Databricks Spark engine on the Databricks cluster to run profiles. When you run a profile in the Databricks environment, the Analyst tool submits the profile

jobs to the Profiling Service Module. The Profiling Service module then breaks down the profile jobs into a set of mappings. The Data Integration Service pushes the mappings to the Databricks Spark engine through the Hadoop connection. The Databricks Spark engine processes the mappings and the Data Integration Service writes the profile results to the profiling warehouse.

Configure the following properties when you choose **Data Domain Discovery** or **Column Profile and Data Domain Discovery** option:

Property	Description
Data Domain Discovery Type	Choose one of the following data domain discovery types: - Run Discovery on Source Data. - Run Discovery on Source Metadata. - Run Discovery on both Source Metadata and Data. - Run Discovery on Source Data Where Metadata Matches.
Select Data Domain	Choose one of the following data domain options: - All Data Domains. - Specific Data Domain Groups. In the **Data Domain Groups** field, choose one or more data domain groups in the **Data Domain Groups** dialog box. - Specific Data Domains. In the **Data Domains** field, choose one or more data domains in the **Data Domains** dialog box.
Use Conformance from	Choose one of the following values: - Data Domain - Custom
Data Domain Match Criteria	Choose one of the following values: - Percentage - Rows
Exclude Null Values from Data Domain Discovery	Select the option to exclude the null values in the data source when you run data domain discovery.

Similarity Profile and Value Frequency Settings

Configure the column similarity properties to identify similar columns and value frequency in the resource.

The following table describes the properties that you can configure in the **Metadata Load Settings** > **Similarity Profile Data Preparation and Value Frequency Settings** tab:

Property	Description
Run Similarity Profile	Choose one of the following options: - Yes. The profiling scanner scans the data source and prepares data to perform the following tasks: - Discover similar columns. The algorithm discovers similar columns in the resource based on column names, column patterns, and unique values. - Identify business terms. The algorithm identifies and recommends the business terms for a column based on the accepted data domains and similar columns. - No.
Save Source Data	Choose one of the following options: - Yes. Persists the computed information about similar columns, column patterns, and unique values in the resource in PostgreSQL database. - No.
Sampling Options	Choose one of the following sampling options: - Reuse Basic Profile Settings. - All Rows. - Auto Random Rows. - Random N Rows. - First N Rows.
Domain Connection Settings	Choose one of the following options: - Use Profile Configuration Settings. - Specify Domain Connection Settings. For information about domain connection settings properties, see the *Domain Connection Settings* section.

Unique Key Inference Settings

You can configure the following settings to generate unique key candidate from selected columns.

Property	Description
Run Unique Key Inference	Choose one of the following options: - Yes. The profile scanner scans and infers unique keys from the data source. - No
Null Threshold % in Unique Key Inference	The threshold for null values in unique key inference. You can enter a value between 0 and 1.
Skip Unique Key Inference When Accepted or Documented Unique Key Exists	Choose one of the following options: - Yes. The profiling service skips unique key inference for columns with documented or accepted keys. - No
Unique Key Sampling Option	Choose one of the following options: - All Rows. Chooses all the rows in the data object for unique key inference. - First N Rows. Chooses only the first N rows in the data object for unique key inference.

See "Unique Key Inference Settings" on page 39 for the list of the supported resources and file type.

Composite Data Domain Discovery

As a one-time prerequisite step, verify that you run profiling before you run composite data domain discovery. Alternatively, Enterprise Data Catalog discovers composite data domains if inferred data domains exist in your environment. Enterprise Data Catalog discovers data domains that are not enabled for discovery, during the composite data domain discovery.

Enterprise Data Catalog discovers composite data domains for the following resources:

- Amazon Redshift
- Amazon S3
- Azure Microsoft SQL Data Warehouse
- Azure Microsoft SQL Server
- File System
- HDFS
- Hive
- IBM DB2
- IBM DB2 for zOS
- IBM Netezza
- JDBC
- Microsoft SQL Server
- Oracle
- Salesforce
- SAP R/3
- Sybase
- Teradata

The following options in the **Composite Domain Discovery** section help you to enable and configure composite data domain discovery for the resource:

Property	Description
Enable Composite Domain Discovery	Enables composite data domain discovery for the resource.
Select Composite Data Domain	Select the composite data domains that the resource must use for inference. Select one of the following options to specify the composite data domains: - All Composite Data Domains. Selects all composite data domains. - Specific Composite Data Domains. Selects specific composite data domains. When you select this property, the **Composite Data Domains** text box appears. Click **Select...** to select a list of composite data domains from the **Select Composite Data Domains** dialog box.

Note: If you enable composite data domain discovery and data domain discovery for the resource, Enterprise Data Catalog discovers all data domains included in the composite data domains for the resource.

Enable Business Term Association

Configure the **Business Term Association** settings for a resource to automatically associate or recommend the business terms as a business title for tables, columns, views, files, and field assets.

If you choose the **Metadata Load Settings** > **Business Term Association** > **Enable Business Term Association** option to configure business term propagation, you need not configure the **Run Similarity Profile** option for a resource to automatically accept or recommend the business term for columns or field assets. Configure the **Run Similarity Profile** and **Enable Business Term Association** options for a resource if you want to improve the confidence score of the business term propagation.

Enterprise Data Catalog supports the **Enable Business Term Association** option for the following resources:

- Amazon S3
- Amazon Redshift
- Azure Data Lake Store
- Azure Microsoft SQL Data Warehouse
- Azure Microsoft SQL Server
- File System
- Google BigQuery
- HDFS
- Hive
- IBM DB2 for z/OS
- IBM DB2
- IBM Netezza
- JDBC
- Microsoft Azure Blob Storage
- Microsoft SQL Server
- One Drive
- Oracle
- Salesforce
- SAP
- SharePoint
- Snowflake
- Sybase
- Teradata
- Workday
- MongoDB

Note: After you configure the business term association settings, run the Data Domain Propagation system resource if you want to view business term recommendations immediately in Enterprise Data Catalog Application.

Configure the following properties in the **Business Term Association** section of **Metadata Load Settings** page to automatically associate business terms:

Property	Description
Enable Business Term Association	Enables business term association for columns and fields in the resource.
Select Glossaries	Select business glossaries or Axon glossaries from which you want Enterprise Data Catalog to automatically associate business title to assets.
Use synonyms and stop words	Select this option to use synonyms and stop words during the business term propagation process. Some examples of stop words are 'the', 'a', 'an', and 'in'. For example, consider a resource that contains a column named **Profit and Loss Report**. The word **'and'** is a stop word in the column name, and you want Enterprise Data Catalog to assign or recommend the business term named **ProfitLossReport** for the column. After you select the **Use synonyms and stop words** option, Enterprise Data Catalog avoids the stop word **'and'** in the column name, and then assigns or recommends the business term **ProfitLossReport** for the column. Upload the synonym definitions list in Catalog Administrator if you want Enterprise Data Catalog to use synonyms to propagate business terms. For example, Enterprise Data Catalog contains a business term named **Social Security Number** and column named **SSN**. You want the Catalog to automatically associate or recommend the business term **Social Security Number** for the column **SSN**. After you define the synonyms definition list in the Catalog Administrator, Enterprise Data Catalog recommends or assigns **Social Security Number** as a business title to the column. For more information about the synonyms definitions, see the Chapter 13, "Manage Synonym Definitions" on page 115 section.
Avoid prefixes	Select this option if you want to avoid prefixes in the asset name during the business term propagation process. For example, consider a resource that contains a column named as **Reorderdetails**. You want Enterprise Data Catalog to recommend or automatically assign the business term **orderdetails** to the column. After you select the **Avoid prefixes** option, Enterprise Data Catalog ignores the prefix **Re** in the column name, and then automatically assigns the business term **orderdetails** to the column **Reorderdetails**.
Avoid suffixes	Select this option if you want to avoid suffixes in the asset name during the business term propagation process. For example, consider a resource that contains a table named as **EMPLOYEE_amr_test**. You want Enterprise Data Catalog to recommend or automatically assign the business term **EMPLOYEE_amr** to the table. After you select the **Avoid suffixes** option, Enterprise Data Catalog ignores the suffix **_test** in the table name, and then automatically assigns the business term **EMPLOYEE_amr** to the table **EMPLOYEE_amr_test**.

Property	Description
Business Term Status	Enables status-based filtering for business term imports. After you enable the option, select the corresponding status value to apply the import filter: - Active - Imports active business terms. - Inactive - Imports inactive business terms. - Both - Imports business terms regardless of status.
Custom options	Parameters that you can set to configure the business term propagation. Use the feature weight custom options to customize the priority of the data domains, column similarity, and name match conditions during the business term propagation process. Use the threshold custom options to customize the threshold limit for automatic business term associations and recommendations. Based on the threshold limit that you set, Enterprise Data Catalog assigns business terms or recommends business terms to assets. To configure the feature weight, add the following custom options to the Data Domain Propagation system resource: - `-Dglossary.propagation.domain.curation.feature.weight=<value>` - The default value is 3. - `-Dglossary.propagation.column.similarity.feature.weight=<value>` - The default value is 1. To configure thresholds, add the following custom options to the Data Domain Propagation system resource: - `- Dglossary.propagation.autoaccept.score.threshold=<value>` - The default value is 80. - `- Dglossary.propagation.inference.score.threshold=<value>` - The default value is 60.

For more information about business term propagation, see the "Business Term Propagation" on page 45 topic.

Enable Data Provisioning

As a one-time prerequisite step, verify that you have imported the Informatica Intelligent Cloud Services organization into Catalog Administrator. Configure the **Data Provisioning** settings for a resource to create data provisioning tasks in Enterprise Data Catalog. Enterprise Data Catalog supports the **Enable Data Provisioning** option for Oracle and Microsoft SQL Server resources.

Configure the following properties in the **Data Access** page to enable data provisioning for a resource:

Before you enable data provisioning, ensure you have met the perquisites in the Data Preview and Provisioning Prerequisites topic.

Property	Description
Enable Data Provisioning	Enables data provisioning for relational tables in the resource.
Org Name	Choose the Informatica Intelligent Cloud Services organization in which you have created the source and target connections. For more information about Informatica Intelligent Cloud Services organizations, see the Managing Cloud Organization chapter in the *Catalog Administrator* guide.
Connection	Select the connection that you created in Informatica Intelligent Cloud Services for the resource. Ensure the connection that you select has the same schema which is configured for a resource.

For more information about data provisioning, see the "Data Provisioning " on page 50 topic.

Editing a Resource

You can make changes to a resource after you create it. You can change the settings, such as the connection properties, custom attributes, source metadata settings, profile metadata settings, and the attached schedule. You cannot change the name of the resource and its resource type after you create it.

1. From the Catalog Administrator header, click **Library**.

   The **Library** workspace opens.
2. In the resource list, point to a resource, and select **Edit** from the control menu.

   The resource details appear on a new tab.
3. Make changes to the description, connection properties, custom attributes, source metadata settings, profile metadata settings, and schedule as required.
4. To save the changes without running the resource again, click **Save**.
5. To save the changes and run the scan, click **Save and Run**.

Cloning a Resource

You can create a copy of a resource by cloning the resource. This feature is useful if you want to copy the configuration of an existing resource, and then modify the configuration as required. When a resource is cloned, certain resource configuration such as passwords, file uploads, and connection assignments are not copied. You cannot clone a system resource.

1. From the Catalog Administrator header, click **Library**.

   The **Library** workspace opens.
2. In the resources list, point to the resource that you want to clone, and select **Clone** from the control menu.

A message appears asking you to confirm if you want to continue.

3. Select **OK**.

   The resource details appear.

   Note: You can also open a resource and click on the **Clone** button to go to the resource details page, and create a resource from the clone.

4. Edit the resource name and make changes to the configuration of the copied resource as required.
5. To save the changes without running the resource again, click **Save**.
6. To save the changes and run the scan, click **Save and Run**.

Running a Scan on a Resource

You can run a scan on a resource either as part of the resource schedule or manually as a one-time task based on your requirement.

1. From the Catalog Administrator header, click **Library**.

   The **Library** workspace opens.

2. In the resource list, point to a resource, and select **Run** from the control menu.

   The resource details appear on a new page with the **Monitoring** tab enabled.

System Resources

Enterprise Data Catalog creates the following system resources to perform various internal jobs when you enable the Catalog Service:

DataDomain

Internal system job that is auto-scheduled to run every 45 minutes. The scanner synchronizes the data domains between the Model repository and catalog.

DataDomainPropagation

Internal system job that is auto-scheduled to run at regular intervals. The scanner uses the inferred similar columns to propagate the smart data domains to other similar columns. The scanner also identifies and propagates the business terms to other similar columns.

DomainUsers

Internal system job that synchronizes Informatica domain users between the Informatica domain and catalog.

Health Check

Internal system job that scans the Solr index to detect and report duplicate documents. The job identifies resource IDs with duplicates, retrieves associated metadata, and a console summary to help identify data inconsistencies. The job creates detailed CSV reports and saves them to remote storage.

SolrReconciliation

Internal system job that identifies and deletes obsolete duplicate documents in Solr. The job processes duplicate IDs from the Health Check job, compares the data with MongoDB data, deletes the obsolete Solr documents, and updates synchronization metrics.

You can configure the following settings for the resources based on your requirements similar to how you configure any other resources that you create:

- Modify the memory required to run the resource.
- Assign custom attributes to the resource.
- Modify the schedule for the resource.

Note: You can modify the description for the scanners, but you cannot modify the name or resource type details for system resources.

Viewing a Resource

You can view the list of resources on the Library tab. You can launch a read-only view of a specific resource. You can also edit a resource from the resource list.

1. From the Catalog Administrator header, click **Library**.

   The **Library** workspace opens.
2. In the resource list, point to a resource, and select **Open** from the control menu.

   A read-only view of the resource details appears on the **General** tab. You can see multiple tabs for the resource details.
3. Click each tab to view more information about the resource.

CHAPTER 5

Managing Resource Security

This chapter includes the following topics:

- Managing Resource Security Overview, 81
- Configuring Default Permissions for Resources, 82
- Configuring Permissions for Specific Users and User Groups, 83
- Applying Restrictions on a Resource, 85
- Assigning Configuration Permissions to Resources, 85

Managing Resource Security Overview

You can configure specific permissions on resources for users and user groups configured in Informatica domain.

As a catalog administrator, you can specify access permissions on resources for specific users and user groups. The type of access permissions depends on the specific security and privilege requirements in your enterprise. For example, in a financial institution, apart from the data steward who validates the integrity, consistency, and quality of the data, no one in the institution must be able to view the details of the data sources that store confidential customer details. Identification of data sources that store customer details by unauthorized personnel might lead to hacking of the data sources and leaking of confidential information.

You can specify permissions using the Catalog Administrator in the following ways:

- Specify default permissions on all resources or specific resources for users and user groups.
- Select a specific resource and specify the permissions for the users and user groups.
- Select specific users or user groups and configure permissions on the resources.

As a catalog administrator, you can assign users and user groups the following permissions on resources:

Read

View the details of the resource and assets in Enterprise Data Catalog.

Read and Write

Allows the user or users included in the user group to enrich the assets in the Enterprise Data Catalog in addition to the read permission. You can enrich assets by assigning custom attributes, business terms, or data domains to the asset. Enriching assets helps you search for the asset using the assigned custom attribute, business term, or data domain.

Note: When you configure data permissions for relational sources such as Oracle, the users or users included in the user groups have permissions to all the assets including assets such as tables, views,

and synonyms. However, when you configure data permissions for the following assets, the users or users included in the user groups have permissions to access only the following assets:

- Tables
- Views
- Synonyms

Metadata and Data Read

Allows the user or users included in the user group to view the value frequency for an asset in Enterprise Data Catalog. You can assign this permission to resources that support profiling. For the list of resources that support profiling, see "Data Discovery" on page 29.

All Permissions

Allows the user or users included in the user group to enrich the assets in the Enterprise Data Catalog and view the value frequency for the asset. You can enrich assets by assigning custom attributes, business terms, or data domains to the asset. Enriching assets helps you search for the asset using the assigned custom attribute, business term, or data domain.

Configuring Default Permissions for Resources

You can specify the default permissions on resources for users and user groups. You can modify the default permissions later as required for specific users and user groups. Enterprise Data Catalog applies the default permissions for all users or user groups till you specify permissions for a user or user group. The specific permissions that you configure for users or user groups take effect only on the resources for which you grant permissions by modifying the default permissions.

1. Click **Manage** > **Security**.

   The **Security** tab page opens.

2. Click **Set Default Permissions**.

   The **Default Permissions** dialog box appears listing the users and groups configured on the Informatica domain.

3. Select the users and user groups for whom you want to configure permissions on the resources and then click **Next**.

   The **Specify permissions for selected users and groups** section appears.

4. Select one of the following options to specify the resources on which you want to assign permissions for the selected users or user groups:

   - **All resources**
     Select this option if you want to specify default permissions for all the resources in Enterprise Data Catalog. Select **Read** or **Read and Write** from the drop-down list to assign the selected permission to all the resources.
   - **Custom**
     Select this option if you want to select the resources for which you want to specify permissions. Select the required resources and click the drop-down list adjacent to the resource. Select the required permission for the resource from the drop-down list. If you select multiple resources, click the drop-down list next to***Inherited Permissions** and select the required permission that you want to apply for all the resources.

Note:

- Resource Types. Type in the name of the resource in the text box under the **Resource Types** section and press **Enter**. Enterprise Data Catalog lists the resources with matching patterns.
- Permissions. Select the permission from the **Permissions** drop-down list. Enterprise Data Catalog lists the resources that have the matching permission.

Click **Clear Filter** to clear the filter options you specified.

5. Click **OK**.

Configuring Permissions for Specific Users and User Groups

You can configure specific permissions for users and user groups to access resources.

Enterprise Data Catalog retains the default permissions (inherited permissions) configured for the users or user groups till you configure specific permissions. Enterprise Data Catalog adds an asterisk (*) to the permission name to indicate that the selected user or user group has the default permissions.

Note: By default, the users included in a user group inherit permissions that you assign to the user group. If you configure specific permissions for a user in a user group, Enterprise Data Catalog applies the user-specific permissions configured for the user. Enterprise Data Catalog displays the permission status as **Mixed** for a user who has different permissions configured at the user level and at the user group level.

To select users and user groups to assign permissions on the resource, perform the following steps:

1. Click **Manage** > **Security** from the Catalog Administrator header.

   The **Security** tab page opens with the **Users and Groups** option selected in the **View** section.

2. On the **Users and Groups** panel, select the required user or user group.

   You can use the following filters to list the users or user groups that match the specific criteria:

   - Name. Type in the name of the user or user group in the text box under **Name** and press **Enter**. Enterprise Data Catalog lists the matching users and user groups.
   - Type. Select **User** or **Group** from the drop-down list under **Type** to specify if you want to view a list of users or groups.
   - Security Domain. Type in the security domain configured for the user or the user group in the text box under **Security Domain**. Enterprise Data Catalog lists all the users and user groups configured with the specified security domain.

   Click **Clear Filter** to clear the filter options you specified.

3. Select the required resources from the **Resources** panel, click the permission column for each resource, and select the required permission that you want to configure on the resource.

   You can use the following filters to list the resources accordingly:

   - Name. Type in the name of the resource in the text box under the **Name** section and press **Enter**. Enterprise Data Catalog lists the resources with matching names.
   - Data Permissions. Select the permission from the **Data Permissions** drop-down list. Enterprise Data Catalog lists the resources for which the selected data permission is configured.
   - Resource Type. Type in the type of the resource in the text box under the **Resource Type** section and press **Enter**. Enterprise Data Catalog lists the matching resource types.

Click **Clear Filter** to clear the filter options you specified.

Selecting Resources to Assign Permissions for Specific Users or User Groups

As an alternative to selecting users or user groups and assigning permissions for resources, you can select multiple resources and assign permissions for specific users or user groups.

1. Click **Manage** > **Security**.

   The **Security** tab page opens with the **Users and Groups** option selected in the **View** section.

2. Select **Resources** from the **View** section.

   You can use the following filters to list the users or user groups required:

   - Name. Type in the name of the resource in the text box under **Name** and press **Enter**. Enterprise Data Catalog lists the matching resources.
   - Resource Type. Type in the type of the resource in the text box under **Resource Type** and press **Enter**. Enterprise Data Catalog lists the list of matching resource types.
   - Click **Clear Filter** to clear the filter options you specified.

   The list of resources configured in Enterprise Data Catalog appears.

3. Select the resource for which you want to assign data and configuration permissions for the users or user groups.
4. Perform this step if you want to assign data and configuration permissions for user groups. Select the required user group from the **Groups** tab page, click the data and configuration permission columns for that user group, and select the required data and configuration permissions that you want to configure for the user group.
5. Perform this step if you want to assign data and configuration permissions for users. Select the required user from the **Users** tab page, click the data and configuration permission columns for that user, and select the required data and configuration permissions that you want to configure for the user.

   You can use the following filtering options on the **Groups** or **Users** tab pages to filter the list of users and user groups based on the name, security domain, data permissions, and configuration permissions configured for the users or user groups:

   - Name. Type in the name of the user or the user group in the text box under the **Name** section and press **Enter**. Enterprise Data Catalog lists the users or user groups with matching names.
   - Configuration Permissions. Select the permission from the **Configuration Permissions** drop-down list to filter users or user groups based on the configuration permissions assigned. Enterprise Data Catalog lists users or user groups for which the selected configuration permission is configured.
   - Data Permissions. Select the permission from the **Data Permissions** drop-down list to filter users or user groups based on the data permissions assigned. Enterprise Data Catalog lists the users or user groups for which the selected data permission is configured.
   - Security Domain. Type in the security domain in the text box under the **Security Domain** section and press **Enter**. Enterprise Data Catalog lists the users or user groups configured in the specified security domain.

   Click **Clear Filter** to clear the filter options you specified.

Applying Restrictions on a Resource

As a catalog administrator, you can control the operations performed on a resource by restricting a resource. By default, all the resources are unrestricted.

1. Click **Manage** > **Security**.

   The **Security** tab page opens with the **Users and Groups** option selected in the **View** section.

2. Select **Resources** from the **View** section.

   The list of resources configured in Enterprise Data Catalog appears.

3. To apply restrictions on a resource, click the **Actions** menu, and choose **Restrict**.

   The resource status changes to **This is a restricted resource.**

   Note: Users or user groups with read, scan, and write configuration permissions on a resource can unrestrict a resource.

Assigning Configuration Permissions to Resources

As a catalog administrator, you can assign configuration permissions to resources to control the operations performed on a resource.

You can assign the following configuration permissions to resources:

Read

The user or users included in the user group can view the details of the resource, associated jobs, connections and data permissions in Catalog Administrator.

Read and Scan

The user or users included in the user group can view the details of the resource, associated jobs, connections and data permissions, run the metadata scan, and perform rollback and purge operations on the resource.

Read, Scan, and Write

The user or users included in the user group can view the details of the resource, associated jobs, connections and data permissions, run the metadata scan, edit the resource, delete the resource, assign or remove connections, update data permissions, and perform rollback and purge operations on the resource.

No Access

The user or users included in the user group cannot access the resource.

Note: The user or users included in the user group must have Read, Scan, and Write configuration permissions on both resources participating in the connection to view, assign, or remove connections.

CHAPTER 6

Managing Schedules

This chapter includes the following topics:

- Managing Schedules Overview, 86
- Schedule Types, 86
- Creating a Schedule, 87
- Viewing the List of Schedules, 87

Managing Schedules Overview

Schedules determine when scanners extract metadata from sources. You can have recurring daily, weekly, and monthly schedules to extract metadata at regular intervals. Create a reusable schedule if you want to assign multiple resources to the same schedule. If you choose to have a reusable schedule for metadata extraction, you can select from a list of existing schedules or create a different reusable schedule that meets your requirements.

Create custom schedules that you can assign to specific resources. You can assign separate schedules to resources to extract source metadata and profiling metadata. When you create a schedule, you can choose to have a schedule without an end date or that recurs until a specific date.

Schedule Types

You can create reusable or custom schedules that meet the frequency requirements for each resource to extract metadata.

You can attach more than one resource to a reusable schedule. You can create a custom schedule if you need a separate schedule specific to a single resource. Refer to the server time zone when you select or create custom or reusable schedules.

Reusable Schedules

Source systems might have changes to the metadata at different times. The changes can include newer data assets being added to the source or updates to the existing data assets. You can set up a reusable schedule that you can assign to multiple resources so that you continue to extract these source changes at regular intervals.

Custom Schedules

Create a custom schedule if none of the existing reusable schedules match the metadata extraction schedule for the resource.

You can create a custom schedule when you create a resource. Create a daily, weekly, or monthly custom schedule for a resource. You can create an indefinite custom schedule or a schedule that ends by a specific date.

Creating a Schedule

You can create a schedule when you configure a resource. You can create a reusable schedule using the **New** menu on the Catalog Administrator header.

1. Click **New** > **Reusable Schedule**.

   The **New Reusable Schedule** wizard appears on the **Schedule** workspace.
2. Enter a name and an optional description for the schedule.
3. Verify that the server time zone displayed is accurate.
4. Click the **Starts on** field to open a calendar, and choose a start date for the schedule.
5. Use the fields to the right of the **Starts on** field to set up the start time.
6. Choose whether you want to create a daily, weekly, or monthly schedule.
7. Configure the recurrence settings, such as every *n* days for a daily schedule or day of the week for a monthly schedule.

   You have different recurrence settings based on the schedule frequency.
8. Choose either an end date or set up the schedule without an end date.
9. Click **Save**.

Viewing the List of Schedules

You can view the list of schedules on the **Library** workspace.

1. From the Catalog Administrator header, click **Library**.

   The **Library** workspace opens.
2. On the left pane, click **Schedule**.

   The list of schedules appears on the right pane.
3. To view the schedule frequency, mouse over the icon at the beginning of the schedule name.
4. To view the complete information or edit the schedule, click the schedule name.

   The schedule opens in the **Schedule** workspace.
5. To make changes to the schedule, click **Edit**.

CHAPTER 7

Managing Attributes

This chapter includes the following topics:

- Managing Attributes Overview, 88
- System Attributes, 88
- Custom Attributes, 89
- General Attribute Properties, 89
- Search Configuration Properties, 89
- Creating a Custom Attribute, 90
- Editing a Custom Attribute and System Attribute, 90
- Assigning Custom Attributes to Classes and Resources, 91

Managing Attributes Overview

Attributes are metadata properties that scanners extract from different source systems. System attributes are predefined properties that scanners use for default resource types. You can create custom attributes that you can configure and assign to specific resources.

Based on the business requirements, you can choose to assign custom attributes to resources. For example, you might want to assign a business glossary term or category titled City or Department to a resource. When you create a custom attribute, you can configure the basic attribute properties and search behavior. For example, you can select a specific data type, such as Data, Decimal, City, Department, or User. You can also make the attribute a search filter in Enterprise Data Catalog where users search for the required enterprise metadata.

System Attributes

System attributes represent the different types of metadata that scanners extract from source systems. For example, Author is a system attribute of the String data type that you can assign to a resource. You can configure these predefined attributes in Catalog Administrator.

You can use system attributes to filter the search results that you are looking when you use Enterprise Data Catalog to search the metadata. Use the Catalog Administrator to configure the search ranking of a system attribute based on the requirement. You can also set up the system attribute so that Enterprise Data Catalog includes the attribute in search filters.

Custom Attributes

You can create custom attributes based on the search filters that you need to use in Enterprise Data Catalog where you search for metadata. Custom attributes help you quickly find specific metadata . For example, you might want to create a custom attribute named Data Center Location in Catalog Administrator and assign it to some of the resources. You can then use the custom attribute Data Center Location in Enterprise Data Catalog to quickly filter resources associated with a specific location.

You need to specify a name and data type when you create a custom attribute. You can choose a core data type, such as Decimal, Integer, Date, and Boolean or extended data type, such as User.

General Attribute Properties

The general properties for both system attributes and custom attributes constitute the basic properties, such as name and description.

The following table describes the general properties for both system attributes and custom attributes:

Property	Description
Name	Name of the system attribute or custom attribute.
Description	Descriptive text about the attribute.
Data Type	Basic or extended data type for the attribute. Examples are basic data types, such as String and Boolean and extended data types, such as user and CSV.
Allow Multiple Value Selection	Displays a multivalued list for the attribute when you use the attribute for metadata search in Enterprise Data Catalog. You can simultaneously select multiple values from the list. **Note:** This property does not appear for or apply to the Boolean data type.

Search Configuration Properties

The search configuration properties define how Enterprise Data Catalog uses attributes in metadata search.

The following table describes the search configuration properties for both system attributes and custom attributes:

Property	Description
Search Rank	Indicates the level of search ranking associated with the attribute. This setting determines the position of the attribute in the search query results of Enterprise Data Catalog.
Allow filtering	Determines whether Enterprise Data Catalog can use the attribute as a search filter.
Analyzer name	Name of the analyzer associated with string values. **Note:** Applies only to String data type.

Creating a Custom Attribute

Create custom attributes that you want Enterprise Data Catalog users to add to the search filters. Add search filters based on custom attributes in Enterprise Data Catalog to quickly categorize metadata search results.

1. Click **Manage** > **Attributes**.

   The **Attributes** workspace appears.
2. From the **Actions** menu, select **New**.

   The **New Custom Attribute** dialog box appears.
3. Enter the name and description for the custom attribute.
4. In the **Data Type** list, select a data type, such as Integer, String, Boolean, or Date.

   The data type determines the valid type of values for the custom attribute.
5. In the **Search Rank** field, choose the level of search ranking for the custom attribute.
6. Choose whether you need to display the custom attribute as a search filter in Enterprise Data Catalog.
7. Optionally, choose the analyzer name for a String data type. The analyzer name determines the analysis method that the Enterprise Data Catalog search engine uses when the search engine performs indexing of string values for the attribute. You can choose **STRING**, **TEXT_ GENERAL**, or **TEXT_TECHNICAL**.
8. Next, select the object types that you want to assign to the custom attribute.
9. Click **OK** to save the changes.

Alternatively, you can perform this task using the REST API. For more information, see the Create Custom Attributes topic in the *Enterprise Data Catalog REST API Reference guide*.

Editing a Custom Attribute and System Attribute

You can make changes to the search configuration properties of custom attributes and system attributes. You cannot edit the remaining properties, such as Name, Description, and Data type.

1. From the Catalog Administrator header, click **Manage** > **Attributes**.

   The **Attributes** workspace opens.
2. Click **Custom Attributes** or **System Attributes** tab depending upon the attribute you want to edit.
3. In the **Custom Attributes** or **System Attributes** panel, select the attribute, and then click **Edit**.

   The fields in the **Search Configuration** section appear in the edit mode.
4. Make the required changes to the Search Rank, Allow filtering, and Analyzer name properties.
5. Click **Save**.

Alternatively, you can update and delete a custom attribute using the REST APIs. For more information, see the Update Custom Attribute and Delete Custom Attribute topics in the *Enterprise Data Catalog REST API Reference* guide.

Assigning Custom Attributes to Classes and Resources

You can assign custom attributes to existing resources and classes in the resource model when you create or modify custom attributes. You can either assign custom attributes to specific classes and resources or all classes and resources in the catalog.

Perform the following steps to assign custom attributes to resources and classes when you modify an existing custom attribute:

1. From the Catalog Administrator menu, click **Manage** > **Attributes**. The **Custom Attributes** page appears.
2. From the **Custom Attributes** pane, select the custom attribute that you want to associate with resources and classes.
3. Click **Edit**.
4. From the **Class Assignment** section, select one of the following options:
   - All Classes. Assigns the custom attribute to all classes.
   - Specific Classes. Assigns the custom attribute to specific classes. Click **Show object types in hierarchy** to view the object types included in the classes.

   Note: In 10.5.8, there are new class types added for Schema and Database objects because of MITI 11.1 changes. When you upgrade to 10.5.8, the upgrade utility assigns new class types to custom attributes with existing class types. After you upgrade to 10.5.8, if you create or edit a custom attribute, ensure that you select both the existing and new class types for Schema and Database objects. If you don't assign both the class types for Schema and Database objects, the Catalog Service prompts for an upgrade when you restart the Catalog Service. The following list contains the new and existing class types for Schema and Database objects:
   - Schema.
     - Existing class type: com.infa.ldm.relational.Schema
     - New class type: com.infa.ldm.relational.DatabaseSchema
   - Database.
     - Existing class type: com.infa.ldm.relational.Database
     - New class type: com.infa.ldm.relational.DatabaseServer
5. From the Resource Assignment section, select any of the following options:
   - All Resources. Assigns the custom attribute to all resources and reference resources.
   - Specific Resources. Assigns the custom attribute to specific resources and reference resources. Use the **Show selected resource** option to toggle between the resources you selected and all the resources.
   - None. Does not associate the custom attribute to any resource.
6. Click **Save**.

CHAPTER 8

Assigning Connections

This chapter includes the following topics:

- Assigning Connections Overview, 92
- Auto-assigned Connections, 92
- User-assigned Connections, 93
- Managing Connections, 93
- Missing Links Report , 94
- Missing Links Report Columns , 95
- Generating Missing Links Report , 95

Assigning Connections Overview

When you run a scan on resources for some of the resource types, you need to ensure that the source connection maps accurately to the schemas from the resource. Enterprise Data Catalog can automatically detect how the database schemas are assigned to the resources after you run a scan on the resources. You can assign and unassign schemas from resource to connections based on your requirements. The connection management tasks that you perform in Catalog Administrator apply only to SAP Business Objects, Informatica Platform, PowerCenter, IBM Cognos, Informatica Cloud Services, and Cloudera Navigator resource types.

If the data asset lineage information does not look accurate in Enterprise Data Catalog, you can troubleshoot the assigned and unassigned connections and make the required corrections in Catalog Administrator. You can then verify that the lineage flow is accurate in Enterprise Data Catalog. When you create resources in Catalog Administrator, you can choose to automatically assign database schemas to resources. You can also manually assign the schemas to specific connections.

Auto-assigned Connections

When you create a resource for an Informatica Platform, SAP Business Objects, IBM Cognos, Informatica Cloud Services, Cloudera Navigator, and PowerCenter source, you can choose to automatically assign the database schemas to the resource. You can view the list of automatically assigned schemas and their connections for each resource. You can assign or unassign schemas in the auto-assigned connections.

For Oracle mappings for PowerCenter, the connection is extracted as `<connectionname>.<schemaname>`. The Catalog Administrator assigns the connections at database level.

Note: When you rescan a resource with the auto-assignment option disabled, connections which were auto-assigned during the first scan remain in the **Auto Assigned Connections** tab. All other connections move to the **User Assigned Connections** tab. For more information about user-assigned connections, see "User-assigned Connections" on page 93. .

User-assigned Connections

After you create a resource for an Informatica Platform, SAP Business Objects, IBM Cognos, Informatica Cloud Services, Cloudera Navigator, and PowerCenter source connection and run a scan on it, you can view the resource as a user-assigned connection in Catalog Administrator. You can manually assign or unassign schemas to resources based on your requirements.

When you manually assign or unassign connections, the status of the connection changes to **In Progress**. You can refresh the **Connection Assignment** workspace to view the latest status.

Note: When a connection for a PowerCenter resource is extracted as `<connectionname>.<schemaname>` in the **Connection Assignment** workspace, you need to assign a database to the connection. For PowerCenter Oracle resources, connection is extracted as `<connectionname>.<schemaname>` .

Managing Connections

You can assign or unassign connections one at a time or select multiple connections to make the changes.

1. From the Catalog Administrator header, click **Manage** > **Connection Assignment**.

   The **Connection Assignment** workspace opens. The **User Assigned Connections** tab is displayed.
2. Use the filters at the top of the page to view the required connections based on the resource type and assignment type.
3. To assign a schema to a resource, select the connection and click **Assign** on the control menu.

   The **Assign Connection** dialog box appears.
4. Select the schema that you want to assign, and click **Select**.

   The Assignment Type status changes to **In Progress**.
5. On the control menu at the top of the page, click **Refresh** to view the latest status under the Assignment Type column. You can view the latest user-assigned connections and auto-assigned connections when you click **Refresh** from the control menu.
6. To unassign a connection, select the connection, and click **Unassign**.

   The **Unassign connection** dialog box appears.
7. Click **OK**.
8. To reassign a connection with another schema, select the connection, and click **Reassign**.
9. To assign or unassign multiple connections, select the connections, and select **Manage Multiple Connections** from the control menu at the top of the page.

   The **Assigned Connections** dialog box appears.

10. Make the required changes to the connections, and click **OK**.

Assigning Connections to Reference Resources

You can assign connections from reference resources to resources or other reference resources using the **Connection Assignment** workspace.

Follow the steps listed in the *Managing Connections* section to assign connections. You can generate a missing links report after performing the connection assignment to view the details of source and target objects that are not linked. For more information, see the *Missing Links Report* section.

Alternatively, you can automatically assign connections between the reference resource and resource from the **Auto Assigned Connections** tab. After automatic connection assignment, you can view the list of connections assigned to manually assign or unassign connections.

When you perform connection assignment, you can select the **Retain Unresolved Reference Assets** option in the **Assign Connection** dialog box to retain unresolved reference assets in the catalog after you assign connections. Unresolved reference assets include tables, columns, and any other assets that are not present in the actual resource, but need to be retained to view the complete lineage. The unresolved reference assets appear under the resource to which the connection assignment is performed for the reference resource.

After you unassign connections between reference resources and resources, Enterprise Data Catalog retains the reference resources and assets in the catalog with the custom attributes and business terms you had associated with the reference assets before you performed connection assignment. After you unassign the connection, Enterprise Data Catalog does not retain any custom attributes or business terms propagated from the reference asset to the asset.

Missing Links Report

The Missing Links report helps you identify connection links that are missing after you assign schemas from a resource to connections. Missing links can occur when you have incorrect connection assignments or when resources have outdated metadata.

You can generate a Missing Links report in an Excel sheet and view the details of source and target objects that are not linked. You can also identify missing connection links that are resolved to reference assets in the Missing Links report.

Missing Links Report Columns

The **Missing Links** report contains details of all the links that are missing after you assign connections to the resource. The report includes columns, such as the Source Object Connection, Source Object ID, Destination Object Connection, and Destination Object Connections.

The following table describes the columns in the **Missing Links** report:

Column	Description
Source Object Connection	Name of the source object connection.
Source Object ID	ID of the source object.
Source Object Name	Name of the source object in the resource.
Source Object Resolved	Indicates if the source object is available in the catalog.
Reference Source	Indicates if the source object is resolved to a reference asset.
Destination Object Connection	Name of the target object connection.
Destination Object ID	ID of the target object.
Destination Object Name	Name of the target object in the resource.
Destination Object Resolved	Indicates if the target object is available in the catalog.
Reference Destination	Indicates if the target object is resolved to a reference asset.
Type	Type of the data flow between source and target objects.

Generating Missing Links Report

Generate a missing links report to find the missing connection links after you assign connections to the resource.

1. Open the **Connection Assignment** workspace.
2. From the **User Assigned Connections** or **Auto Assigned Connections** tab, select a resource for which you want to generate the report.
   - Optionally, you can select a resource, and then select the connections for which you want to generate the report.
3. Click **Generate Missing Links Report**.

   Catalog Administrator downloads the report to your machine.

   Note: You can generate a missing links report for only assigned connections.

CHAPTER 9

Configuring Reusable Settings

This chapter includes the following topics:

- Reusable Configuration Overview, 96
- General Configuration Properties, 96
- Data Integration Service Connection Properties, 97
- Setting Up a Reusable Data Integration Service Configuration, 97

Reusable Configuration Overview

You need to configure the Data Integration Service settings for a resource to extract profile metadata from source systems. You can create a reusable configuration for scanners to extract profile metadata that you can reuse for multiple resources.

A reusable configuration helps you quickly configure multiple resources for extraction of profile metadata. Specify settings, such as the domain name, Data Integration Service Name, and user credentials.

General Configuration Properties

The general properties for reusable configuration include name, description, and the profiling configuration type.

The following table describes the general properties for a reusable configuration:

Property	Description
Name	Name of the reusable configuration for profile metadata extraction.
Description	Descriptive text about the reusable configuration.
Profiling	Indicates the Data Integration Service configuration for profiling.

Data Integration Service Connection Properties

Data Integration Service connection properties include the domain information, domain user information, Data Integration Service information, and Model repository information.

The following table describes the Data Integration Service properties for a reusable, global configuration:

Property	Description
Domain Name	Name of the domain. The name must not exceed 128 characters and must be 7-bit ASCII. It cannot contain a space or any of the following characters: ` % * + ; " ? , < > \ /
Data Integration Service	Name of the Data Integration Service associated with the Catalog Service.
User Name	Username to access the Model Repository Service.
Password	Password to access the Model Repository Service.
Security Domain	Name of the security domain to which the Informatica domain user belongs.
Host	Host name of the node running the Model Repository Service.
Port	Port number of the node running the Model Repository Service.

Setting Up a Reusable Data Integration Service Configuration

Use the **Manage** menu to create a reusable configuration to extract profile metadata from the source systems.

1. From the Catalog Administrator header, click **Manage** > **Reusable Configuration**.

   The **Reusable Configuration** workspace opens.
2. From the control menu, click **New**.

   The **New Reusable Configuration** dialog box appears.
3. Enter the general properties, such as name and description.

   The DISOptions option is selected by default in the Profiling field. The Profiling field indicates the configuration type.
4. In the **Domain Connection Settings** section, configure the domain information, Data Integration Service information, and Model Repository Service information.

CHAPTER 10

Monitoring Enterprise Data Catalog

This chapter includes the following topics:

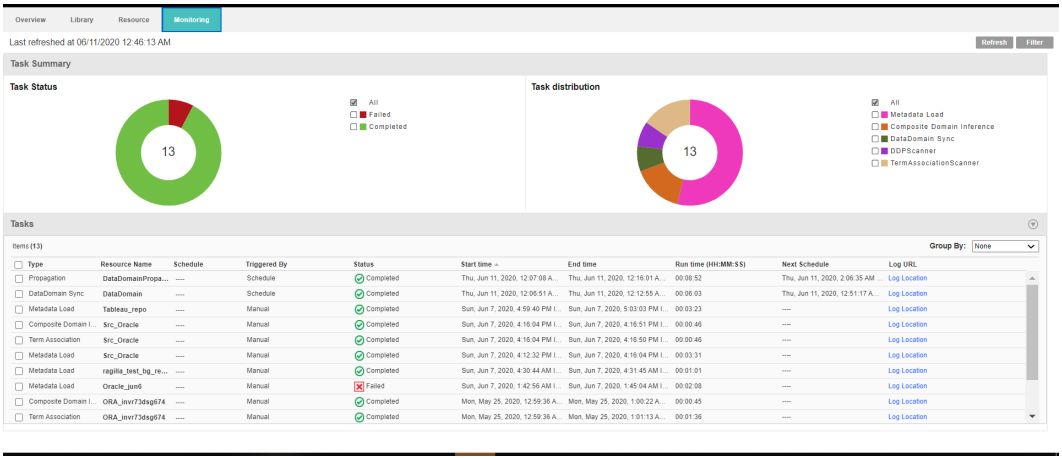
- Monitoring Enterprise Data Catalog Overview, 98
- Task Status, 99
- Task Distribution, 99
- Monitoring by Resource, 100
- Monitoring by Task, 100
- Managing Tasks , 101
- Applying Filters to Monitor Tasks, 101

Monitoring Enterprise Data Catalog Overview

Monitoring Enterprise Data Catalog includes tracking the status and schedule of tasks. You can monitor the duration of the tasks that are running. You can also monitor the resource distribution in terms of the number of resources for each resource type.

The **Overview** workspace displays an overview of the monitoring statistics. You can view the number of resources for each resource type, task status details, and task schedule. To perform a detailed analysis of Enterprise Data Catalog performance, you can open the **Monitoring** workspace. The task status that you can monitor includes the number of tasks and their statuses, such as Complete, Failed, and Running. You can also view the number of tasks for each phase of the metadata extraction, such as metadata load, profile executor, and profile result fetcher. Open the log files for troubleshooting Enterprise Data Catalog tasks and further scrutiny. You can also filter and group by the jobs and tasks based on multiple factors.

The following image shows the Monitoring workspace:



Task Status

Tasks can have different statuses based on what stage of the metadata extraction process the tasks are in. The task status pie chart in the **Monitoring** workspace represents different task statuses and the number of tasks in each task status. Each task status has a different color in the chart.

Place the pointer on the different sections of the pie chart to view the number tasks for each task status. Click any section on the pie chart to view more task details.

The task status chart displays the following task statuses:

- Canceled. Number of canceled tasks.
- Failed. Number of failed tasks.
- Queued. Number of tasks that are in queue for run.
- Running. Number of tasks that are running.
- Complete. Number of tasks that have been successfully completed.

Task Distribution

Tasks distribution pie chart in the **Monitoring** workspace displays a summary of the task types and the number of tasks for each task type. The task types are Metadata Load, Profile Executor, and Profile Result Fetcher. Each task type has a different color in the chart.

Place the pointer on the different sections of the pie chart to view the number tasks for each task type. Click any section on the pie chart to view more task details. Use the filters to the right of the pie chart to filter specific task types.

Monitoring by Resource

After scanners start extracting metadata from resources, you can monitor the resource statistics in the **Monitoring** tab. You can view a consolidated summary of resources with different task statuses.

The monitoring statistics include resource schedule, task type, task status, and the details of the scheduled next run. You might want to monitor specific resources based on certain conditions. For example, view the resources that contributed to the metadata extraction for the last 12 hours or resources that had failed tasks as part of troubleshooting issues. Use the filters on the **Monitoring** tab to view specific resource statistics.

Monitoring Profiles

When you enable data discovery and run a resource, the profiling scanner performs the Profile Run job. In the Monitoring tab, when you select the Profile Run job, the Logs section displays the statistics and progress of the profile run.

When you run a resource, the profiling scanner performs the following validation steps:

Data Integration Validation

1. Verify that the Data Integration Service configured for the resource is enabled.
2. Verify that the profiling warehouse has been properly configured.
3. Verify that the profiling warehouse connection is properly configured.

Profile Validation

1. Chooses a random data source in the resource to run the profile on. If the profile run is successful, the scanner continues to run the profile on the rest of the data sources.
2. If the profile run fails on the first data source, the profiling scanner proceeds to choose another random data source to run the profile on. If the profile run is successful, the scanner continues to run the profile on the rest of the data sources.
3. If the profile run fails on the first and second data sources, the profiling scanner proceeds to choose another random data source in the resource to run the profile on. If the profile run is successful, the scanner continues to run the profile on the rest of the data sources.

If the profile run fails for all the three data sources, the scanner stops the Profile Run job and the Profile Validation property in the Statistics and Progress tabs display the `Validatation failed` value.

When the validation fails, click the **Log URL** for to view the validation summary in the profile scanner application log to verify the reason for validation failure.

Monitoring by Task

As part of Enterprise Data Catalog administration activities, you can monitor the tasks for each scan job. To effectively troubleshoot issues, you need to closely monitor the tasks. You might also want to focus on the tasks that failed or that are running for a long time.

The monitoring statistics include the task type, resource name, schedule, task status, and the start and end time. To view the log file for a task, click **Open Log** under the **Log URL** column.

You can use the control menu for each task to cancel, pause, or resume each task. To cancel, pause, or resume multiple tasks, select the tasks and use the control menu at the top of the **Tasks** grid.

Managing Tasks

You can manage the tasks listed in the **Tasks** panel on the **Monitoring** workspace.

Managing tasks includes the following actions that you can perform by clicking the drop-down list next to the **Log URL** column for each task:

Cancel

Select this option to stop a task in progress.

Pause

Select this option to pause a task in progress.

Resume

Select this option to resume a task that was paused.

Retry

Select this option to retry a task that failed.

Applying Filters to Monitor Tasks

The **Tasks** grid lists all the tasks in the Enterprise Data Catalog repository. Use the **Filter** panel on the Monitoring workspace to filter the task list. You can filter the tasks based on multiple criteria, such as resource name, resource type, and the time of resource creation. You can also view tasks of resources that were created at a specific time period.

1. On the **Monitoring** workspace, click **Filter** at the top, right corner.

   The **Filter** panel appears.
2. To filter the task list based on a resource name, in the **Resource Name** field, enter the required text.
3. To filter tasks based on one or more resource types, select the appropriate check boxes next to the **Resource Type** field.
4. To filter the tasks based on a user who created a resource, in the **Created By** field, enter the username.
5. To filter the tasks based on the time period when a resource was created, in the **Resource Create Time** field, specify the appropriate filter criteria.
6. Click **Apply**to view the filtered task list.
7. To clear the filter condition, click **Clear**.
8. To refresh the list of tasks in the **Tasks** grid, click **Refresh**.

CHAPTER 11

Managing Data Domains

This chapter includes the following topics:

- Managing Data Domains Overview, 102
- Best Practices for Creating Data Domains Using Rules, 103
- Creating a Rule-based Data Domain, 103
- Creating a Data Domain Group, 105
- Viewing Data Domains and Data Domain Groups, 106
- Filtering Data Domains, 106
- Filtering Data Domain Groups, 106
- Modifying a Data Domain or Data Domain Group, 107
- Permissions and Privileges, 107
- Data Domain Discovery on the Spark Engine, 108
- Data Domain Discovery on the Databricks Cluster, 109

Managing Data Domains Overview

Data domain discovery is the process of discovering the functional meaning of data in data sources based on the semantics of the data. To perform data domain discovery, you need to create a rule-based data domain.

To manage data domains and data domain groups, you can perform the following tasks :

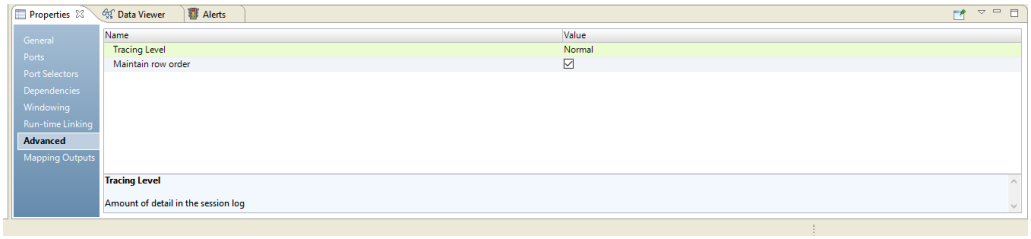
- Create new data domains and data domain groups.
- Modify a data domain or data domain group.
- Add data domains to a data domain group.
- Delete data domains and data domain groups.

You need the correct permissions and privileges to access data domains and data domain groups in Enterprise Data Catalog.

Best Practices for Creating Data Domains Using Rules

Before you create a rule-based data domain, you need to create rules in the Developer tool. Consider the following best practices for the rules that you define in the Developer Tool:

- If the mapplet contains an Expression transformations, you must enable the **Advanced** > **Maintain row order** property in the Developer Tool as shown in the following image:



- The mapplet output must be of type integer.
- Do not link the incompatible ports in the mapplet. For example, do not connect a string type port to an integer type port.
- Expression transformations must not contain multiple variable ports. Instead of creating multiple variable ports, you can use Java transformations in the mapplet.
- You need to create a mapping with the expression rule and run the mapping in the Spark or Blaze runtime environment. You can then verify the mapping log files before you use the rule to create a data domain in Catalog Administrator.
- The output transformation must return integer 1 or 0 as the output instead of boolean or string.

Creating a Rule-based Data Domain

You can create a rule-based data domain in Catalog Administrator. After you create a data domain, you can discover the data domain in data sources. You can edit predefined data domains and smart data domains in Catalog Administrator

1. Click **New** > **Data Domain**.

   The **New Data Domain** wizard appears.
2. In the **Name** field, enter a name for the data domain.

   Note: Do not include whitespace, numbers, or special characters except the at (**@**), hash (**#**), or underscore (**_**) in the data domain name.
3. In the **Description** field, enter a brief description of the data domain.
4. Select one of the following options for the **Specify Rules for this data domain** field:
   - **Yes**. Specify data rules, column name rules, or both the rules for the data domain.
   - **No**. If you do not specify rules in the data domain, then it is a smart data domain. Smart data domains are user-defined data domains that can be manually curated and propagated to similar columns.
5. To choose a data rule for the data domain, click **Browse**.
6. In the **Data Rule** dialog box, choose one of the following rules:

- **Select Reference Table**. Select the option, and choose a reference table in the list of records. When you choose a reference table, the column data in the reference table is used to associate data domains to assets.
- **Specify Regular Expression**. Select the option, and enter a regular expression to match string data. A regular expression is a specialized formula for matching text strings that follow a pattern. The syntax used for regular expression must conform to the grammar rules and tokens described here: https://docs.informatica.com/data-catalog/shared-content-for-data-catalog/10-5/transformation-language-reference/functions/reg_extract.html.
- **Select Rule**. Select the option, and choose a rule in the list of records. You can choose mapplet-based rules that were created using Informatica Analyst or Informatica Developer client. For more information about using mapplet based rules, see: https://docs.informatica.com/data-engineering/shared-content-for-data-engineering/10-5/developer-mapping-guide/mapplets/mapplet-validation/mapplet-as-a-rule-validation.html.

7. Click **Select**.

   The data rule appears in the **Data Rule** field.

8. To choose a column name rule for the data domain, click **Browse**.
9. In the **Column Name Rule** dialog box, choose one of the following rules:
   - **Select Reference Table**. Select the option, and choose a reference table in the list of records. When you choose a reference table, the column data in the reference table is used to associate data domains to assets.
   - **Specify Regular Expression**. Select the option, and enter a regular expression to match string data. A regular expression is a specialized formula for matching text strings that follow a pattern. The syntax used for regular expression must conform to the grammar rules and tokens described here: https://docs.informatica.com/data-catalog/shared-content-for-data-catalog/10-5/transformation-language-reference/functions/reg_extract.html.
   - **Select Rule**. Select the option, and choose a rule in the list of records. You can choose mapplet-based rules that were created using Informatica Analyst or Informatica Developer client. For more information about using mapplet based rules, see: https://docs.informatica.com/data-engineering/shared-content-for-data-engineering/10-5/developer-mapping-guide/mapplets/mapplet-validation/mapplet-as-a-rule-validation.html.
10. Click **Select**.

    The column rule appears in the **Column Rule** field.

11. In the **Conflict Resolution** field, choose one of the following options:
    - **If rules conflict, data overrides column name rule**. Uses data rule result to auto-accept data domains.
    - **If rules conflict, column name overrides data rule**. Uses column name rule result to auto-accept data domains.
    - **Match data and column name rule**. Data domains are auto-accepted when the data and column name rule results match.
    - **Match data or column name rule**. Data domains are auto-accepted when data or column name rule results match.

    If you do not select any option from the **Conflict Resolution** field, Enterprise Data Catalog auto-accepts data domains based on the data rule results.

12. Configure the following data conformance properties if you specify rules for the data domain:
    - **Minimum conformance**. Enter a minimum conformance percentage for the data domain.

- **Auto accept if more than**. Enter a percentage. Enterprise Data Catalog accepts the data domain automatically if the data domain match exceeds the configured percentage.
- **Row Count**. Enter the number of minimum rows for data conformance.

13. To add the data domain to multiple data domain groups, perform the following steps:
    a. Click  in the **Data Domain Groups** pane.
       The **Data Domain Groups** dialog box appears.
    b. Select the data domain groups, and click **Select**.
       The list of data domain groups appears in **Data Domain Groups** pane.
    c. After you choose the data domain groups for the data domain, you can also remove a data domain group. Click to remove a data domain group.
14. If you specify rules for the data domain, you can also add multiple proximity data domains for the data domain.
    Perform the following steps to add a proximity data domain:
    a. In the **Proximity Data Domains** section, click  .
       The **Data Domains** dialog box appears.
    b. In the **Data Domains** dialog box, choose a data domain, and click **Select**.
    c. In the **Proximity Data Domains** section, click  to add another proximity data domain.
    d. In the **Proximity Data Domains** section, click to remove a proximity data domain.
15. Click **Save** to save the data domain.

Creating a Data Domain Group

You can create a data domain group and add one or more data domains to the group.

1. Click **New** > **Data Domain Group**.
   The **New Data Domain Group** wizard appears.
2. In the **Name** field, enter a name for the data domain group.
   Note: In the **Name** field, do not include whitespace characters, or start the name of the data domain with special characters or numbers except the at (**@**), hash (**#**), or underscore (**_**) in the data domain group name.
3. In the **Description** field, enter a brief description for the data domain group.
4. In the **Data Domains** pane, click  to add data domains to the group.
   The **Data Domains** dialog box appears.
5. Select the data domains, and click **Select**.
   The list of data domains appear in the **Data Domains** pane.
6. To remove a data domain in the group, select the data domain in the **Data Domains** pane, and click  .
7. Click **Save** to save the new data domain group.

Viewing Data Domains and Data Domain Groups

You can view the list of data domains and data domain groups in the Library workspace. When you open a data domain or data domain group, the domain properties appear in read-only mode. When you view a data domain, Enterprise Data Catalog shows the composite data domains that include the data domain.

1. In the header area, click **Library**.

   The **Library** workspace appears.
2. To view the data domain groups, click **Data Domain Groups** in the **Assets** pane.
3. To view the data domains, click **Data Domains** in the **Assets** pane.
4. To view a brief summary of the data domain or data domain group, click ▸.
5. To open, edit, or delete a data domain group or data domain, select the group or data domain, and click the **Actions** (⊙) menu.

   The **Open**, **Edit**, **Delete** options appear.
6. To view the data domain group or data domain, click **Open**. Alternatively, you can click a data domain group or data domain to view the details.

Filtering Data Domains

You can filter the data domains to open, edit, or delete specific data domains.

1. In the header area, click **Library**.

   The **Assets** pane appears in the **Library** workspace.
2. Click **Data Domains** in the **Assets** pane.

   The data domains appear in the **Data Domains** pane. The **All** filter is selected by default in the **Filter** pane and the list of data domains appears in the **Data Domains** pane.
3. To view the data domains in a data domain group, select a data domain group in the **Filter** > **Domain Filters** list. Click **Apply**.

   The data domains in the domain group appear in the **Data Domains** pane.
4. To apply a filter on the data domains, enter a term in the **Domain Filters** field. Click **Apply**.

   The filter results appear in the **Data Domains** pane.
5. To clear the filter and filter results, click **Clear All**.

Filtering Data Domain Groups

You can filter the data domain groups to open, edit, or delete specific data domain groups.

1. In the header area, click **Library**.

   The **Assets** pane appears in the **Library** workspace.
2. Click **Data Domain Groups** in the **Assets** pane.

   The data domain groups appear in the **Data Domain Groups** pane.

3. To apply a filter on the data domain groups, enter a term in the **Domain Group Name** field. Click **Apply**.

   The filter results appear in the **Data Domain Groups** pane.
4. To clear the filter and filter results, click **Clear All**.

Modifying a Data Domain or Data Domain Group

You can modify the properties for a data domain or data domain group. You cannot delete a data domain if the data domain is included in a composite data domain.

1. In the header area, click **Library**.

   The **Library** workspace appears.
2. In the **Assets** pane, select **Data Domains**.

   The data domains appear in the **Data Domains** pane.
3. Select the data domain, and click the **Actions** ( ) menu.
4. To modify the data domain properties, click **Edit**.

   The **Data Domain** wizard appears.
5. Edit the properties for the data domain based on your requirement.

   Note: You cannot modify the name of a data domain or data domain group.
6. Similarly, you can modify the data domain group properties.

Permissions and Privileges

You need permissions and privileges to access data domains and data domain groups in Enterprise Data Catalog.

Assign Privileges in Informatica Administrator to Access Data Domains

In Informatica Administrator, you need to assign privileges for users to view, edit, or delete data domains and data domain groups.

Configure the following privileges in the Administrator tool to access data domains and data domain groups:

Application Service	Privilege	Description
Model Repository Service	Manage Data Domains	User can create, edit, and delete data domains in the data domain glossary. This privilege is part of the **Data Domain Administration** privilege group.
Catalog Service	Catalog Management: Domain Creation	Users can perform the following actions: - Create data domains - Update data domains - View data domains - Delete data domains

Application Service	Privilege	Description
	Catalog Management: Domain Curation	Users can perform the following actions: - View data domains - Accept or reject data domains
	Domain Management: Admin - ViewDomain and Domaingroup	Users can perform the following actions: - View data domain and data domain group
	Domain Management: Admin - EditDomain and Domaingroup	Users can perform the following actions: - Create data domains and data domain groups - Update data domain and data domain group - View data domains and data domain group - Delete data domain and data domain group

Assign Permissions on Resources in Catalog Administrator to Access Data Domains

You can specify default permissions for one or more resources, users, user groups, or specify resources, permissions, users, and user groups as required.

As a catalog administrator, you can assign the following permissions on resources to users and user groups:

Permission	Description
Read	View data domains and data domain groups in Enterprise Data Catalog.
Read and Write	View data domain and data domain group assets and assign data domains to assets. You can curate (accept or reject) data domains.
All Permissions	View data domains and data domain groups, assign data domains to assets, and create smart data domains. You can perform all data domain-related tasks.

Data Domain Discovery on the Spark Engine

When you run a profile to perform data discovery on the Spark engine, reference tables are staged on the Hadoop cluster. To make sure that reference tables for all the data domains are staged on the cluster, you can perform the following steps:

Prerequisite:

You must have a permission to impersonate HDFS user when you perform a data domain discovery.

Download the JDBC .JAR Files

1. Obtain the JDBC .jar files of the reference database that you use. You can download the files from the database vendor web site.
2. Copy the files that you download to the following location:`<INFA_HOME>/externaljdbcjars`

Configure Custom Properties on the Data Integration Service

1. Launch Informatica Administrator, and then select the **Data Integration Service** in the **Domain Navigator**.

2. Click the **Custom Properties** option on the **Properties tab**.
3. Set the following custom properties to stage reference tables for the data domains:

Property Name	Property Value
AdvancedProfilingServiceOptions.ProfilingSparkReferenceDataHDFSDir	/tmp/cms
ExecutionContextOptions.SparkRefTableHadoopConnectorArgs	--connect <JDBC thin driver connection URL>

4. Make sure `/tmp/cms` directory exists on the cluster. If the directory is not present, create the `/tmp/cms` directory or a custom directory where you want to stage the data. The reference data is staged at `/tmp/cms` directory by default.
5. Recycle the Data Integration Service.
6. Open the Catalog Administrator and make sure you run a first profile with all the data domains to stage the reference data.

Note: If you do not select all the data domains in the first profile run and then select additional data domains in the next profile run, the profile run may fail.

Data Domain Discovery on the Databricks Cluster

Use the Databricks cluster to perform data discovery on the Spark engine. The Databricks cluster is an environment to run the spark jobs. You can run a profile to perform data discovery for the Azure sources using the Databricks cluster.

You need to perform the following steps to connect to the Azure sources in the Databricks cluster:

Prerequisite

Add the following advanced Spark configuration parameters for the Databricks cluster and restart the cluster:

- `fs.azure.account.auth.type OAuth`
- `fs.azure.account.oauth.provider.type org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider`
- `fs.azure.account.oauth2.client.id <your-service-client-id>`
- `fs.azure.account.oauth2.client.secret <your-service-client-secret-key>`
- `fs.azure.account.oauth2.client.endpoint https://login.microsoftonline.com/<directory-ID-of-Azure-AD>/oauth2/token`

If you use Azure as the Databricks cluster and run the profile on a Microsoft Azure Data Lake Store resource, then add the following advanced Spark configuration parameter for the Databricks cluster and restart the cluster: `spark.hadoop.fs.azure.account.key.<<ACCOUNT_NAME>>. dfs.core.windows.net <<VALUE>>`

Download and Copy the JAR files for the Profiling Warehouse

1. Get the Oracle DataDirect JDBC driver JAR files for the profiling warehouse. You can copy the files from the following location: `<INFA_HOME>/services/shared/jars/thirdparty/com.informatica.datadirect-dworacle-6.0.0_F.jar`.

2. Place the Oracle DataDirect JDBC driver JAR files in the following locations:
   - `<INFA_HOME>/connectors/thirdparty/informatica.jdbc_v2/spark`
   - `<INFA_HOME>/connectors/thirdparty/informatica.jdbc_v2/common`
   - `<INFA_HOME>/services/shared/hadoop/<DataBricksversion>/runtimeLib`

Download and Copy the JAR files for the JBDC Delta Objects

1. Get the JDBC .jar files for JDBC delta objects. You can download the files from the database vendor website.
2. Update the genericJDBC.zip with the JDBC delta JAR files in the following location: `INFA_HOME/services/CatalogService/ScannerBinaries .`
3. Recycle the Catalog Service.

Configure Custom Properties on the Data Integration Service

1. Launch Informatica Administrator, and then select the **Data Integration Service** in the **Domain Navigator**.
2. Click the **Custom Properties** option on the **Properties** tab.
3. Set the following custom property to perform automatic installation of the Informatica libraries into the Databricks cluster:
   `ExecutionContextOptions.databricks.enable.infa.libs.autoinstall: true`
4. Recycle the Data Integration Service.

Supported sources for data domain discovery on the Databricks Cluster

- JDBC Delta
- Azure Data Lake Store Gen2

CHAPTER 12

Managing Composite Data Domains

This chapter includes the following topics:

- Managing Composite Data Domains Overview, 111
- Creating Composite Data Domains, 111
- Viewing Existing Composite Data Domains, 112
- Filtering Composite Data Domains, 113
- Modifying Existing Composite Data Domains, 113
- Deleting Existing Composite Data Domains, 113

Managing Composite Data Domains Overview

As a catalog administrator, make sure that you identify the existing data domains, create additional data domains or composite data domains if required for the new composite data domain, and understand the combination of rules required to create the composite data domain. Make sure that data domains or other composite data domains exist before you create a new composite data domain.

Managing composite data domains involves the following tasks:

- Creating composite data domains.
- Modifying an existing composite data domain.
- Deleting composite data domains.

Creating Composite Data Domains

Make sure that you create data domains before creating composite data domains.

1. Click **New** > **Composite Data Domain**.

   The **Composite Data Domain** page appears.
2. In the **Name** box, specify the name for the composite data domain.

Note:

- Verify that you provide a unique name for each composite data domain.
- Make sure that you do not include whitespace characters in the **Name** for the composite data domain.
- You must not start the name of the composite data domain with special characters or numbers. You can include the following special characters in the composite data domain name: at (@) hash (#), or underscore (_).

3. In the **Description** box, specify the description for the composite data domain.
4. Specify the data domains that you want to include within the composite data domain in the **Specify Match Criteria** section. You can click  (**Select Data Domain**) to select the required data domains from the **Data Domains** dialog box.

   - Click  (**Add**) to add rules that include data domains or composite data domains that you want to include in the composite data domain definition. Each text box in the **Specify Match Criteria** section that appears when you click **Add** signifies a rule. Enterprise Data Catalog connects the rules using the OR operator.
   - Click  (**Remove**) to remove data domain rules from the composite domain definition.
5. Click **Save**.

Viewing Existing Composite Data Domains

1. Click **Library**.

   The **Library** workspace opens with **Composite Data Domains** option selected under the **Assets** panel by default.
2. Point the mouse to the composite data domain that you want to view from the list.

   The drop-down menu option () appears.
3. Click the drop-down menu option.

   The following list of actions appears:

   - Open
   - Edit
   - Delete
4. Click **Open** to view the details of the selected composite data domain. Alternatively, you can click a composite data domain to view the details.

   If you want to view a summary of the composite data domain with the name and the description, click ▸ adjacent to the composite data domain.

Filtering Composite Data Domains

You can filter existing composite data domains from the **Library** workspace to open, edit, or delete specific composite data domains.

1. Click **Library**. The **Library** workspace opens with **Composite Data Domains** option selected under the **Assets** panel by default.
2. Specify the composite data domain name in the **Name** text box in the **Composite Data Domain Filters** section.
3. Click **Apply** to filter the list of composite data domains.

Modifying Existing Composite Data Domains

1. Click **Library**.

   The **Library** workspace opens with **Composite Data Domains** option selected under the **Assets** panel by default.

2. Point the mouse to the composite data domain that you want to view from the list.

   The drop-down menu option ( ) appears.

3. Click the drop-down menu option.

   The following list of actions that you can perform on the composite data domain appears:

   - Open
   - Edit
   - Delete

4. Click **Edit** to modify the details for the selected composite data domain.
5. Update the data domains and rules associated with the composite data domain as required, and click **Save**.

   Note: You cannot modify the name of a composite data domain.

Deleting Existing Composite Data Domains

You can delete existing composite data domains from the **Library** workspace. When you delete a composite data domain, Enterprise Data Catalog removes the composite data domain from the model repository and the associated search results from the catalog.

1. Click **Library**.

   The **Library** workspace opens with **Composite Data Domains** option selected under the **Assets** panel by default.

2. Point to the composite data domain that you want to delete from the list of existing composite data domains.

   The drop-down menu option ( ) appears.

3. Click the drop-down menu option.

   The following list of actions that you can perform on the composite data domain appears:

   - Open
   - Edit
   - Delete

4. Click **Delete** to delete the selected composite data domain.

   The delete confirmation screen appears.

5. Click **Yes** to confirm deleting the composite data domain.

CHAPTER 13

Manage Synonym Definitions

This chapter includes the following topics:

- Manage Synonym Definitions Overview, 115
- Synonym Definition File Validation, 115
- Uploading Synonym Definition Files, 116

Manage Synonym Definitions Overview

Synonym definitions for assets in the catalog help you search for corresponding assets.

You can define synonyms in a synonym definition file and upload the file in Catalog Administrator. Make sure that the synonym definition file is saved as a text file with a .txt extension.

Define synonyms separated by the symbol (=>) in the synonym definition file as shown in the following sample:

social=>SSN

security=>SSN

uniqueID=>SSN

number=>SSN

customer=>SSN

Note: Consider the following rules and guidelines for valid synonyms:

- Define the synonyms that correspond to each asset in a separate line in the synonym definition file.
- Ensure that each synonym that you define is a single word without white spaces and special characters.

The synonyms social, security, uniqueID, number, and customer correspond to the asset named SSN in the catalog. You can search for a column, table, data domain, or any other asset named SSN using the synonyms.

Synonym Definition File Validation

Catalog Administrator validates a synonym definition file when you upload the file.

Synonym file upload fails with warnings or errors if validation issues exist. Catalog Administrator displays the errors onscreen and writes the errors and warnings to the LDM.log file.

Catalog Administrator records warnings and errors for the following scenarios:

Error scenarios in LDM.log and user interface

- Null asset mapping of a synonym definition
- Special characters and spaces in the synonym definition

Warning scenarios in LDM.log

- Duplicate synonyms

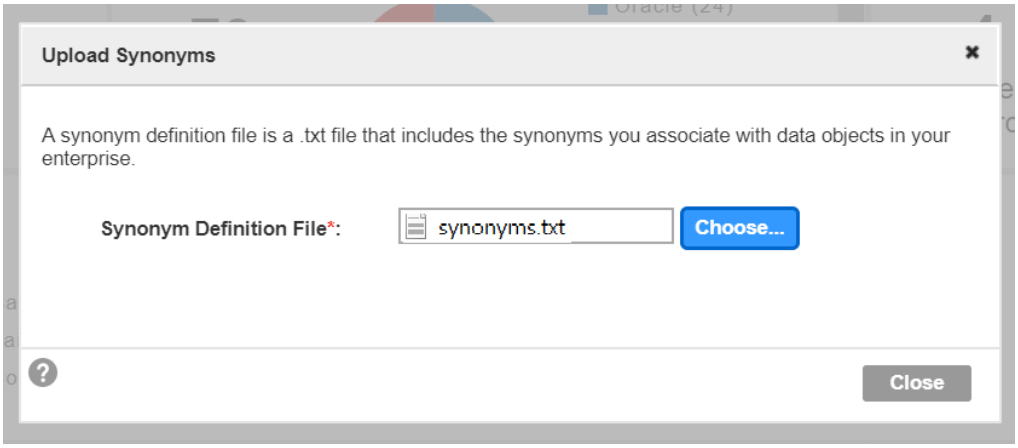
Navigate to the following path to view the LDM.log file:

```
<Informatica installation directory>/logs/<node_name>/services/CatalogService/
<Catalog_service_name>/LDM.log
```

Uploading Synonym Definition Files

1. Click **Manage** > **Synonym Definition File Upload**.

   The **Upload Synonyms** dialog box appears as shown in the following image:



2. Click **Choose**.

   The **Open** dialog box appears.

   Note: Alternatively, you can select a file from your machine and place the file in the **Synonym Definition File** box using a drag-and-drop operation.

3. Select the synonym definition file that you created and click **Open**.
4. Click **Close**.

CHAPTER 14

Managing Cloud Organization

This chapter includes the following topics:

- Managing Cloud Organization Overview , 117
- Cloud Organization Properties , 117

Managing Cloud Organization Overview

Use an organization to access the Informatica Intelligent Cloud Services from Catalog Administrator. Catalog Administrator connects to Informatica Intelligent Cloud Services to import connections into the catalog. Enterprise Data Catalog uses the connections to preview source data and transfer data from source to target.

An organization is a secure area in the Informatica Intelligent Cloud Services repository that stores your licenses, user accounts, data integration assets such as mappings and tasks, and information about jobs and security.

You can add, edit, or delete an organization in Catalog Administrator. Add an existing organization to configure data provisioning when you create a resource. You can add multiple organizations.

Cloud Organization Properties

After you add an organization, the Catalog Administrator displays the properties of the organization. The organization properties are based on properties that you specified for an organization in Informatica Intelligent Cloud Services.

The following table describes the organization properties:

Property	Description
Org Name	Name of the Informatica Intelligent Cloud Services organization.
Org Id	Unique id that is assigned to the organization when you created the organization in Informatica Intelligent Cloud Services.
Username	Username to log in to the Informatica Intelligent Cloud Services.

Adding an Informatica Intelligent Cloud Services Organization

Add an organization in Catalog Administrator to access the Informatica Intelligent Cloud Services.

1. From the Catalog Administrator header, click **Manage** > **IICS orgs**.

   The **IICS Orgs** workspace appears.
2. From the actions menu, click **New**.

   The **Add IICS Org** dialog box appears.
3. Enter the username and password that you used to connect to the Informatica Intelligent Cloud Services.
4. Click **Validate**.

   The Catalog Administrator validates the credentials.
5. Click **OK**.

Editing Informatica Intelligent Cloud Services Organization

Edit the organization properties to update the password of the Intelligent Cloud Services instance.

1. In the **IICS Orgs** workspace, select the organization for which you want to update the password.
2. In the organization properties section, enter the new password for the organization.
3. Click **Validate**.

   The Catalog Administrator validates the new password for the organization.
4. Click **Save**.

CHAPTER 15

Custom Metadata Integration Overview

This chapter includes the following topics:

- Custom Metadata Integration Overview, 119
- Custom Metadata Integration Workflow, 120
- Downloading the Model Definition Files for Validation, 121
- Creating Custom Models, 121
- Updating Custom Models, 122
- Exporting Models, 122
- Deprecating Custom Models, 123
- Custom Resource Type Overview, 123
- Creating Custom Resource Types, 124
- Creating Custom Resources, 124
- Metadata Ingestion Overview, 126
- Customize Icons and Configure Relationships Views for Custom Resources, 129
- Context Lineage for Custom Resources, 140

Custom Metadata Integration Overview

Custom metadata integration signifies ingesting metadata from custom data sources into the catalog.

You can use custom metadata integration to ingest metadata from data sources for which Enterprise Data Catalog does not provide a resource. Based on the model for the data source, Enterprise Data Catalog extracts metadata from the data source. Enterprise Data Catalog, by default, provides models for multiple data sources from which you can extract metadata. These models are referred to as system models. To ingest custom metadata in the catalog, you must define a custom model as the first step.

Custom metadata represents metadata that you define for custom data sources. Custom metadata includes custom properties, relationships, attributes, classes, and context information in the custom data source. For more information about configuring a custom resource to extract context information associated with the lineage, see "Context Lineage for Custom Resources" on page 140

You can define a custom model in Catalog Administrator. The model defines the structure of the metadata stored in the catalog. You can reuse a system model and update the model to create and define a custom model. Alternatively, you can create a model on your own.

After you define the custom model, you must create a custom resource type in Catalog Administrator based on the custom model. A custom resource type can reuse existing models or you might want to create a new model. You can associate multiple models, both system and custom models, to a custom resource type. Based on the custom resource type, you can create a custom resource in Catalog Administrator. You can configure custom icons to represent data assets from a custom resource in Enterprise Data Catalog. You can configure the **Relationships Views** page in Enterprise Data Catalog for custom resources. As part of the configuration, you can define a set of configurations for classes in the custom model. Based on the definitions, you can filter or group related objects for each class type and view the objects on the **Relationships Views** page.

You can configure a custom resource to extract detailed lineage or summary lineage from an ETL resource.

Custom metadata integration involves the following high-level steps:

1. Create the custom model.
2. Create the custom resource type.
3. Create the custom resource.
4. Provide the metadata that you extract from the source as a ZIP file when you configure the custom resource.

Custom Metadata Integration Workflow

The custom metadata integration workflow starts with the definition of the custom model.

The workflow includes the following tasks:

1. Verify if you can reuse an existing model. For example, for a relational data source, you can use any of the relational models included in Enterprise Data Catalog.
2. Creating custom models that involve the following steps:
   a. Define the model for the source metadata in an XML file.

      Note: The custom metadata model XML schema utilizes the Java library package, `jakarta.xml.bind`. The schema enforces strict validation rules that prohibit nested `<parent>` elements within `<parents>` containers and disallow empty self-closing elements, such as `<item/>`.

   b. Use the `custom_model.xsd` model definition file to create and validate the model XML file that you created.

   c. Specify the classes, attributes, and associations that you want to ingest in the catalog.

   d. Configure the search, sort, and filter options for the ingested custom metadata objects.

      Note: Search, sort, and filtering options are resource-intensive operations. Verify that you configure the options that you require.
3. Register the custom model with the Catalog Administrator.
4. Create a custom resource type.
5. Select the models that you want to assign to the custom metadata template.

6. Use the CSV files in the custom resource type template to fill metadata and association details of the data source. You can also fill the transformation mapping details if you plan to view the transformation details between multiple data sources in the lineage diagram.
7. Create a custom resource and upload a ZIP file that includes the CSV files with data when you configure the resource.
8. Run the resource.

When the resource runs, Enterprise Data Catalog processes the metadata from the CSV files and ingests the metadata into the catalog.

Downloading the Model Definition Files for Validation

Enterprise Data Catalog provides model definition files in XSD format that you can use to create and validate the XML files that you create.

To download the model definition files, perform the following steps:

1. Click **Manage** > **Models**. The **Models** page appears.
2. Click  from the **Models** pane.
   The following options appear:
   - **New Custom Model**
   - **Download Model Definition File**
   - **Export Model Content Template**
3. Click **Download Model Definition File**. The `SchemaDefinitions.zip` file is downloaded to your machine. The ZIP file includes the following files that you can use to create and validate the XML files that you create:
   - `custom_model.xsd`. Use the file to create and validate the custom model XML file that you create.
   - `view_configuration.xsd`. Use the file to create and validate the `ViewConfiguration.xml` file that you create to define a set of configurations for classes in the custom model.

Creating Custom Models

Verify that you have created and validated the model XML file before you perform the following steps to create a custom model:

1. Click **Manage** > **Models**.
   The **Models** page appears.
2. Click  from the **Models** pane.
   The menu appears with the following options:
   - **New Custom Model**

- **Download Model Definition File**
- **Export Model Content Template**

3. Click **New Custom Model**.

   The **Upload Custom Model Definition File** dialog box appears.

4. Click **Browse** and select the custom model XML file that you had created and saved.

   The custom model is registered with the catalog.

Updating Custom Models

You can make changes to a custom model based on your requirements.

- Ensure that you increment the version number of the model by one.
- Verify that you exported the custom model XML file to your machine, made the required changes, and saved the file.
- See the following sections to know more about the properties that you must not modify when you update custom models:

  - *Association Properties*
  - *Attribute Properties*
  - *Class Properties*

1. Click **Manage** > **Models**.

   The **Models** page appears.

2. Select the custom model that you want to edit and click **Update**.

   The **Upload Custom Model** dialog box appears.

3. Click **Browse** to select the modified custom model XML file.

The custom model updates take effect during the subsequent scan by the resource.

Alternatively, you can perform this task using the REST APIs. For more information, see the Update the Class topic in the *Enterprise Data Catalog REST API Reference* guide.

Exporting Models

You can export a system model or a custom model XML file to your machine. You can then use the exported XML file to create a custom model.

1. Click **Manage** > **Models**.

   The Models page appears.

2. Select the model that you want to export and click **Export**.

   Enterprise Data Catalog downloads the model XML file to your machine.

Deprecating Custom Models

You can deprecate a custom model. After you deprecate a custom model, you cannot create custom resource types based on the deprecated custom model or associate the deprecated custom model to existing custom resource types. When you deprecate a custom model, Enterprise Data Catalog places the model in the **Deprecated Models** section on the **Models** page. Verify that the custom model is not associated to any custom resource type. If a custom resource type is associated with the custom model that you want to deprecate, remove the association and associate the resource type with another model.

1. After you deprecate a custom model, you cannot change it back to a model that you can use to create custom resource types.

1. Click **Manage** > **Models**.

   The **Models** page appears.

2. Select the custom model that you want to deprecate and click **Deprecate**.

   The confirmation dialog box appears.

3. Click **OK** to deprecate the custom model.

   The custom model appears in the **Deprecated Models** section on the **Models** page.
   After you deprecate a custom model, Enterprise Data Catalog continues to scan and ingest metadata into the catalog using the custom resource that you created based on the custom model.

Custom Resource Type Overview

Custom resource type represents the resource type that you want to create based on the custom model that you defined.

In a custom resource type, you define the models that you want to associate with the custom resource type. In addition, you also specify the associations in the selected models that you want to associate with the custom resource type.

After creating a custom resource type, you can create a custom resource based on the custom resource type.

Custom ETL Resources

To view the transformation logic between multiple data sources in the lineage diagram, you can create and configure a custom ETL resource.

You can configure the custom resource to view the detailed lineage or the summary lineage. For example, you have data from an Oracle table that is transformed in Informatica PowerCenter before being loaded to another Oracle database. For this scenario, you can configure a custom PowerCenter ETL resource to capture the data transformation details in the lineage diagram.

You can use the system model, **com.infa.ldm.etlcore** in Enterprise Data Catalog to create a custom model for the ETL resource.

Creating Custom Resource Types

Perform the following steps to create a custom resource type based on the custom model you created:

1. Select **Manage** > **Custom Resource Types**.

   The **Custom Resource Types** page appears.

2. Click  (Add).

   The **New Custom Resource Type** dialog box appears.

3. Provide the following details for the new custom resource type:
   - Name. Verify that the name starts with a letter. The name can include letters, numerals, underscore (_), hyphen (-), period (.), and white space.
   - Description.
   - Model. Click **Browse**. The **Select Models** dialog box appears. Select the custom model that you created and click **OK**.

     Note: You can select multiple system or custom models that you want to associate with the custom resource type.
   - Connection Types. Click **Select**. The **Select Connection Types** dialog box appears. Select the classes that you can assign as connection endpoints in the custom resource type and click **OK**.
4. Click **OK**.

   The custom resource type is created based on the custom model that you defined.

Creating Custom Resources

Perform the following steps to create a custom resource:

1. Click **New** > **Resource**.

   The **New Resource** wizard appears.
2. Enter a name and an optional description for the resource.
3. Click **File** > **Choose** to open the **Select Resource Type** dialog box.
4. Select the custom resource from the **Custom Resources** section and click **Select**.

   More fields appear in the wizard based on the resource type you created.
5. Configure the following options under the **Connection Properties** section:
   a. **ETL Resource**. If you are configuring an ETL resource to view the transformation lineage, Select **Yes**.

      Make sure that you include the `lineage.csv` file with the lineage details for the custom data source in the ZIP file. See "Entering Transformation Details for Custom ETL Resources" on page 128 for more information.

   b. **Auto Assign Connections**. This option appears if you had selected **Yes** for the **ETL Resource** option. If you want to automatically assign connections to the resource, select this option.

   c. **Run Script**. If you have automated the process to collect metadata from the data sources and generate the ZIP file using a script or a sequence of commands, choose **Yes**.

Provide the following details if you selected **Yes** for the **Run Script** option:

- Script Directory. The absolute path to the directory from where you want to run the commands.
- Script or Commands. Specify the fully qualified path to the script that you want to run with all the arguments. Alternatively, you can specify a sequence of commands separated by semicolons (;) to generate the ZIP file.
- Output File. The fully qualified path and the name of the ZIP file that you want the script to generate.

d. **Files present in remote path**. If you want to select the ZIP file that includes the custom metadata from a remote file location, choose **Yes**. Specify the absolute path to the file in the **Remote File Path** text box.

e. **File**. This option is not applicable if you selected **Yes** for the **Files present in remote path** option. Click **Choose** to select the ZIP file that you created for the custom data source. Alternatively, you can place the file in the **File** box using a drag-and-drop procedure.

Verify the following prerequisites for the ZIP file:

- The file is located on a cluster node.

  Note: The path to the file must be the same on all cluster nodes.
- The required access permissions are configured for the file.
- The file can be accessed by the custom resource.

6. Click **Next** to move to the **Metadata Load Settings** page.
7. Select **Enable Source Metadata** to extract metadata from the ZIP file.
8. Select one of the values for **Memory** to specify the memory required to run the scanner job:

   - Low
   - Medium
   - High

   Select the required memory based on the size of the data set that you want to import. See the *Tuning Enterprise Data Catalog Performance* How-to-Library article for more information about memory values.

   Note: Enterprise Data Catalog restricts the size of the `lineage.csv` file based on the following memory size you specify:

   - Low. Maximum file size is 500 MB
   - Medium memory. Maximum file size is 1024 MB
   - High memory. Maximum file size is 2048 MB

   To override the restriction on the file size, use the `-DcustomScannerMaxFileSize` JVM option. If you specify a very large file size, you might encounter a Java out-of-memory error.
9. Click **Next** to move to the **Custom Attributes** page, and configure the attribute settings. You can select the custom attributes that you want to associate with the resource.
10. Click **Next** to go to the **Schedule** page.
11. Optionally, select schedules for source metadata load and profiling metadata. You can create a global schedule if required.
12. Click **Save**, or Click **Save and Run**.

After you create a custom resource, ensure that you refresh the Enterprise Data Catalog search before you search for custom metadata in the catalog.

Metadata Ingestion Overview

To ingest custom metadata into the catalog, you must enter the data from the custom data source into a template following the template structure. The custom resource uses the data provided in the template to extract and ingest metadata into the catalog.

After creating a custom resource type based on the custom model, you must perform the following steps to ingest metadata into the catalog:

1. Export the custom resource type ZIP file.
2. Extract the CSV files included in the ZIP file.
3. Fill the details in the CSV files according to the structure.
4. Include the updated CSV files in the ZIP file.
5. Provide the ZIP file when you configure the custom resource.

Exporting the Custom Resource Type Template

You can use the custom resource type template to add the object details of a specific class defined in the model. Enterprise Data Catalog uses the details entered in the template to ingest metadata into the catalog.

1. Click **Manage** > **Custom Resource Types**.

   The Custom Resource Types page appears.
2. Select the custom resource for which you want to add details from the **Custom Resource Types** section and click **Export Template**.

   The custom resource type template is downloaded to your machine as a ZIP file.
3. Extract the following files in the ZIP file to your machine and enter the required details:
   - objects.csv. Use this file to enter data based on the class details from the data source in the CSV file.
   - links.csv. Enter association details of objects in the data source in the CSV file.
   - lineage.csv. Applicable if you want to configure a custom ETL resource to show the transformation details between multiple data sources in the lineage diagram. Enter the transformation mapping details in the CSV file.
4. Replace the CSV files in the ZIP file with the updated CSV files.

   You can include multiple `objects.csv`, `links.csv`, and `lineage.csv` files in the ZIP file. Ensure that you name the files in the following format:
   - objects*.csv
   - links*.csv
   - lineage*.csv

   Note: * indicates the custom file names that you want to define after the mandatory name if you are uploading multiple files. For example, you can have multiple `objects.csv` files in the ZIP file such as `objects1.csv`, `objectscritical.csv`, and `objects12July.csv`.

Entering Association Details

To enter the association details for objects in the custom data source, perform the following steps:

Verify that you do not include white spaces in the headers.

1. Open the `links.csv` file in any text editor.

2. Enter the association details for the data in the custom data source in the following format:

   ```
   association, from object, to object
   ```

   See the following table for more information about the association details that you must enter:

Field	Description
association	Mandatory. Represents the name of the association in the data source. To enter object association data for objects in a single resource, verify that you enter the `association` entries in the following format:`com.example.packageName.associationName` You can use the custom lineage scanner to specify the association details with other resource types.
fromObjectIdentity	Mandatory. Unique identity of the object from the `from` side of the association. Verify that you enter the complete path to the object. Make sure that you specify the identity of the of the object in a parent-child format. Verify that the identity of the object matches with the identity of the object provided in the `objects.csv` file. For example, to refer to a column, provide the identity with the full path that includes the schema, the table, and the column as shown: `Schema/Table/Column` Entering the complete path to the objects ensures that Enterprise Data Catalog displays the lineage for the objects.
toObjectIdentity	Mandatory. Unique identity of the object from the to side of the association. Make sure that you specify the identity of the of the object in a parent-child format. Verify that the identity of the object matches with the identity of the object provided in the `objects.csv` file. For example, to refer to a column, provide the identity with the full path that includes the schema, the table, and the column as shown: `Schema/Table/Column` Entering the complete path to the objects ensures that Enterprise Data Catalog displays the lineage for the objects.

   Note: Verify that you enter each entry on a separate line.

3. Save the CSV file with the updates.

Entering Class Details

To enter the class details for the data in the custom data source, perform the following steps:

1. Open the `objects.csv` file in any text editor.
2. Enter the class details for the data in the custom data source in the following format:

   ```
   class, identity, core.name, core.description, attributes of the class
   ```

See the following table for more information about the class details that you must enter:

Field	Description
class	Mandatory. Class type of the object in the following format: `com.example.packageName.ClassName`
identity	Mandatory. Unique identity of the class with the path to the class. For example, to refer to a column, provide the identity of the object in the parent/child format: `Schema/Table/Column`
core.name	Mandatory. Name of the object.
core.description	Optional. Description for the object.
com.infa.ldm.etlcore.transformationType	Optional. Define the type of transformation if you want to show the transformation type in the lineage diagram.
attributes of the object	Optional. Data for the attributes associated with the object, separated by commas.

Note: Verify that you enter each entry on a separate line.

3. Save the CSV file with the updates.

Entering Transformation Details for Custom ETL Resources

To enter the transformation details for a custom ETL resource, perform the following steps:

1. Open the `lineage.csv` file in any text editor.
2. Enter the transformation details in the following format:

   `association, from connection, to connection, from object, to object`

   See the following table for more information about the class details that you must enter:

Field	Description
Association	Mandatory. Represents the name of the dataflow association in the core model. To enter object association data for objects in a single resource, verify that you enter the association entries in the following format: `com.example.packageName.associationName` . **Note:** Ensure that you provide only dataflow associations and not any other association types such as a parent-child associations.
From Connection	Optional. The name of the source for the transformation.
To Connection	Optional. The name of the target for the transformation.
From Object	Mandatory. Unique identity of the object from the `from` side of the association. Verify that the identity of the object matches with the identity of the object provided in the `objects.csv` file.

Field	Description
To Object	Mandatory. Unique identity of the object in the `to` side of the association (transformation applied for the object in the **From Object** column). Verify that the identity of the object matches with the identity of the object provided in the `objects.csv` file. **Note:** You can use the `$etlRes` variable at the start of the identity.

com.infa.ldm.etl.ETLContext	Optional. Use the following variables located in the **Association** column of the CSV file to view detailed lineage or summary lineage of the transformations:
	- core.DataSetDataFlow. Links the source and target objects at the data set level. For example, for a relational source, the data flow is at the table level.
	- core.DirectionalDataFlow. Links the source and target objects at a data element level. For example, for a relational source the data flow is at the column level.
	Add the variables in the **Association** column based on the source and target objects for which you want to view detailed or summary lineage.
	Enter **True** in the `com.infa.ldm.etl.ETLContext` column for the variables if you want to view detailed lineage for the transformations. Leave the option blank or enter **False** if you want to view the summary lineage for the transformations.
	In the following sample, detailed lineage is enabled for the tables source table SUPPLIER and the target table SALES:
	Sample 1
	`core.DataSetDataFlow,Oracle_source,Oracle_wembley,SUPPLIER,SALES,TRUE`
	In the following sample, detailed lineage is enabled for the source and target columns ITEMSID and SALESID in the source and target tables:
	Sample 2
	`core.DirectionalDataFlow,Oracle_source,Oracle_wembley,SUPPLIER/ITEMSID,SALES/SALESID,TRUE`

For the `From Object` and `To Object`, fields, you can use the `$etlRes://` variable at the start of the identity to represent the custom ETL resource that you created. Enterprise Data Catalog replaces `$etlRes` with the name of the custom ETL resource to form the complete path to the identity of the object. For example, you can specify the identity as `$etlRes://SUPPLIER/QUANTITY`. If you created a custom ETL resource named `test_etl_res`, Enterprise Data Catalog replaces the variable as shown in the following sample: `test_etl_res://SUPPLIER/QUANTITY`.

Note: Verify that you enter each entry on a separate line.

3. Save the CSV file with the updates.

Customize Icons and Configure Relationships Views for Custom Resources

You can configure custom icons to represent data assets from custom resources in Enterprise Data Catalog. You can also configure the **Relationships Views** page in Enterprise Data Catalog for custom resources.

After you configure custom icons, the icons appear in Enterprise Data Catalog to represent the data assets from a custom resource. You can use custom icons to distinguish custom resource data assets in Enterprise Data Catalog.

As part of the configuring the **Relationships Views** page, you can define a set of configurations for classes in the custom model.

The steps to include custom icons for the custom resource and defining configurations for classes is integrated in the process to create custom models. The steps are optional when you define the custom model. You can configure the features based on your requirement.

Workflow

The workflow to customize icons for custom resource data assets and configure the **Relationships Views** page in Enterprise Data Catalog for custom resources includes the following steps:

1. Exporting the model template. The step involves downloading the following files that you can use to create the custom model, associate custom icons, and configure **Relationships Views** for custom resources:
   - `model.xml`
   - `ViewConfiguration.xml`
   - `icons.zip`
2. Creating the custom resource model in the `model.xml` file as described in the "Creating the Custom Model XML File" on page 131 section.
3. Optional. Placing the custom icons in the `icons.zip` file as described in the "Creating the Icons ZIP File" on page 131 section.
4. Optional. Creating the `ViewConfiguration.xml` file to define a set of configurations for classes in the custom model as described in the "Creating the View Configuration XML File" on page 131 section.
5. Registering the custom resource model as described in the "Creating the Custom Model with Custom Icons" on page 140 section.
6. Preparing the metadata to be ingested as described in the "Metadata Ingestion Overview" on page 126 section.
7. Creating the custom resource as described in the "Creating Custom Resource Types" on page 124 section.

Exporting the Model Template

To download the template to configure the custom icons, perform the following steps:

1. Log in to Catalog Administrator.
2. Click **Manage** > **Models**. The **Models** page appears.
3. Click  from the **Models** pane.
   The following options appear:
   - **New Custom Model**
   - **Download Model Definition File**
   - **Export Model Content Template**
4. Click **Export Model Content Template** .
   The `ModelContentTemplate.zip` file that includes the following files is downloaded to the machine:
   - `model.xml`
   - `ViewConfiguration.xml`

- icons.zip

Creating the Custom Model XML File

Use the model.xml file to create a custom model that defines the structure of metadata ingested into the catalog by the custom resource.

For more information about custom models and creating custom models, see the "Custom Metadata Integration Workflow" on page 120 and the "Creating Custom Models" on page 121 sections.
You can use the custom_model.xsd file to create and validate the model.xml file.

Creating the Icons ZIP File

To create the icons.zip file that includes the custom icons, perform the following steps:

1. Unzip the icons.zip file to the local machine.
2. Rename the directory named default in the unzipped location with the model name. For example, if the name of the model is **com.infa.access**, rename the directory to com_infa_access. Ensure that you replace the dot (.) in the package name with underscore (_) when you rename the directory.
3. Verify that the icon files are named based on the classes in the model. For example, if the model includes classes such as Tables, Views, and Reports, name the icons as Tables.svg, Views.svg, and Reports.svg.

   Note: Ensure that you match the icon names and class names for case sensitivity.
4. Place the icon files in .svg format in the directory you renamed.
5. Zip the file and name the file as icons.zip.

Creating the View Configuration XML File

Use the ViewConfiguration.xml file to define a set of configurations for classes in the custom model.

You can use the ViewConfiguration.xsd file to create and validate the XML file. For more information about downloading the file, see the "Downloading the Model Definition Files for Validation" on page 121 section.
To identify and use the tags that you can use in the XML file, see the following table:

Tag	Description	Sample
<ViewConfigurationCollection> </ ViewConfigurationCollection>	The root element for the viewConfiguration.xml file. Include all the configuration information within the <viewConfigurationCollection> </ viewConfigurationCollection> tag.	<ViewConfigurationCollection> <viewConfiguration <viewConfiguration> <classId>com.infa.db.Table</ <associationConfigs> <entry> <key>core.DataSet</ <value> <name>Views</name> </value> </entry> </ associationConfigs> </viewConfiguration> </ viewConfigurations> </ViewConfigurationCollectio
<viewConfigurations> </ viewConfigurations>	The parent tag that includes a set of <viewConfiguration> </ viewConfiguration> tags that you use to define the configurations for the classes in the model.	See the sample in the previous row.

Tag	Description	Sample
<viewConfiguration> </viewConfiguration>	Defines the required configurations for the class that you want to customize. As shown in the sample, the configurations are defined in the tags specified within the `<viewConfiguration> </viewConfiguration>` tags.	`<viewConfiguration> <classId>com.infa.db.Table</ <associationConfigs> <entry> <key>core.DataSet</ <value> <name>Views</name> </value> </entry> </ associationConfigs> </viewConfiguration>`
<classId> </classId>	The tag to specify the class for which you want to define the configurations. As shown in the sample, `com.infa.db.Table` is the name of the class for which you are defining the configurations.	`<classId>com.infa.db.Table</classId>`
<associationConfigs> </associationConfigs>	The tag that defines a configuration for every association where you want to include the class. As shown in the sample, the association configured is `core.DataSet` and the class included is `com.infa.db.Table`	`<viewConfiguration> <classId>com.infa.db.Table</ <associationConfigs> <entry> <key>core.DataSet</ <value> <name>Views</name> </value> </entry> </ associationConfigs> </viewConfiguration>>`
<entry> </entry>	The tag to define the configuration for a specific association where the specified class can be included.	`<entry> <key>core.DataSet</key> <value> <name>Vi </value> </entry>`
<key> </key>	The tag to define the association for which you want to define the configuration.	`<key>core.DataSet</key>`
<value> </value>	The tag to define the configuration details for the association.	`<entry> <key>core.DataSet</key> <value> <name>Vi </value> </entry>`
<name> </name>	The tag to create a group based on the specified name. Based on the association type with the seed object in the class, Enterprise Data Catalog displays all the objects in the group name on the **Relationship Views** page. The name specified for the group in the sample is `Views`.	`<classId>com.infa.db.Table</classId> <associatio <entry> <key>core.DataSet</key> <value> <name>Vi </value> </entry> </associationConfigs>`
<type> </type>	The tag to define all the objects that you want to collect based on the class type and association configured. In the sample, all the objects of type `com.infa.ldm.relational.View` associated with the class `infa.ldm.relational.Table` through the association `core.DataSetDataFlow` are collected.	`<viewConfiguration><classId>com.infa.ldm.relatio classId><associationConfigs> <entry> <key>core.DataSetDataFlow</key> <value><name>Vie name><type>com.infa.ldm.relational.View</type> < entry></associationConfigs></viewConfiguration>`

Tag	Description	Sample
<classTypes> </classTypes>	Use the tag to configure associations for every class type associated with another class through a specific association. For example, a data domain can be associated with relational tables and views in the catalog. The requirement is to classify the objects of type relational tables and relational views associated with the data domain separately in the relationship diagram. In the sample, the class that represents that objects of type data domain is `com.infa.ldm.profiling.DataDomain`. The objects of type tables is represented by the class type `com.infa.ldm.relational.Table`. The objects of type views is represented by the class `com.infa.ldm.relational.View`. The association between the data domain and the tables and views is represented by the association `com.infa.ldm.profiling.DataDomainTable`.	`<viewConfiguration> <classId>com.infa.ldm.profiling.DataDomain</clas <associationConfigs> <entry> <key>com.infa.ldm.profiling.DataDomainTable</key <classTypes> <entry> <key>com.infa.ldm.relationa key> <value>Tables</value> </entry> <entry> <key>com.infa.ldm.relational.View</key> <value>V value> </entry> </classTypes> </value> </entry> associationConfigs> </viewConfiguration>`
<incomingName> </incomingName> <outgoingName> </outgoingName>	Use the `<incomingName> </incomingName>` and `<outgoingName> </outgoingName>` tags to classify subsets of objects from a group based on specific association types with the seed object. In the sample, objects with the following association types are associated with the seed object class `com.infa.ldm.relational.Table`: - `core.LookupRelatedDataSets` - `core.JoinRelatedDataSets` - `core.PkFkRelatedDataSets` All objects related to the seed object with the defined associations types appear within the name Tables. You can use the `<incomingName> </incomingName>` and `<outgoingName> </outgoingName>` tags to configure specific names for subsets of objects based on association types with the class. The subset names appear within the `Tables` group name. The specific names assigned to classify subsets of objects with the specific association types are the following: - `core.LookupRelatedDataSets.Lookup`	`<viewConfiguration> <classId>com.infa.ldm.relational.Table</classId> <associationConfigs> <entry> <key>core.LookupRelatedDataSets</key> <value> <name>Tables</name> <incomingName>Lookup</incomi <outgoingName>Lookup</outgoingName> </value> </e <entry> <key>core.JoinRelatedDataSets</key> <val <name>Tables</name> <incomingName>Join</incoming <outgoingName>Join</outgoingName> </value> </ent <key>core.PkFkRelatedDataSets</key> <value> <nam name> <incomingName>Primary Key Tables</incoming <outgoingName>Foreign Key Tables</outgoingName> entry> </associationConfigs> </viewConfiguration`

Tag	Description	Sample
	- `core.JoinRelatedDataSets.Join` - `core.PkFkRelatedDataSets.Primary Key Tables, Foreign Key Tables` The specific names appear within the `Tables` name. **Note:** You can use the `<incomingName> </incomingName>` tag if the objects in the association are of the same type. For example, columns in a table related to columns in another table.	

Sample View Configuration XML File

```
<?xml version="1.0" encoding="UTF-8"?>
<ViewConfigurationCollection>
    <ViewConfigurations>
        <viewConfiguration>
            <classId>com.infa.ldm.relational.Table</classId>
            <associationConfigs>
                <entry>
                    <key>core.DataSetDataFlow</key>
                    <value>
                        <name>Views</name>
                    </value>
                </entry>
            </associationConfigs>
        </viewConfiguration>
        <viewConfiguration>
            <classId>com.infa.ldm.relational.Table</classId>
            <associationConfigs>
                <entry>
                    <key>core.DataSetDataFlow</key>
                    <value>
                        <name>Views</name>
                        <type>com.infa.ldm.relational.View</name>
                    </value>
                </entry>
            </associationConfigs>
        </viewConfiguration>
        <viewConfiguration>
            <classId>com.infa.ldm.profiling.DataDomain</classId>
            <associationConfigs>
                <entry>
                    <key>com.infa.ldm.profiling.DataDomainTable</key>
                    <value>
                        <classTypes>
                            <entry>
                            <key>com.infa.ldm.relational.Table</key>
                            <value>Tables</value>
                            </entry>
                            <entry>
                            <key>com.infa.ldm.relational.View</key>
                            <value>Views</value>
                            </entry>
                        </classTypes>
                    </value>
                </entry>
            </associationConfigs>
        </viewConfiguration>
        <viewConfiguration>
```

```
            <classId>com.infa.ldm.relational.Table</classId>
            <associationConfigs>
                <entry>
                    <key>core.LookupRelatedDataSets</key>
                    <value>
                    <name>Tables</name>
                        <incomingName>Lookup</incomingName>
                        <outgoingName>Lookup</outgoingName>
                    </value>
                </entry>
                <entry>
                    <key>core.JoinRelatedDataSets</key>
                    <value>
                    <name>Tables</name>
                        <incomingName>Join</incomingName>
                        <outgoingName>Join</outgoingName>
                    </value>
                </entry>
                <entry>
                    <key>core.PkFkRelatedDataSets</key>
                    <value>
                    <name>Tables</name>
                        <incomingName>Primary Key Tables</incomingName>
                        <outgoingName>Foreign Key Tables</outgoingName>
                    </value>
                </entry>
            </associationConfigs>
        </viewConfiguration>
    </ViewConfigurations>
</ViewConfigurationCollection>
```

Sample Use Cases and Relationship Diagrams

For more information about defining tags in the `ViewConfiguration.xml` file based on specific use cases, see the following sections . Each section also includes a relationship diagram at the end of the section to show how the group names appear based on the definitions.

Using the <name> Tag

In the use case scenario, the objective is to group all objects of type synonym associated with the table under the group name `Custom Synonyms`.

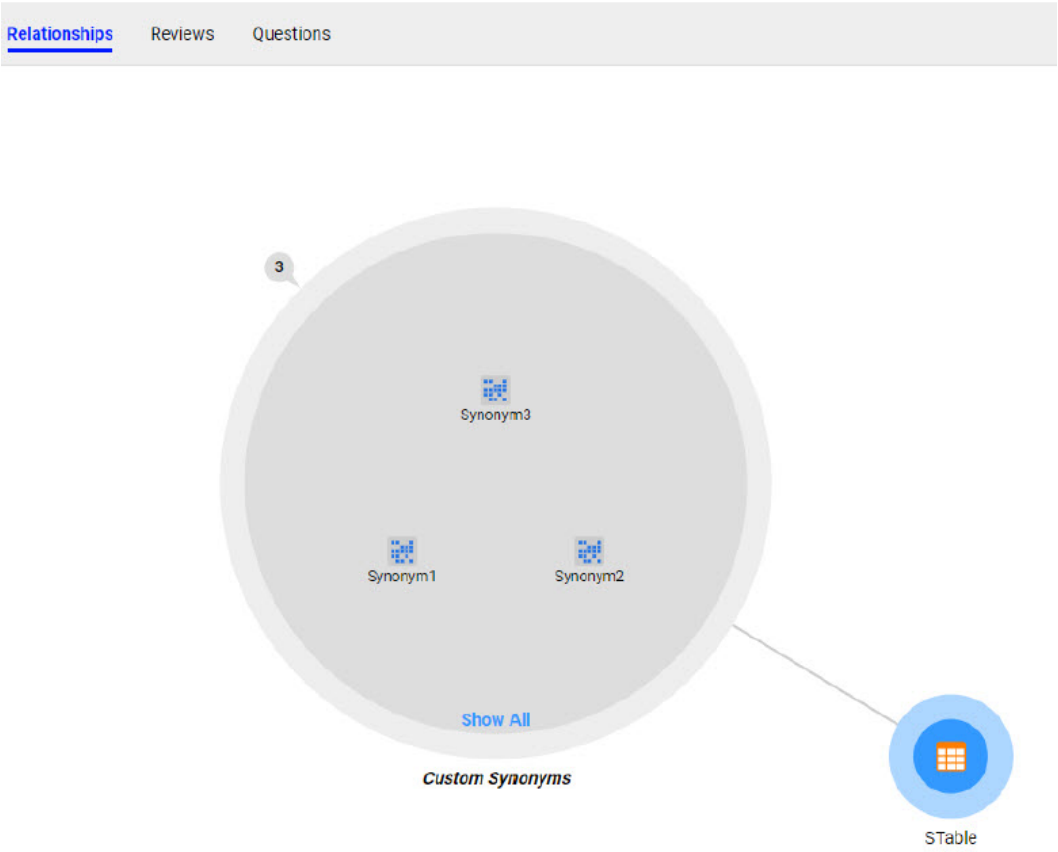
The details of the table, synonyms, and association are provided in the following list:

- `STable`, a table object from a relational database included in the class with the ID `org.custom.relational.Table`
- Synonyms with the names `Synonym1`, `Synonym2`, and `Synonym3` associated with `STable` through the association, `core.TableSynonym`.
- Name for the group that you want to define to include all objects of type synonyms with the specific association type with the class is `Custom Synonyms`.

To address the use case, you can add the following tags in the `ViewConfiguration.xml` file:

```
<viewConfiguration>
                                        <classId>org.custom.relational.Table</classId>
                                        <associationConfigs>
                                                    <entry>
                                                            <key>core.TableSynonym</key>
                                                                    <value>
                                                                        <name>Custom
Synonyms</name>
                                                                    </value>
                                                    </entry>
                                        </associationConfigs>
                                </viewConfiguration>
```

After you run the custom resource and view the **Relationships Views** tab page for the data asset `STable`, you can see all the synonyms in the `Custom Synonyms` group name as shown in the following image:



Using the <type> Tag

In the use case scenario, the objective is to show only specific types of objects that are involved in a lineage with the table. The table can be associated with other tables and views. In the sample, the requirement is to show only views associated with the table under the group name `Custom Views`.

The details of the table, types of views, and association are provided in the following list:

- `TableA`, a table object from a relational database included in the class with the ID `org.custom.relational.Table`
- View `ViewA` is associated with `TableA` through the association, `core.DataSetDataFlow`.
- Name for the group that you want to define to include all objects of a specific type `Custom Views`

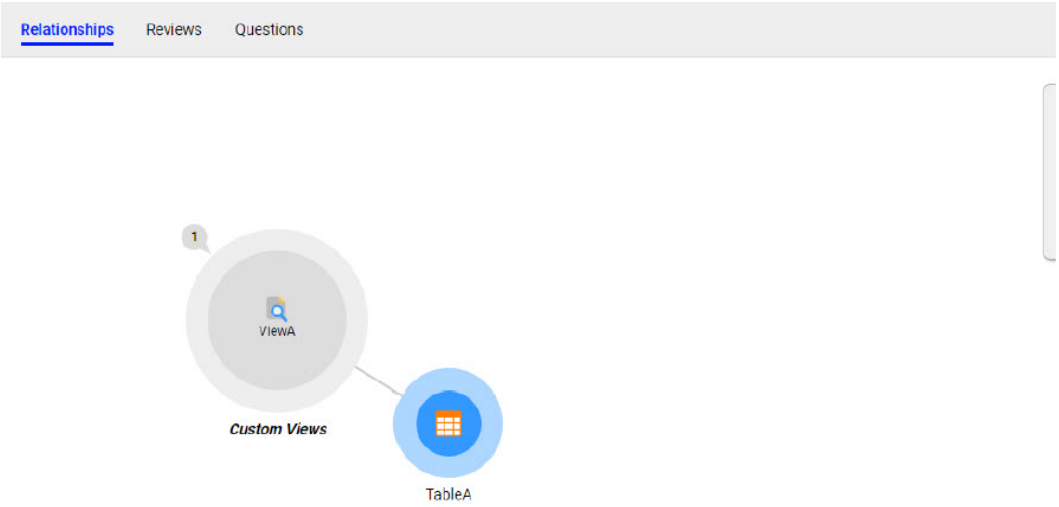
To address the use case, you can add the following tags in the `ViewConfiguration.xml` file:

```
<viewConfiguration>
                                        <classId>org.custom.relational.Table</classId>
                                        <associationConfigs>
                                                <entry>
                                                        <key>core.DataSetDataFlow</key>
                                                        <value>
                                                                <name>Custom Views</name>

<type>org.custom.relational.View</type>
                                                        </value>
                                                </entry>
```

```
        </associationConfigs>
    </viewConfiguration>
```

After you run the custom resource and view the **Relationships Views** tab page for the data asset `TableA`, you can see all the views in the `Custom Views` group name as shown in the following image:



Using the <classTypes> Tag

In the use case scenario, for a table, there are views and tables associated with the table using the same association type. We want to group all the tables that are involved in a lineage with the table under the `Tables` group name and all the views that are associated in a lineage with the table under the `Views` group name.

The details of the data assets, the specific table, and association are provided in the following list:

- `TableM`, a table object from a relational database included in the class with the ID `org.custom.relational.Table`
- View `ViewO` is associated with `TableM` through the association, `core.DataSetDataFlow`.
- Table `TableN` is associated with `TableM` through the association, `core.DataSetDataFlow`.
- Names for the groups that represent the specific tables associated with `TableM` is `Tables` and views associated with the table is `Views`.

To address the use case, you can add the following tags in the `ViewConfiguration.xml` file:

```
<viewConfiguration>
                                        <classId>org.custom.relational.Table</classId>
                                        <associationConfigs>
                                                <entry>
                                                        <key>core.DataSetDataFlow </key>
                                                        <value>
                                                                <classTypes>
                                                                        <entry>

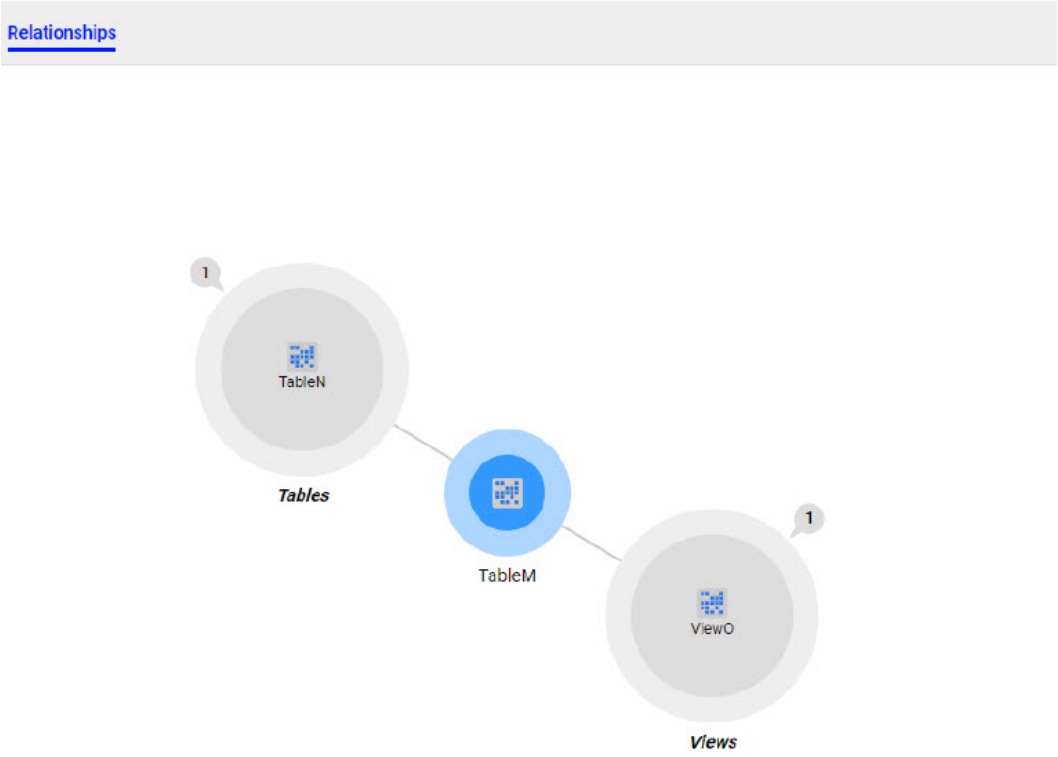
<key>com.infa.ldm.relational.Table</key>

<value>Tables</value>
                                                                        </entry>
                                                                        <entry>

<key>com.infa.ldm.relational.View</key>
```

```
    <value>Views</value>
                                                                    </entry>
                                                                </classTypes>
                                                            </value>
                                                        </entry>
                                                    </associationConfigs>
                                                </viewConfiguration>
```

After you run the custom resource and view the **Relationships Views** tab page for the data asset `TableM`, you can see all the tables and views in the group names `Tables` and `Views` as shown in the following image:



Using the <incomingName> and <outgoingName> Tags

In the use case scenario, we have a table and we have a set of tables related to the table with Join, Lookup and primary key-foreign key associations. We want to join all the tables under the group name Tables, but want to group the tables specifically based on the association such as Join, Lookup or primary key-foreign key. The sample lists the steps to view the tables with the primary key-foreign key association.

The details of the tables and association are provided in the following list:

- `TableB`, a table object from a relational database included in the class with the ID `org.custom.relational.Table`
- `TableB` is associated with `TableG` through the primary key-foreign key association, `core.PkFkRelatedDataSets`. `TableB` holds the primary key.
- `TableB` is associated with `TableH` through the primary key-foreign key association, `core.PkFkRelatedDataSets`. `TableB` holds the primary key.
- Name for the group that represents the specific association when `TableB` is the seed object or source object in a relationship diagram is `Foreign Key Tables`.

- Name for the group that represents the specific association when `TableG` or `TableH` is the seed object or target object in a relationship diagram is `Primary Key Tables`.

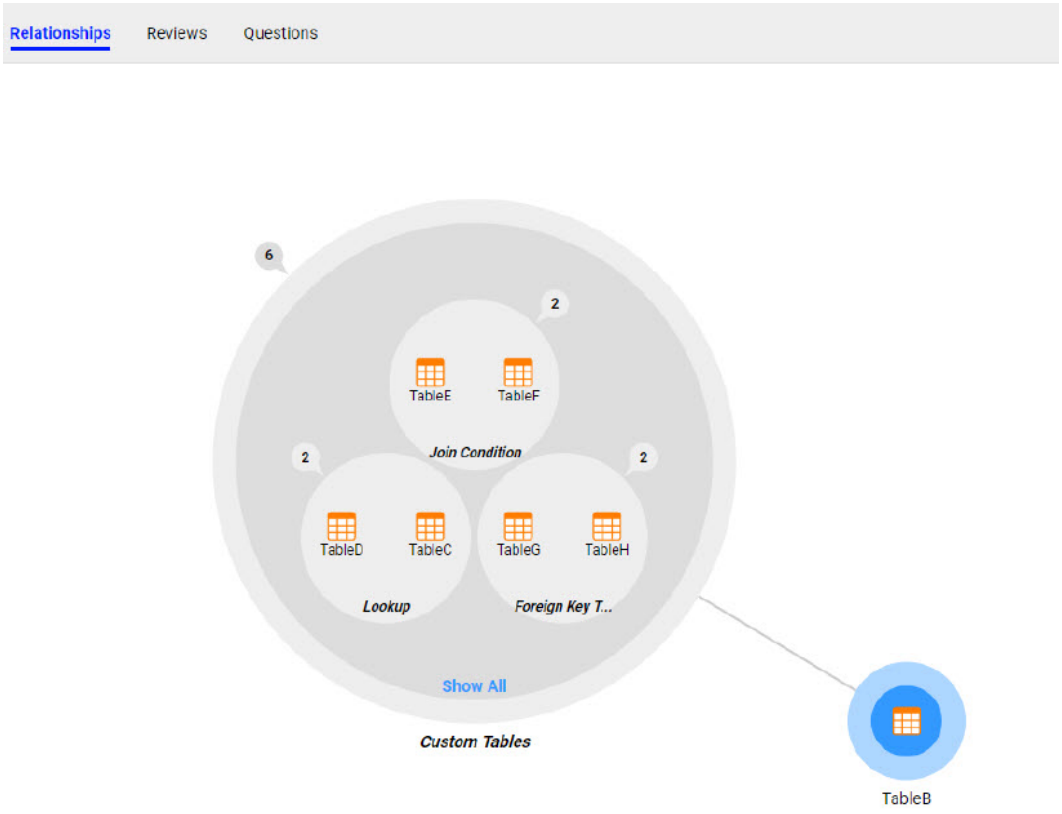
To address the use case, you can add the following tags in the `ViewConfiguration.xml` file:

```
<viewConfiguration>
                                        <classId>org.custom.relational.Table</classId>
                                        <associationConfigs>
                                                <entry>
                                                        <key>core.PkFkRelatedDataSets</
key>
                                                        <value>
                                                                <name>Custom Tables</
name>

<incomingName>Primary Key Tables</incomingName>

<outgoingName>Foreign Key Tables</outgoingName>
                                                        </value>
                                                </entry>
                                        </associationConfigs>
                                </viewConfiguration>
```

After you run the custom resource and view the **Relationships** tab page for the data asset `TableB`, you can see that `Custom Tables` is the group that includes TableG and Table H under the sub group `Foreign Key Tables` as shown in the following image:



Creating the Custom Model with Custom Icons

After you create the `model.xml`, `ViewConfiguration.xml`, and `icons.zip` files, include the files in a ZIP file and perform the following steps to create the custom model:

1. Click **Manage** > **Models**. The **Models** page appears.
2. Click  from the **Models** pane.
   The menu appears with the following options:
   - **New Custom Model**
   - **Download Model Definition File**
   - **Export Model Content Template**
3. Click **New Custom Model**. The **Upload Custom Model Definition File** dialog box appears.
4. Click **Browse** and select the ZIP file that includes the following files:
   - `model.xml`.
   - `ViewConfiguration.xml`.
   - `icons.zip`.

The custom model is registered with the catalog. A successful registration of the custom model is mandatory to view the custom resource association details and the custom icons.

Context Lineage for Custom Resources

The lineage for an asset such as a mapping or stored procedure varies based on the execution instances. A context is an execution instance of an asset. Viewing the details of such instances in the catalog ensures accuracy of the lineage and impact analysis, and maintains the correct dependencies between various assets in the catalog.

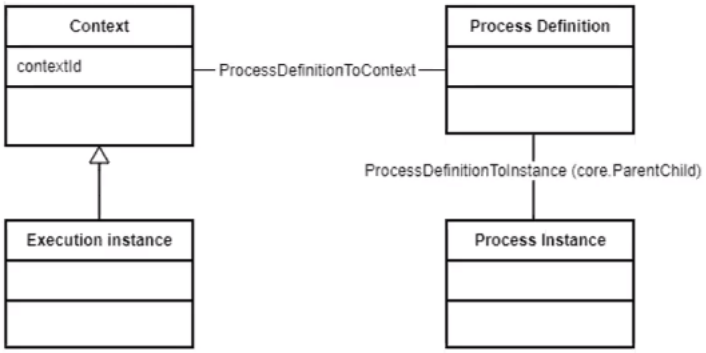
For example, the following table shows multiple data sets consumed by the same stored procedure in different execution instances:

Data Sets	Mapping	Execution Instance	Context
Data set A	Stored Procedure 1	Execution Instance A	Context 1
Data set B	Stored Procedure 1	Execution Instance B	Context 2

The data set A is consumed by stored procedure 1 in execution instance A, and the data set B is consumed by stored procedure 1 in execution instance B. In the Lineage and Impact diagram, you can see all the contexts in which the stored procedure is executed. In this example, context 1 refers to execution instance A, and context 2 refers to the execution instance B. You can set the context for each execution instance so that the Lineage and Impact diagram lists the contexts for you to identify the instances.

You can use Enterprise Data Catalog to view the execution context of a process definition in the Lineage and Impact diagram using custom resources. The parameters that define the dataflow of multiple source and target assets is defined as a process definition. The process definition types include stored procedure or mapping definitions. The execution of a process definition is defined as a process instance.

The following image shows the links or associations that must be included in the custom model to view context metadata in the lineage:



You must use the Process Definition and Process Instance abstract classes to define the custom classes, and the Context class to define the execution contexts. For more information, see the "Context Lineage Resource CSV File Structure" on page 142 topic.

Ingesting Context Metadata using Custom Resources

To ingest context metadata using custom resources in the Catalog Administrator, perform the following steps:

1. Derive an out-of-the-box custom model from the `com.infa.ldm.processflow` system model. Include the `processDefinition` and `processInstance` classes from the `com.infa.ldm.processflow` system model as parent or superclasses in the derived custom model.
   The custom model creation includes the following steps:
   - Define the model for the source metadata in an XML file.
   - Use the `custom_model.xsd` model definition file to create the model XML file that you created.
   - Specify the classes, attributes, and associations that you want to ingest context metadata in the catalog. For more information about context metadata ingestion, see the "Metadata Ingestion Overview" on page 126 and "Context Lineage Resource CSV File Structure" on page 142 sections.

   For more information about custom models and creating custom models, see the "Custom Metadata Integration Workflow" on page 120 and the "Creating Custom Models" on page 121 sections.
2. Register the custom model with the Catalog Administrator. For more information about registering a custom model, see the "Creating the Custom Model with Custom Icons" on page 140 topic.
3. Create a custom resource type. For more information, see the "Creating Custom Resource Types" on page 124 topic.
4. Select the models that you want to assign to the custom metadata template.
5. Use the CSV files in the custom resource type template to fill metadata and association details of the data source. You can also fill the transformation mapping details if you plan to view the transformation details between multiple data sources in the lineage diagram.
6. Create a custom resource and upload a ZIP file that includes the CSV files with data when you configure the resource. For more information, see the "Creating Custom Resources" on page 124 topic.
7. Run the resource.

   Note: To view the `processInstance` class within the `processDefinition` in the custom model, you must derive the `processInstance` from the `core.DataSet`.

   For more information about viewing context information in the lineage diagram, see the *Informatica 10.4.1 Enterprise Data Catalog User Guide*.

Context Lineage Resource CSV File Structure

You can configure a custom resource to view the context information associated with a process definition, or mapping definition.

The following table describes the elements that must be included in the `objects.csv` file to view context information in the lineage:

Element	Description
Classes	Include the following classes in the custom model: - **com.infa.ldm.processflow.ProcessDefinition**. Links the source and target objects at the process flow level. You must define an instance or a subclass of this class in the custom model. - **com.infa.ldm.processflow.ProcessInstance**. The execution of a process definition is called a process instance. The process flow links are established at the process instance level. - **com.infa.ldm.processflow.Context** and **com.infa.ldm.processflow.ExecutionContext**. Contains all the execution context objects.
com.infa.ldm.processflow.contextId	Mandatory. Specify the context ID for all the context objects. The length of the context ID must be 32 characters, and unique. You can use the MD5 Hash algorithm to generate unique context IDs. **Note:** The context information is not displayed in the detailed lineage if the context ID is not assigned to the context objects.

For more information about including the classes in the custom data source, see the "Entering Class Details" on page 127 topic.

The following table describes the associations that must be included in the `links.csv` file to view context information in the lineage:

Association	Description
com.infa.ldm.processflow.ProcessDefinitionToInstance	Links the process definitions to the process instances.
com.infa.ldm.processflow.ProcessDefinitionToContext	Links the context objects to the process definitions.

For more information about including associations in the custom data source, see the "Entering Association Details" on page 126 topic.

The following table describes the elements that must be included in the `lineage.csv` file to view context information in the lineage:

Element	Description
Association	Include the following dataflow associations: - **core.DataSetDataFlow**. Optional. Links the source and target object at data set level. - **com.infa.ldm.etl.DetailedDataSetDataFlow**. Links the data sets and the process flows, or transformation objects and creates lineage links between data sets and process flows. - **com.infa.ldm.etl.DetailedDataFlow**. Links data element and process flows at the column level. Include the following process flow associations: - **com.infa.ldm.processflow.DataSetContextAttributeProjection**. Projects the data set attributes such as source resource name, and target resource name as context information in the lineage. - **com.infa.ldm.processflow.DataSetToContext**. Links the source objects to the context objects. - **com.infa.ldm.processflow.ContextToDataSet**. Links the context objects to the target objects.
core.targetAttribute	A source or a target data set creates two dataflow links to context. For a source data set, the core.targetAttribute element must include com.infa.ldm.processflow.sourceName, and com.infa.ldm.processflow.sourceResourceName as the context dataflow links. For a target data set, the core.targetAttribute element must include com.infa.ldm.processflow.targetName, and com.infa.ldm.processflow.targetResourceName as the context dataflow links. **Note:** Specify the core.targetAttribute only for the com.infa.ldm.processflow.DataSetContextAttributeProjection association.
com.infa.ldm.processflow.contextId	Mandatory. Specify the context ID for all the dataflow links. The length of the context ID must be 32 characters, and unique. You can use the MD5 Hash algorithm to generate unique context IDs. **Note:** The context information is not displayed in the detailed lineage if the context ID is not assigned to the dataflow links. You must include the context ID for the following associations: - com.infa.ldm.etl.DetailedDataFlow - core.DataSetDataFlow - core.DirectionalDataFlow - com.infa.ldm.etl.DetailedDataElementDataSetDataFlow - com.infa.ldm.etl.DetailedDataSetDataElementDataFlow

For more information about including the transformations in the custom data source, see the "Entering Transformation Details for Custom ETL Resources" on page 128 topic.

CHAPTER 16

Catalog Backup and Log Collection Utilities

This chapter includes the following topics:

- Overview, 144
- Accessing Utilities, 145
- Performing a Catalog Backup, 145
- Log Collection Utilities, 146
- Monitoring Utility Run Status, 147

Overview

The utilities help you to back up the catalog and collect the required log files to analyze and troubleshoot issues that you might encounter when you use Enterprise Data Catalog.

Catalog Backup

You can backup the catalog data if you want to troubleshoot issues based on the data in the catalog without stopping the Catalog Service.

Service Logs

The logs for the Enterprise Data Catalog services (Catalog Service, Informatica Cluster Service, EBF history, and PhoneHome), MongoDB, Apache Solr, Nomad service and scan jobs, Apache ZooKeeper (ZK), and PostgreSQL help you analyze and troubleshoot issues related to metadata ingestion in the catalog, issues related to search, or issues related to service failures and stability.

Resource Logs

Resource logs help you troubleshoot issues related to failure of specific resources.

Accessing Utilities

You can use the backup and log collection utilities after you install and configure Enterprise Data Catalog. Make sure that you use a user account configured with Informatica Administrator privileges to access the utilities.

To access the utilities, perform the following steps:

1. Click on the Help menu, and click **Log Interface**.
2. In the login page, enter the user name and password. The **Monitoring Job Details** page appears by default.
3. Click **Back Up Catalog** to access catalog backup, or click **Collect Service Logs** or **Collect Resource Logs** to access the service and resource log collection utilities.

Performing a Catalog Backup

You can back up the catalog without stopping the Catalog Service. Also, you need not stop using Enterprise Data Catalog when you take a backup.

You can back up the catalog to a directory of your choice. You can perform the backup from the **Catalog Backup Settings** page.
When you perform a backup, you can view the catalog data and the catalog configuration details using the Enterprise Data Catalog applications. However, you cannot perform any configuration changes, updates, or add new resource or assets in Enterprise Data Catalog. The restricted tasks during backup include running resources, creating new resources, adding enrichments to resources, creating new connection assignments, and performing bulk imports.

Perform the following steps to back up the catalog:

1. Click **Back Up Catalog** from the left panel. The **Catalog Backup Settings** page appears.
2. Configure the following optional properties as required:
   a. In the **Backup Path** text box, enter the path to the directory where you want the catalog backup to be saved. By default, the backup is saved in `<INFA_HOME>/logs/<NODE_NAME>/services/CatalogService/<SERVICE_NAME>/JobsStore/`.

      Note: The content in the default backup directory is retained only for 24 hours. Also, the content in the default directory is lost when you recycle the Catalog Service. Make sure that you specify a separate directory to store the backup or download the backup from the default directory.

   b. Select the store that you want to backup.
3. Click **Download** to start the backup job. The **Monitoring Job Details** page appears from where you can download the backup.

Note: When you perform a catalog backup, the status of running resource scans changes to System Cancelled. The scan resumes after the backup job is completed.

The backup file is encrypted using the Advanced Encryption Standard (AES) algorithm.

Log Collection Utilities

Enterprise Data Catalog provides utilities that you can use to collect logs from the Catalog Service, Informatica Cluster Service, and specific resources.

You can collect logs to analyze and troubleshoot issues related to services and resources if they do not work as expected. The utilities do not require you to stop the services or resources when the logs are collected. After you run the utilities to collect the logs, make sure that you download the logs within 24 hours. The logs are removed from the directory after 24 hours. Also, if you recycle the Catalog Service after you run the REST APIs, the logs are removed from the directory.

Collect Enterprise Data Catalog Service Logs

You can collect Enterprise Data Catalog service logs to identify and troubleshoot issues related to Enterprise Data Catalog services. You can use the **Collect Service Logs** page to collect the service logs.

To collect service logs, perform the following steps:

1. Click **Collect Service Logs** on the left panel. The **Collect Service Logs** page appears.
2. From **Select Logs**, choose any of the following options for which you want to collect the logs:
   MONGO

   Collect MongoDB logs on data ingestion.

   SOLR

   Collects Apache Solr logs to troubleshoot issues with data asset search.

   NOMAD

   Collects logs related to the Nomad orchestration service and scan jobs.

   ZK

   Collects logs related to the Apache ZooKeeper service.

   POSTGRES

   Collects logs related to PostgreSQL database.

   SERVICE

   Collects logs related to the Catalog Service, Informatica Cluster Service, EBF history, and PhoneHome.

   ALL

   Collect logs for all the above listed items.
3. From the **Start Date** box, select the date from which you want to collect the service logs.
4. From the **End Date** box, select the date till which you want to collect the service logs.
5. Click **Download**. The **Monitoring Job Details** page appears from where you can download the log file.

Collect Resource Logs

You can collect specific resource Nomad logs and Data Integration logs for profiling to identify and troubleshoot issues related to resources. You can use the **Collect Resource Logs** page to collect the logs.

To collect resource logs, perform the following steps:

1. Click **Collect Resource Logs** on the left panel. The **Collect Resource Logs** page appears.

2. Select **SCANNER**, **PROFILING**, or both the options to collect the logs.
3. From the **Start Date** box, select the date from which you want to collect the logs for the resource.
4. In the **Resource** box, click **Add item** to specify the names of the resources for which you want to collect the log files.
5. Click **Download**. The **Monitoring Job Details** page appears from where you can download the log file.

After you download and extract the log files, the resource log files appear in the <resource name> directory in the Informatica logs directory.

Note: If there are no resources corresponding to the resource name that you specified, the resource log collection job fails. Click **Log Location** to download the log file.

Monitoring Utility Run Status

You can monitor the status of the catalog backup and log collection utility runs that you initiated, on the **Monitoring Job Details** page. You can also download the logs and the backup from this page.

Click **Monitoring** on the left panel to launch the **Monitoring Job Details** page.
The page lists the status of the utility runs using the following labels:

- Submitted
- Running
- Completed
- Failed

You can click **Refresh** to refresh the **Monitoring Job Details** page with the updated status.

To download the required log or backup, click the **Download** link present in the **Result** column for the job.

If a job fails, it displays the **Failed** status. You can click the **Log Location** link in the **Log URL** column for the job to view the log generated for the job failure.

APPENDIX A

Registering Unknown Data Types for Profile Run

This appendix includes the following topics:

- Overview, 148
- Registering an Unknown Data Type as a Supported Data Type, 149

Overview

When you enable data discovery and run a resource, the profiling scanner excludes the columns with unknown data types, such as user-defined data types, and runs the profile on the remaining source columns that have supported data types. The excluded columns do not appear in the profiling results. To mitigate this issue for relational sources, you can register the unknown data type as varchar data type. After you register the data type, the profiling scanner uses the registered data type when you run the resource.

You can map the data types for relational data sources such as Oracle, Microsoft SQL Server, IBM DB2, and Teradata. You cannot register BLOB, CLOB, image, and binary data types in the XML file. You cannot register custom data types in the Oracle data source.

Example

In the Customer table of a Microsoft SQL Server data source, you concatenate the FirstName and LastName columns to create a new column named CustomerFullName. You configure a user-defined data type named user_name for the column CustomerFullName. You want to run the column profile on the Customer table. When you run the resource, the profiling scanner excludes the CustomerFullName column as the column has a user-defined data type. To resolve this issue, you can register the user_name data type as varchar data type.

To register the unknown data type, perform the following steps:

1. Navigate to the `<Enterprise_Data_Catalog Installation Directory>/services/CatalogService/ScannerBinaries` location in the Enterprise Data Catalog installation directory.
2. Unzip the catalogMapping.zip file. The zip file contains a sample **catalog-mapping.xml** file.
3. In the catalog-mapping.xml file, enter the source attribute as `user_name` and target attribute as `varchar`. The following code snippet shows the changes that you need to make to the file:

```
<ColumnDataTypes>
      <!--<item source="CustomerFullName" target="varchar"/>-->
</ColumnDataTypes>
```

4. Zip the file as **catalogMapping.zip**.
5. Upload the file to the `<Enterprise_Data_Catalog Installation Directory>/services/CatalogService/ScannerBinaries` location.
6. Navigate to the `<Enterprise_Data_Catalog Installation Directory>/.../CatalogService/ScannerBinaries/CustomDeployer/` location.
7. In the scannerDeployer.xml file, enter the following code snippet:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="catalogMapping.zip">
<PropertyName>CATALOG_MAPPING_LOC</PropertyName>
<PropertyValue>scanner_agents/profileMapping</PropertyValue>
</ExecutionContextProperty>
```

8. Save the file.
9. In Informatica Administrator, recycle the Catalog Service.
10. In the Catalog Administrator, run the resource.

The profiling scanner uses the varchar data type for the column CustomerFullName during profile run.

Registering an Unknown Data Type as a Supported Data Type

In Enterprise Data Catalog, you can register unknown data types as varchar data type. The profiling scanner uses the registered data types for columns that have unknown data types.

1. Unzip the catalogMapping.zip file located in the `<Enterprise_Data_Catalog Installation Directory>/services/CatalogService/ScannerBinaries` path in the Enterprise Data Catalog installation directory.

   The ZIP file contains a sample catalog-mapping.xml file.
2. Open the catalog-mapping.xml file.

   The XML file contains the following XML elements:

XML Element	Description
`<CatalogMapping>` `</CatalogMapping>`	Root XML element.

`<Connection>`
`<Name>`
`</Name>`
`</Connection>`
	Enter the data type mapping information for a relational source. You can enter a generic connection type, such as ODBC or relational database-specific connection type, such as Oracle.

`<source>`
`<Name>`
`</Name>`
`</source>`
	If you enter the generic connection type, you must enter `global` for all the relational databases that you can use for mapping. For database-specific connection type, enter the name of the relational database. For example, for Oracle connection type, enter Oracle as the source database value.

XML Element	Description
`<ExcludeDataTypes> </ExcludeDataTypes>`	Enter the data type that you want to exclude from mapping.
`<ColumnDataTypes> </ColumnDataTypes>`	Enter the data type mapping.
`<item source=" " target=" " />`	Enter the source data type attribute and target data type attribute within quotes. You can enter one or more item elements.

3. Enter the source data type attribute, and enter the target data type attribute as **varchar**. You can enter multiple source and target attributes.
4. Save the XML file, and zip the file as catalogMapping.zip.
5. Upload the ZIP file to the `<Enterprise_Data_Catalog Installation Directory>/services/CatalogService/ScannerBinaries` location in the Enterprise Data Catalog installation directory.
6. Navigate to the `<Enterprise_Data_Catalog Installation Directory>/.../CatalogService/ScannerBinaries/CustomDeployer/` location.
7. Open the scannerDeployer.xml file.
8. Enter the following content in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="catalogMapping.zip">
    <PropertyName>CATALOG_MAPPING_LOC</PropertyName>
    <PropertyValue>scanner_agents/profileMapping</PropertyValue>
</ExecutionContextProperty>
```

If the content already exists in a comment element, remove the comment element for the

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="catalogMapping.zip"> ... </ExecutionContextProperty>
```

element.

The

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="catalogMapping.zip"> ... </ExecutionContextProperty>
```

element enables the profiling scanner to read the ZIP file.

9. Save and close the file.
10. In Informatica Administrator, recycle the Catalog Service.

Note: After you change or update the XML file, recycle the Catalog Service and run the resource in Catalog Administrator.

APPENDIX B

Managing Import from ServiceNow

This appendix includes the following topics:

- Managing Import from ServiceNow Overview, 151
- Importing Connections from ServiceNow , 151

Managing Import from ServiceNow Overview

Catalog Administrator connects to ServiceNow to import connections and extract the configuration metadata into the catalog. The configuration metadata that you import include the instance name, host name, port number, and database name. You can import the configuration metadata for Oracle, IBM DB2, and Microsoft SQL Server resource types.

After the import, Catalog Administrator creates a resource for each imported connection. The resource name is the same as the connection name that you imported.

Importing Connections from ServiceNow

Import connections from ServiceNow to extract configuration metadata into the Catalog.

1. Click **Manage** > **ServiceNow Import**.

   The **Import Connections from ServiceNow** window appears.
2. Enter the following properties to connect to ServiceNow:
   - **Service URL**. URL to connect to the ServiceNow instance.
   - **User Name**. ServiceNow instance user name.
   - **Password**. Password to the ServiceNow instance.
3. Click **Next**.

   The **Import Connections from ServiceNow** window displays available connections from ServiceNow.

4. Select the connections that you want to import from the available connections list, and then click **Import**. Optionally, click the connection name to change the name of the connection.

   Catalog Administrator displays a notification about the status of the import job.

   Note: The Import from ServiceNow feature is available for technical preview. Technical preview functionality is supported but is unwarranted and is not production-ready. Informatica recommends that you use these features in non-production environments only.

APPENDIX C

Custom Lineage Resource CSV File

This appendix includes the following topic:

- Custom Lineage Resource CSV File , 153

Custom Lineage Resource CSV File

Create a custom lineage resource to view the data lineage from data sources, such as the relational databases and flat file databases. The custom lineage resource uses a csv file to extract the lineage information. The elements in the csv file include Association, From Connection, To Connection, From Object, and To Object.

The following table describes the elements in the csv file:

Element	Description
Association	Indicates the type of association. Specify correct values in the association column based on the type of source and target object that you want to link. The following are a few types of dataflow associations: - **core.DataSourceDataFlow**. To link source and target object at the data source level. The dataflow is at the schema level for the relational sources. - **core.DataSetDataFlow**. To link source and target object at data set level. The dataflow is at the table level for the relational sources. - **core.DirectionalDataFlow**. To link source and target object at data element level. The dataflow is at the column level for the relational sources. - **core.DirectionalControlFlow**. To link source and target object at data element level. The control flow is at the column level for the relational sources. - **core.DataSetControlFlow**. To link source and target object at data set level. The control flow is at the table level for the relational sources - **core.DataSourceControlFlow** To link source and target object at the data source level. The control is at the schema level for the relational sources. **Note:** If you do not enter values in the association column, the custom lineage resource extracts lineage information only between relational and file sources.
From Connection	Connection parameter for the source object.
To Connection	Connection parameter for the target object.

Element	Description
From Object	Partial ID of the source object after you perform the connection assignment.
To Object	Partial ID of the target object after you perform the connection assignment.

Custom Lineage Resource CSV File Examples

You can enter values in the csv file for the custom lineage resource using connection parameters and absolute object UID.

Connection Parameters

You can provide connection parameter values in the csv file to extract the lineage information. After you run the custom lineage resource, you must perform connection assignment to display the lineage in catalog.

The following sample shows a .csv file to import lineage between two relational objects:

```
 Association,From Connection,To Connection,From Object,To Object
core.DirectionalDataFlow,<FromConnection>,<ToConnection>,<FromTableName>/
<FromColumnName>,<ToTableName>/<ToColumnName>
core.DataSetDataFlow,<FromConnection>,<ToConnection>,<FromTableName>,<ToTableName>
```

The following sample shows a csv file to import lineage between two structured file objects:

```
 Association,From Connection,To Connection,From Object,To Object
core.DirectionalDataFlow,<FromConnection>,<ToConnection>,<SourceObjectFilePath>/
<FromColumnName>,<TargetObjectFilePath>/<ToColumnName>
core.DataSetDataFlow,<FromConnection>,<ToConnection>,<SourceObjectFilePath>,<TargetObject
FilePath>
```

The following sample shows a csv file to import lineage between relational and structured file objects:

```
 Association,From Connection,To Connection,From Object,To Object
core.DirectionalDataFlow,<FromConnection>,<ToConnection>,<SourceObjectFilePath>/
<FromColumnName>,<ToTableName>/<ToColumnName>
core.DataSetDataFlow,<FromConnection>,<ToConnection>,<SourceObjectFilePath>,<ToTableName>
```

Absolute Object UID

Provide the absolute object UID in the csv file to extract the lineage information. To identify the absolute object UID of an asset, navigate to the asset in the catalog, and then copy the object UID from the URL. For example, the object UID for a file system asset is `ResourceName://FileServer/<Absolute Path>/File`.

The following sample shows a csv file to import lineage from relational and structured file objects using the object UID:

```
Association,From Connection,To Connection,From Object,To Object
core.DirectionalDataFlow,,,Flatfile://FileServer/test2/cust2.csv/column1,Oracle_nc://
ORCL/NC/TEST/column1
core.DataSetDataFlow,,,Flatfile://FileServer/test2/cust3.dat,Oracle_nc://ORCL/NC/TEST2
```

Note: If you want to extract the lineage information using the absolute object UID, you should not specify values in the FROM CONNECTION and TO CONNECTION columns in the csv file.

APPENDIX D

Enterprise Data Catalog Agent

This appendix includes the following topics:

- Installing Enterprise Data Catalog Agent, 155
- Installing Enterprise Data Catalog Agent as a Windows Service, 156
- Configuring Enterprise Data Catalog Agent for SSL, 157

Installing Enterprise Data Catalog Agent

You must install the Enterprise Data Catalog Agent to extract metadata from data sources that run on Microsoft Windows.

Make sure that you install the Enterprise Data Catalog Agent for the following data sources:

- SQL Server Integration Service
- Cognos
- MSTR
- SAP BO
- Erwin
- SAP Power Designer
- QlikView
- Microsoft SQL Server

  Note: You must install the Enterprise Data Catalog Agent when you use Windows authentication for the Microsoft SQL Server resource.
- Qlik Sense

  Note: To extract metadata from the Qlik Sense resource, you must install the Enterprise Data Catalog Agent on the Qlik Sense server.

The machine on which you install the Enterprise Data Catalog Agent must have a minimum of 512 MB of RAM, 1 CPU, and 600 MB of free disk space.

For high volume use cases such as scanning a large number of SAP BO reports, you must have the following minimum system requirements:

- SAP BO MITI Agent configuration - 32 core CPU, 64 GB RAM, and 500 GB of free disk space.
- SAP BO server configuration - 8 core CPU, 32 GB RAM, and 200 GB of free disk space.

To install the Enterprise Data Catalog Agent, perform the following steps:

1. Download the following file from Akamai download manager: `EDC_Agent_<version>_Windows.zip`
2. Extract the binary files to an installation directory on the machine.
3. Change the connector port number in the `<Installation directory>\tomcat\conf\tomcat.properties` file. Default is 19980.

   If port 19980 is used by another application, you can change the port number in the `tomcat.properties` file. For example, you can set the connector port as 19981 and shutdown port as 19982.
4. To start the Enterprise Data Catalog Agent, run the `startup.bat` file from the following directory: `<Installation directory>\tomcat\bin`

   The Enterprise Data Catalog Agent is started. To restart the Enterprise Data Catalog Agent, run the `RestartServerApplication.bat` file from the installation directory.

When you configure a resource in Catalog Administrator to extract metadata from a data source, use the following format in the Agent URL field:
`http://<hostname>:<connector_port>/MIMBWebServices`

To extract metadata from the Cognos resource, copy the JAR files from the `sdk\java\lib` directory of the Cognos SDK installation to `<Installation directory>\java\CognosRepository`. After you copy the JAR files, restart the Enterprise Data Catalog Agent.

Installing Enterprise Data Catalog Agent as a Windows Service

You can install the Enterprise Data Catalog Agent as a Windows service. The display name of EDC Agent service is Meta Integration Application Server.

To install the Enterprise Data Catalog Agent, perform the following steps:

1. Download the `EDC_Agent_<version>_Windows.zip` file.
2. Extract the binary files to a directory.
3. Run the `RestartServerService.bat` file.

   A Windows service with the display name Apache Tomcat 9.0 MITI and service name MITI is created.
4. Verify that the service is running.
5. Access the following URL to verify the installation: `http://<hostname>:19980/MIMBWebServices/mimb`.

   If the service is installed correctly, a page with the following information appears:

   Currently running MIMB threads:

ID	Start time	Name	Log file	Synchronization

   Active synchronization set: {}

Configuring Enterprise Data Catalog Agent for SSL

By default, the Enterprise Data Catalog agent does not communicate over a secure connection. You must use a CA-signed or self-signed certificate to enable SSL for the Enterprise Data Catalog agent. Also, you must modify the `Setup.bat` script for the script to work as expected. Perform the following steps to enable the agent for SSL:

1. Obtain a CA-signed certificate or create a new self-signed certificate using the following command: .

   ```
   keytool -genkey -keyalg RSA -alias edcagent -keystore agent_keystore.jks -storepass
   <keystore password> -validity 365 -keysize 2048
   ```

2. Modify the `Setup.bat` file and replace the reference to the `MMConfiguration.jar` file in the last line with `*`.

3. Run the `Setup.bat` script using the following command to install the Enterprise Data Catalog agent as a Windows service and set up the service to use the specified SSL certificate.

   ```
   .\Setup.bat -s true -sc agent_keystore.jks -sp <keystore password> --tomcat-service true
   ```

Parameter	Description
-s	Set true or false to enable or disable SSL.
-sc	Specify the X.509 file containing the SSL certificate.
-sp	Specify the password for the SSL key file.
--tomcat-service	Set true or false to install or remove tomcat as a service

Note: To view the list of parameters that can be used with the script, run `.\Setup.bat --help`.

The agent is started and can only accept https connections. Scanners that use the agent must be configured to use the https URL for the agent.

Important: If you use Enterprise Data Catalog agent with SAP BO, you must launch the agent as a regular desktop application, not as a service. To launch the agent as a desktop application, run the `RestartServerApplication.bat` script.

APPENDIX E

Lineage between PowerCenter Resources and Flat Files

This appendix includes the following topics:

- Overview, 158
- Configuring and Running a Flat File Resource on a Windows Machine, 159
- Viewing Lineage between PowerCenter Resources and Flat Files, 160

Overview

In the Catalog Administrator, you can run the PowerCenter resources and flat file resources to view the lineage between the resources in Enterprise Data Catalog.

Sometimes, when a PowerCenter mapping resides on a Windows machine, the lineage between the PoweCenter resources and flat files might not appear in Enterprise Data Catalog. This issue might occur because the file path in the resources might not match. When Enterprise Data Catalog generates the lineage, the file system scanner uses the network shared path and the PowerCenter scanner uses the absolute path. To resolve this issue, enter the mapped path on the Windows machine in the **Physical Location** field of the file system resource. You can enter the network or shared path in the **Path** field.

The **Physical Location** field applies to flat files. When lineage is generated between the PowerCenter files and flat files, Enterprise Data Catalog uses the **Physical Location** field instead of the **Path** field in the file system scanner. In a PowerCenter resource, if you choose the **Auto assign Connections** option, Enterprise Data Catalog assigns the connection based on the absolute path.

Example

Assume that the flat files are located in the `c:\user1\1021\SMB\test_files` folder. You map the `c:\user1\1021\SMB` folder to `\\SMB\` shared directory.

This example includes the following folder paths:

- `c:\user1\1021\SMB\test_files` is the absolute path.
- `c:\user1\1021\SMB` is the mapped path or absolute path of the shared directory.
- `\\SMB\` is the network directory.
- `\\SMB\test_files\` is the network or shared path.

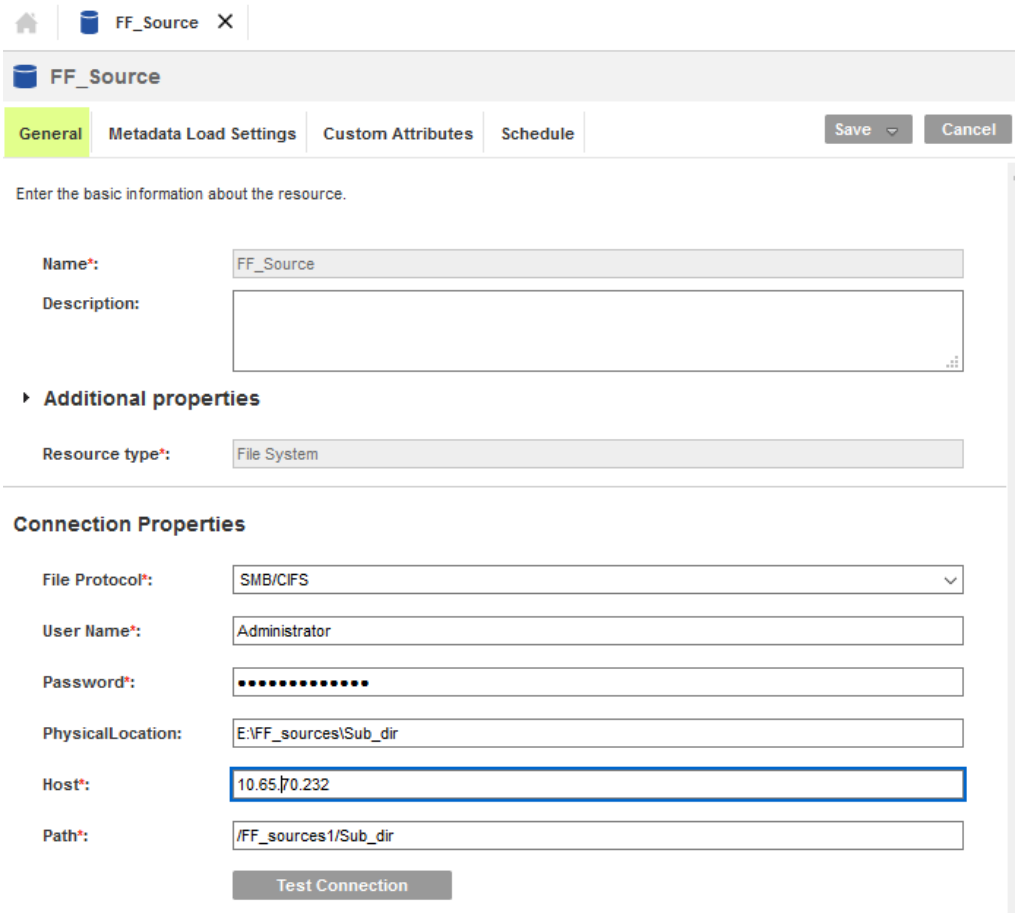
Configuring and Running a Flat File Resource on a Windows Machine

You need to enter the absolute path of a flat file in the file system resource to view the lineage between PowerCenter mappings and flat file sources that reside in the Windows machine.

1. In Catalog Administrator, click **New** > **Resource**.
2. In the **Resource** wizard, enter the following parameters in the **General** tab:

Field	Description
Name	Enter the name for the resource.
Description	Optional. Enter a brief description for the resource.
Resource Type	Choose **File System** as the resource type.
File Protocol	Choose **SMB/CIFS** as the file protocol.
User Name	Enter the user name to access the Windows machine.
Password	Enter the password to access the Windows machine.
Physical Location	Applies to flat files. Enter the absolute path of the shared directory or mapped path on the Windows machine.
Host	Specify the host name or IP address of the machine.
Path	Specify the network or shared path from where you want to import the metadata into the catalog.
Test Connection	Click **Test Connection** to verify the connection to the specified location.

The following image shows a sample General tab of the file system resource:



3. Click **Next**.
4. In the **Metadata Load Settings** tab, set the **Source Metadata**, **Composite Domain Discovery**, and **Data Discovery** options as necessary.
5. Click **Next** to add custom attributes.
6. Click **Save** to save the resource, or click **Save and Run** to save and run the resource.

Viewing Lineage between PowerCenter Resources and Flat Files

After you run the PowerCenter resources and flat file resources, you can view the lineage between the resources in Enterprise Data Catalog.

1. In the Enterprise Data Catalog home page, enter the name of the asset in the **Search** field.
2. Click the **Search** icon.

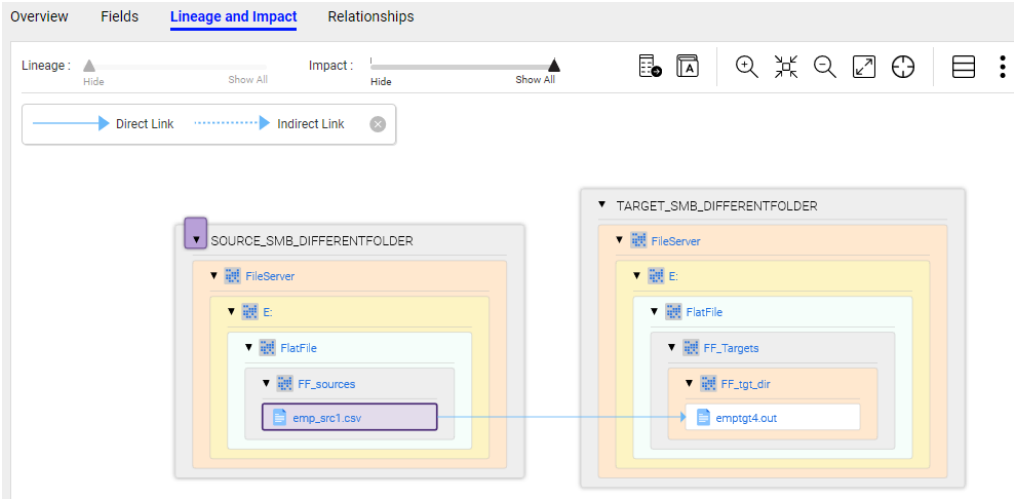
   The results appear in the **Search Results** window.
3. Select the appropriate **Resource Name**, **Asset Type**, and **Resource Type** filters to narrow down the results.

4. Click the asset name.

   The asset details appear in the **Overview** tab.

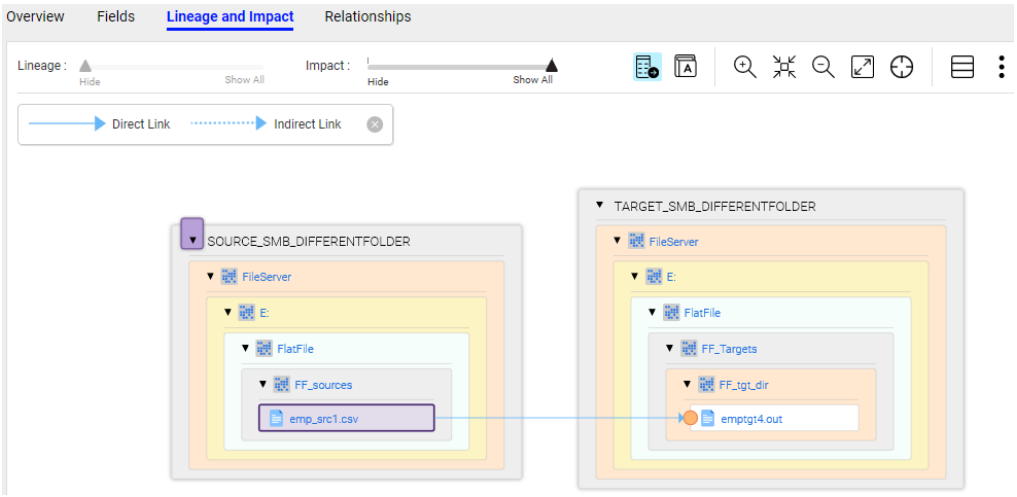
5. Click the **Lineage and Impact** tab.

   The following image shows a sample lineage of an asset:



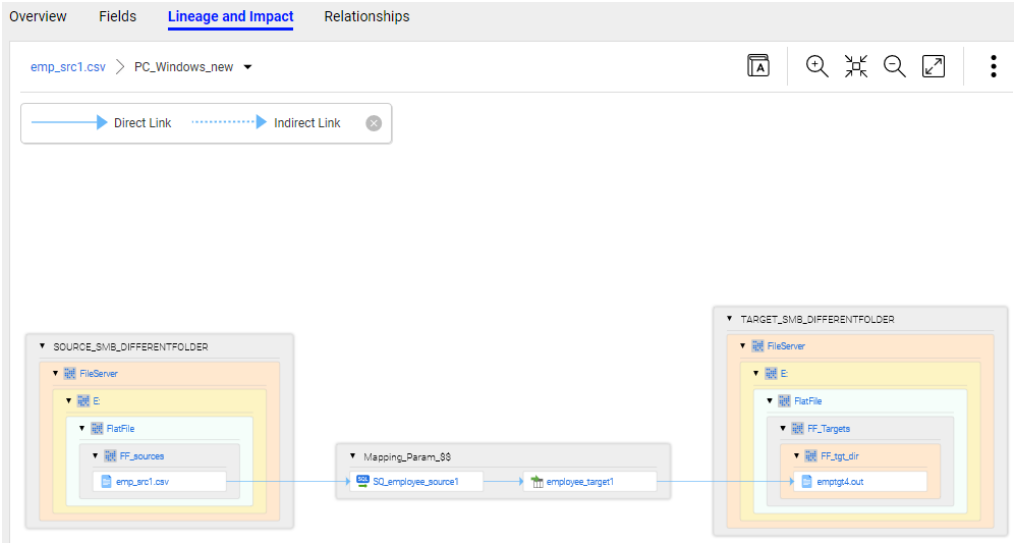
6. Click the  icon.

   The following image shows a sample of the summary view of an asset lineage:



7. Click the  icon.

The following image shows a sample of the detailed view of an asset lineage:



APPENDIX F

Metadata Extraction from Offline and Inaccessible Resources

This appendix includes the following topics:

- Overview, 163
- Process, 165
- Scanner Utility Components, 166
- Prerequisites and Best Practices, 166
- Running the Scanner Script , 167
- Script Command, 168
- Structure of Metadata Archive File, 177

Overview

Informatica provides a standalone scanner utility that you can use to extract metadata from offline and inaccessible resources. The utility contains a script that you need to run along with the associated multiple commands in a sequence.

Catalog Administrator is an administration tool that you can use to manage and monitor the resources, schedules, attributes, and connections. You can extract metadata from all the resources in an enterprise through Catalog Administrator. To extract the metadata, you can create connections, create and run the resources, and create schedules to run the resources at periodic intervals. When the data sources are offline or inaccessible, you cannot use the Catalog Administrator tool to extract metadata.

To extract metadata from external sources that are inaccessible at runtime or offline, you can use the standalone scanner utility. You can run the script and its associated commands to extract metadata from inaccessible or offline data sources and publish it to the catalog. After you publish the metadata to the catalog, you can view the data assets in Enterprise Data Catalog. In Linux, run the **scanner.sh** script and its associated commands.

To run the script, you need appropriate permissions to the folders and files which the script uses to store metadata and Enterprise Data Catalog uses to publish the metadata.

You can extract metadata from the following external sources when they are inaccessible at runtime or offline:

Cloud

- Amazon Redshift
- Amazon S3
- Azure Microsoft SQL Data Warehouse
- Azure Microsoft SQL Server
- Glue
- Google BigQuery
- Google Cloud Storage
- Microsoft Azure Blob Storage
- Microsoft Azure Data Lake Store
- Salesforce
- Snowflake
- Workday

Data Engineering

- Apache Atlas
- Cloudera Navigator
- HDFS
- Hive

Business Glossary

- Axon
- Business Glossary

Business Intelligence

- IBM Cognos
- Microstrategy
- Oracle Business Intelligence
- PowerBI
- QlikView
- SAP BW
- SAP BW/4HANA
- Tableau

Data Integration

- Custom Lineage
- Database Scripts
- Informatica Cloud Service
- Informatica Platform
- PowerCenter
- Microsoft SQL Server Integration Services

Data Modeling

- Erwin
- SAP PowerDesigner

Database Management

- IBM DB2
- IBM DB2 for z/OS
- IBM Netezza
- JDBC
- Microsoft SQL Server
- Oracle
- SAP HANA
- Sybase
- Teradata

File Management

- File System
- OneDrive
- SharePoint

No SQL

Apache Cassandra

Application

SAP S/4HANA

Note: To extract metadata from the SAP S/4HANA data source in the offline mode, specify SAPERP6 as the resource type in the scanner.sh generateResourceConfig command.

Process

You can complete the process to extract and store metadata from offline sources or inaccessible sources and publish it to Enterprise Data Catalog in two stages.

In the first stage, an Enterprise Data Catalog user performs the following steps:

1. Generates a template for the resource configuration file based on the resource type.
2. Encrypts the password in the resource configuration file.
3. Edits the file to enter the connection details of the source and the encrypted password.
4. Validates the file to establish a connection to the source.
5. After the connection is established, the utility extracts the metadata and saves it to a metadata archive file.

In the second stage, an Enterprise Data Catalog administrator performs the following steps:

1. Validates the integrity of the metadata archive file.
2. Publishes the metadata to the catalog.

After the metadata is published to the catalog, you can view the data assets in Enterprise Data Catalog.

Scanner Utility Components

To run the script and its associated commands, you need to unzip the utility and then run the script. The installer ZIP file is located in the `$INFA_HOME/services/CatalogService/ScannerCliUtility/eic-scannercli-utility.zip` location. You can unzip the file on the machine where you installed the Catalog Service and run the scanner.sh and its commands in the `$INFA_HOME/services/CatalogService/ScannerCliUtility/` location.

The installer ZIP file contains the following files and directories:

Files or Directories	Description
/ScannerBinaries	Contains third party and scanner deployment dependencies that are available in `$INFA_HOME/services/CatalogService/ScannerBinaries`.
/ScannerJars	Contains all the jar files in the `bin` folder.
/scanner_depencency	Contains most of the jar files in the access.war file.
/lib	Contains the following jar and log files: - CLI jar - log4j.properties - log4j-publisharchive.xml
/bin/scanner.sh	Contains the main script file the you can run with its associated commands.
/app.properties	Contains the default values for the command parameters.

Prerequisites and Best Practices

You can use the following prerequisites and best practices when you run the script and its associated commands:

Prerequisites

Before you run the script, verify that the following prerequisites are met:

- You must have appropriate permissions to run the **generateResourceConfig**, **encryptPassword**, **validate**, and **scan** script commands.
- Ensure that you have read, write, and execute permissions for the output file. The **generateResourceConfig** command generates the resource configuration file template or output file.
- Ensure that you have read permission for the input file. When you run the **scan** script command, the command uses the resource configuration file as the input file.
- Enterprise Data Catalog administrator must have appropriate permissions to run the **describeArchive** and **publishArchive** script commands.
- Ensure that the machine on which you run the scanner.sh script has a minimum of 1 GB Java heap size when you run the **generateResourceConfig**, **validate**, **cancel**, and **pause** options.

Scanner-specific Prerequisites

Make sure that the following prerequisites are met for the following external resources:

SAP Business Warehouse (SAP BW), SAP BW/4HANA, and SAP S/4HANA resources

- Download the `sapjco3.jar` file and copy the file to `$CLI_HOME/scanner_dependency` location. For example, you can download and copy the file to `<INFA_HOME>/services/CatalogService/ScannerCliUtility/scanner_dependency` location.

  Note: Supported SAP Java Connector version is 3.0.19.
- Include the `libsapjco3.so` file in the `SAPJCO.zip` file and copy the `SAPJCO.zip` file to the `$CLI_HOME/ScannerBinaries` location.

Scanners with external dependencies

Teradata, Sybase, Redshift, and JDBC scanners have external dependencies. Before you run the script to extract metadata from these resources, you need to extract the external dependency libraries for the scanner and place it in their respective folders in the `$CLI_HOME/ScannerJars/externalDependencies/<scanner>` location.

For example, you can unzip the teradataJars.zip file to the `$CLI_HOME/ScannerJars/externalDependencies/TeradataScanner` location before you run the script to extract metadata from Teradata resources.

Best Practices

Consider the following best practices to achieve optimal performance when you run the script:

- Use only one output directory for each scanner execution at a time. Errors appear if you use the same output directory for multiple scanner runs at the same time.
- To cancel or pause a scanner run, open another terminal or console to run the commands.

Running the Scanner Script

To run the script, you can unzip the utility and run the script and its associated commands in a sequence to extract metadata from offline sources or inaccessible sources. After you extract the metadata, you can publish it to the Enterprise Data Catalog to view the data assets.

1. Navigate to the `<INFA_HOME>/services/CatalogService/ScannerCliUtility` directory.
2. Unzip the utility file *eic-scannercli-utility.zip* to a local folder on your machine.
3. Run the **scanner.sh** script command to view the supported commands and resource types.
4. To view the CLI version of the machine, run the **scanner.sh version** command
5. To generate a template of the resource configuration file for a resource, run the **scanner. sh generateResourceConfig (-resourceType < resource_type >) (-output<path>)** command.
6. To encrypt the password, run the **scanner.sh encryptPassword (<password>)** command.
7. To validate the file and establish a connection to the source, run the **scanner. sh validate (- rcf < resource_config_file >)** command.
8. To scan, extract, and store the metadata in a metadata archive file, run the **scanner. sh scan (- rcf <resource_config_file>) (-output <path>)** command.
9. To validate the metadata archive file located in the output folder, run the **scanner. sh describeArchive (- arf < path_to_archive_file >)** command. The output folder is located in the path that you enter for the **-output** parameter in the scan command.

10. To publish the metadata to the catalog, run the **scanner. sh publishArchive (-un < isp_user >) (-pd < isp_user_password >) (- sd < isp_security_domain >) (-dh < isp_gateway_host >) (- dp < isp_domain_port >) (- dn <fully_qualified_domain_name>) (- sn <catalog_service_name>) (- rn <resource_name>) (- arf <path_to_archive_file>) (-csh <catalog_service_hostname>) (-csp <** command.
    - Append **[- dse] [- ts <path_to_truststore>] [-tsp <trust_store_password>]** options to the **publishArchive...** command to enable an SSL domain for Enterprise Data Catalog.
    - Append the **[-f] [-w] [-v]** options to create or update a resource, wait for the ingestion job to complete before the next step in the process initiates, and to display the progress information in verbose, respectively.
11. Open Enterprise Data Catalog to search and view the data assets.

Script Command

Scanner.sh is the script command that you can run to view a list of resource types along with the list of available options. You can run the script along with the options and arguments in a sequence to extract metadata from inaccessible sources and offline sources and publish it to the catalog.

The scanner.sh command uses the following syntax:

```
scanner.sh
version
generateResourceConfig (-resourceType <resource_type>) (-output <path>) [-v]
encryptPassword (<password>)
validate (-rcf <resource_config_file>) [-v]
scan (-rcf <resource_config_file>) (-output <path>) [-v]
cancel (-rn <resource_name>) (-output <path>) [-v]
pause (-rn <resource_name>) (-output <path>) [-v]
resume (-rcf <resource_config_file>)  (-output <path>) [-v]
describeArchive (-arf <path_to_archive_file>) [-v]
publishArchive (-un <isp_user>) (-pd <isp_user_password>) (-sd <isp_security_domain>) (-
dh <isp_gateway_host>) (-dp <isp_domain_port>) (-dn <fully_qualified_domain_name>) (-sn
<catalog_service_name>) [-dse] [-ts <path_to_truststore>] [-tsp <trust_store_password>]
(-rn <resource_name>) (-arf <path_to_archive_file>) [-f] [-w] [-v] (-csh
<catalog_service_hostname>) (-csp <catalog_service_port>)
```

The following table describes scanner.sh options and arguments:

Option	Argument	Description
version	-	Optional. Prints the CLI version number.
generateResourceConfig	(-resourceType <resource_type>) (-output <path>) [-v]	Required. Generates a template of the resource configuration file.
encryptPassword	(<password>)	Required. Prints the encrypted password.
validate	(-rcf <resource_config_file>) [-v]	Required. Reads and prints the resource configuration file to the console and establishes a connection to the source.
scan	(-rcf <resource_config_file>) (-output <path>) [-v]	Required. Prints the job status to the console at regular intervals.

Option	Argument	Description
cancel	(-rn <resource_name>) (-output <path>) [-v]	Optional. Cancels the resource scan.
pause	(-rn <resource_name>) (-output <path>) [-v]	Optional. Pauses a resource scan.
resume	(-rcf <resource_config_file>) (-output <path>) [-v]	Optional. Resumes a paused resource scan.
describeArchive	(-arf <path_to_archive_file>) [-v]	Required. Validates the checksum of the metadata archive file.
publishArchive	(-un <isp_user>) (-pd <isp_user_password>) (-sd <isp_security_domain>) (-dh <isp_gateway_host>) (-dp <isp_domain_port>) (-dn <fully_qualified_domain_name>) (-sn <catalog_service_name>) [-dse] [-ts <path_to_truststore>] [-tsp <trust_store_password>] (-rn <resource_name>) (-arf <path_to_archive_file>) [-f] [-w] [-v] (-csh <catalog_service_hostname>) (-csp <catalog_service_port>)	Required. Publishes the metadata to Enterprise Data Catalog.

Note: The exit code for the script is **0** for Success and **-1** for Fatal Error.

generateResourceConfig

Generates a resource configuration file template based on the resource type. The template includes all the scanner options that apply to the selected scanner.

The scanner.sh generateResourceConfig command uses the following syntax:

```
scanner.sh generateResourceConfig
-resourceType <resourceType>
-output <path>
-v
```

The following table describes scanner.sh generateResourceConfig options and arguments:

Option	Argument	Description
-resourceType	resourceType	Required. Enter a resource type in the following naming convention: *<ResourceType>* For example, Oracle, MSSqlServer, or Teradata. **Note:** For the SAP S/4HANA resource, specify SAPERP6 as the resource type.
-output	path	Required. Path to the output directory where the resource configuration file is generated. For example, you want to generate the scanner configuration file in the `$HOME/resources` location. Enter the path as `$HOME/resources`. **Note:** A [Y/N] prompt appears if the target file already exists. Select **Y** if you want to overwrite the file.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

Troubleshooting

What are the errors that might appear when I run the generateResourceConfig command. How do I resolve it?

The following list consists of potential errors with recommended actions to resolve it:

- The template for the scanner does not exist. Contact Enterprise Data Catalog administrator for more information.
- You do not have appropriate file permissions for the output file. Contact Enterprise Data Catalog administrator to assign the necessary permissions.

encryptPassword

Prints the encrypted password to the console. The encrypted password in the resource configuration file is used for other commands.

The scanner.sh encryptPassword command uses the following syntax:

```
scanner.sh
encryptPassword (<password>)
```

In the **scanner.sh encryptPassword (<password>)** command, enter the password and run the command.

validate

When you run the **scanner. sh validate** command, the command performs the following tasks:

1. Decrypts the password, and tests whether the connection to the data source is valid or not. If the test connection fails, the script stops to run and an error appears.
2. When the test connection is successful, the command reads the resource configuration file and prints the file contents to the console.
3. After the command displays the content, it establishes a connection to the data source.

The scanner.sh validate command uses the following syntax:

```
scanner.sh validate
-rcf <resource_config_file>
-v
```

The following table describes scanner.sh validate options and arguments:

Option	Argument	Description
-rcf	resource_config_file	Required. Enter the resource configuration file name.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

Troubleshooting

What are the errors that might appear when I run the validate command or when the test connection fails. How do I resolve it?

The following list consists of potential errors with recommended actions to resolve it:

- Credentials are not valid. Verify that the credentials that you provide are correct.
- Data source is not reachable. Check the connection details. If unresolved, contact the database administrator for more information.
- Issues related to the data source. Contact database administrator for more information.
- Password has not been encrypted. Run the **scanner. sh encrypt Password (<password>)** command to encrypt the password. Enter the encrypted password in the resource configuration file, if required. Run the **scanner. sh validate (- rcf < resource_config_file >) [-v]**.

scan

When you run the **scanner. sh scan** command, the command performs the following tasks:

1. Runs the resource scan.
2. Prints the job status periodically on the console.
3. Creates an archive file titled *Archive_$resourcename_$version.zip* for the extracted metadata.
4. Saves the archive file in the output folder. The output folder is located in the path that you enter for the **-output** parameter in the scan command.

The scanner.sh scan command uses the following syntax:

```
scanner.sh scan
-rcf <resource_config_file>
-output <path>
-v
```

The following table describes scanner.sh scan options and arguments:

Option	Argument	Description
-rcf	resource_config_file	Required. Enter the resource configuration file name.
-output	path	Required. Path to the output directory where the metadata archive file is generated. For example, you want to generate the metadata archive file in the `$HOME/resources` location. Enter the path as `$HOME/resources`.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

Troubleshooting

What are the errors that might appear when I run the scan command. How do I resolve it?

The following list consists of potential errors with recommended actions to resolve it:

- Insufficient disk space to save the metadata archive file. Increase the disk space and run the command.
- Cannot connect to the data source. Check the connection details. If unresolved, contact the database administrator for more information.
- Credentials are not valid. Verify that the credentials that you provided are correct.

Note: If you can cancel the scan, the metadata archive file will not be created. For more information about the errors, see the log files.

cancel

Cancels the resource scan if the scan is in progress and does not create the metadata archive file.

The scanner.sh cancel command uses the following syntax:

```
scanner.sh cancel
-rn <resource_name>
-output <path>
-v
```

The following table describes scanner.sh cancel options and arguments:

Option	Argument	Description
-rn	resource_name	Required. Enter the resource name.
-output	path	Required. Path to the output directory where the metadata archive file is generated. For example, you want to generate the metadata archive file in the `$HOME/resources` location. Enter the path as `$HOME/resources`.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

pause

Pauses the resource run and does not create the metadata archive file. The command retains the input and output directories.

The scanner.sh pause command uses the following syntax:

```
scanner.sh pause
-rn <resource_name>
-output <path>
-v
```

The following table describes scanner.sh pause options and arguments:

Option	Argument	Description
-rn	resource_name	Required. Enter the resource name.
-output	path	Required. Path to the output directory where the metadata archive file is generated. For example, you want to generate the metadata archive file in the `$HOME/resources` location. Enter the path as `$HOME/resources`.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

Troubleshooting

What are the errors that might appear when I run the pause command. How do I resolve it?

The following list consists of potential errors with recommended actions to resolve it:

- Resource run is not in progress. No action required.

resume

When you run the **scanner. sh resume** command, the scanner resumes the paused resource scan. Afer the resource run completes, the command creates and saves the metadata archive file.

The scanner.sh resume command uses the following syntax:

```
scanner.sh resume
-rcf <resource_config_file>
-output <path>
-v
```

The following table describes scanner.sh resume options and arguments:

Option	Argument	Description
-rcf	resource_config_file	Required. Enter the resource configuration file name.
-output	path	Required. Path to the output directory where the metadata archive file is generated. For example, you want to generate the metadata archive file in the `$HOME/resources` location. Enter the path as `$HOME/resources`.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

Troubleshooting

What are the errors that might appear when I run the resume command. How do I resolve it?

The following list consists of potential errors with recommended actions to resolve it:

- Resource run is in progress. No action required.
- Resource run cannot be resumed due to unavoidable circumstances. To mitigate the issue, you can contact the catalog administrator.

describeArchive

When you run the **scanner. sh describeArchive** command, the command performs the following tasks:

1. Validates the checksum of the metadata archive file.
2. Prints the CLI version, resource configuration details, and timestamp on the console.
3. Displays scan summary that includes scan statistics.

The scanner.sh describeArchive command uses the following syntax:

```
scanner.sh describeArchive
-arf <path_to_archive_file>
-v
```

The following table describes scanner.sh describeArchive options and arguments:

Option	Argument	Description
-arf	resource_config_file	Required. Enter the path to the metadata archive file.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.

publishArchive

When you run the **scanner.sh publlishArchive** command, the command performs the following tasks:

1. Validates the user name and password of the catalog administrator, and gets the API key or authorization token for next API calls.
2. Validates the checksum of the input file.
3. Ensures that the Enterprise Data Catalog version is the same or newer than the offline scanner version.
4. Verifies that the resource configuration version and the Enterprise Data Catalog version are compatible.
5. If required, creates a resource in Enterprise Data Catalog.
6. Verifies that the following are identical in the resource and resource configuration file:
   a. Resource configuration
   b. Resource type and name
7. Ingests the Exchange Documents into the catalog.

You can view the progress of the tasks on the console that include:

1. Connection to domain and Enterprise Data Catalog.
2. Creates a resource configuration file or updates the resource configuration file.
3. File upload.
4. Ingestion job scan.
5. Start time and end time.
6. Monitoring URL.

The scanner.sh publishArchive command uses the following syntax:

```
scanner.sh publishArchive
-un <isp_user>
-pd <isp_user_password>
-sd <isp_security_domain>
-dh <isp_gateway_host>
-dp <isp_domain_port>
-dn <fully_qualified_domain_name>
-sn <catalog_service_name>
[-dse]
[-sse]
[-ts <path_to_truststore>]
[-tsp <trust_store_password>]
-rn <resource_name>
-arf <path_to_archive_file>
[-f]
[-w]
[-v]
-csh <catalog_service_hostname>
-csp <catalog_service_port>
```

The following table describes scanner.sh publishArchive options and arguments:

Option	Argument	Description
-un	isp_user	Required. Enter the user name of the catalog administrator.
-pd	isp_user_password	Required. Enter the encrypted password for the administrator.
-sd	isp_security_domain	Required. Enter **Native** or **LDAP** as the security domain to log in to Enterprise Data Catalog.
-dh	isp_gateway_host	Required. Enter the gateway host name for the domain.
-dp	isp_domain_port	Required. Enter the port number of the gateway node.
-dn	fully_qualified_domain_name	Required. Enter the Informatica node name.
-sn	catalog_service_name	Required. Enter the Catalog Service name.
-dse	<true|false>	Optional. Specify **true** to enable SSL domain. Default is False.
-ts	path_to_truststore	Optional. Enter the path and file name of the truststore file that contains authentication certificates trusted by the Data Integration Service. **Note:** This field is required if you choose **-dse** as true.
-tsp	trust_store_password	Optional. Enter the encrypted password for the truststore file. **Note:** This field is required if you choose **-dse** as true.
-rn	resource_name	Required. Enter the resource name.
-arf	path_to_archive_file	Required. Enter the path to the metadata archive file.
-f	-	Optional. Updates or creates a resource in Enterprise Data Catalog.
-w	-	Optional. Waits for the ingestion job to complete.
-v	-	Optional. Displays information in verbose mode. Verbose mode provides detailed information about object versions, including repository name, folder name, version number, and status.
-csh	catalog_service_hostname	Optional. Name of the host machine on which the Catalog Service runs.
-csp	catalog_service_port	Optional. Port number of the Catalog Service.

Troubleshooting

What are the errors that might appear when I run the publishArchive command. How do I resolve it?

The following list consists of potential errors with recommended actions to resolve it:

- Resource configuration version and Exdocs are not compatible. Implement the required changes, run the commands, and then publish the archive file to the catalog.
- Mismatch in resource type or resource name. Implement the required changes, run the commands, and then publish the archive file to the catalog.

Structure of Metadata Archive File

The metadata archive file or Archive.zip file has the following files:

File or Directory	Description
/EXDOC_STORE	Contains all the Exdocs including TOC.
/XMIFiles	Contains the XMI files located in the `\services\CatalogService\XMIFiles\*` directory.
/logs	Contains log files that include: - Stderr - Stdout - Scanner.log - GC.log
/scanner/ checkpoints	The ingestion job creates checkpoints at every top-level folder during the job scan. These checkpoints are saved in the `/scanner/checkpoints` file.
/scanner/ resourceConfig	Files from EICRepo/*
/scanner/JobStatus	Files from JobStatus/*

File or Directory	Description
/config	Contains a copy of resource configuration file with the file extension `.properties`. The file has the user name **dummy** and password **dummy**. The folder also has the Runtime.log file that has the following information: - Environment variables. - Operating system-level settings. - Hardware configuration, such as available physical memory and diskspace. - Runtime environment.

/META-INF/ metainfo.json	The metainfo.json file contains checksums, which includes metadataArchiveCheckSum. The metadataArchiveCheckSum parameter is updated when the script adds entries to the archive file. After the checksums are generated, the script generates the metainfo.json file and appends it as the very last entry to the Archive.zip file.
	The metadata.json file represents the dates as String in Coordinated Universal Time (UTC) format.
	The following sample shows the content of a metadata.json file:
	`{`
	`"createdOn" : 12345678,`
	`"scanMode" : "FULL",`
	`"cli_version" : "10.2.2",`
	`"summary" : { {"Tables" : "10"}. {"Column" : "20" } }, // updateStats()`
	`"resourceConfigCheckSum" : "abcd",`
	`"metadataArchiveCheckSum" : "xyz",`
	`"startTime" : "YYYY-MM-DDThh:mm:ssZ",`
	`"endTime" : "YYYY-MM-DDThh:mm:ssZ",`
	`"resourceName" : "OracleResource"`
	`...`
	`}`

APPENDIX G

Enterprise Data Catalog Add-Ons

This appendix includes the following topics:

- Enterprise Data Catalog Add-Ons Overview, 179
- Deploy the Add-Ons, 180
- Log in to the Enterprise Data Catalog Add-Ons, 180
- View Resource Schedules and Duplicate Resources, 180
- Accept Data Domains, 181
- Moving Data Domains using the Data Domains Manager Utility, 182
- Associate Business Terms from Synonym File, 183
- Migrate Custom Attributes, 184

Enterprise Data Catalog Add-Ons Overview

Enterprise Data Catalog add-ons extend or supplement the features available in the Catalog tool and the Catalog Administrator tool.

Use the following add-ons to perform additional tasks on the resources and assets in the Catalog:

Catalog Operation Reports

View resource schedules for all resources in the Catalog and the list of duplicate resources.

CurateAssist

Curate data domain assets by moving specified data domains to the accepted state and rejecting other data domains in bulk.

Business Term Association from Synonym file

Associate business terms from an Informatica Axon resource with specific columns in a tabular asset.

Data Domains Manager

Move the accepted resources from one data domain to another in bulk.

Deploy the Add-Ons

Download the Enterprise Data Catalog add-ons installation package from the Informatica Marketplace. The ZIP file includes the files that you need to deploy the add-ons.

Perform the following steps to deploy the add-ons:

1. On the node on which you want to deploy the add-ons, extract the ZIP file.
2. From the directory where you extracted the files, run the following command to start the application: `./startup.sh`
   The host name and the port number to access the application appear in the following format: https://<host>:<port>

Log in to the Enterprise Data Catalog Add-Ons

After you deploy the add-ons, perform the following steps to log in to the Enterprise Data Catalog add-ons:

1. Open a browser.
2. In the address field, type the URL to access the add-ons in the following format: https://<host>:<port>
3. Enter the connection properties to log in to the Catalog tool.
   The following table describes the connection properties:

Property	Description
Catalog host	The gateway node host name.
Catalog port	The port number configured for the Catalog Service.
Protocol	Select **https** if you enabled a secure HTTPS connection for the Catalog tool.
Username	The user name to log in to the Catalog tool.
Password	The password to log in to the Catalog tool.

4. Click **Log In**.

Note: The Enterprise Data Catalog add-ons does not support single sign-on using LDAP authentication. To access the Enterprise Data Catalog add-ons using LDAP authentication, you must have the following Catalog Service privileges on the Informatica domain:

- REST API Privilege
- REST API User Privilege

View Resource Schedules and Duplicate Resources

You can generate the Resource Schedules report to view a list of all resources in the catalog and the schedules attached to each resource. The report helps you determine the best schedule to minimize the

number of simultaneous resource scan jobs when you create a new schedule. You can also generate the Duplicate Resources report to identify duplicate non-JDBC relational database type resources. Use the report to view the resource types and the number of duplicate resource groups for each resource type. You can also download the schedule report in Enterprise Data Catalog Add-On.

View Resource Schedules

1. Access the Enterprise Data Catalog add-ons.
2. In the **My Services** page, click **Catalog Operation Reports**.
3. In the **Catalog Information** page, verify that Resource Schedules is selected as the type of report.
4. Click **Connect** to view the report.
   The Resource Schedule Report displays scan schedules in calendar view or report view. Default is calendar view. Use the toggle icon to switch between the views.

Use the report view to view a list of all resources in the catalog, and the scan schedules. In the report view, click **Resources with schedules** to view only the resources with schedules. Click the link to the resource in the **Schedule details** column to view schedule details for the resource.

Note: The time zone of the resource schedule is the same as the system time zone.

In the report view, you can sort the list of resources, filter the list of resources, download a report containing the list of resources and schedules to a CSV file, and refresh the list. The CSV file includes the following schedules:

- Business Term Scheduler
- Composite Domain Discovery Scheduler
- Metadata Scheduler
- Profile Scheduler

You can also save filter criteria. Saved filters appear in the **Saved filters** menu. Use the calendar view to view the resource scan schedules on a calendar. Customize the view using the options at the top of the calendar.

View Duplicate Resources

1. In the **Catalog Information** page, select Duplicate Resource Groups as the type of report.
2. Click **Connect** to view the report.
3. Click the number corresponding to a resource type to view the duplicate resource groups for the resource type. Resources that are grouped as duplicates have identical connection properties.
4. Click **Duplicate properties** in the **Connection properties** column to view the connection properties for the resource group. The page also indicates if a metadata filter is applied to the resource.

Accept Data Domains

You can move specified data domains to the accepted state and reject other data domains in bulk using the Enterprise Data Catalog Add-on. To perform bulk curation, export the assets to a CSV file and specify the data domains to move to the accepted state. Use the add-on to parse the CSV file and generate an output

file. In the output file, the specified data domains are moved to the accepted state and all other data domains are moved to the rejected state. Import the CSV file into the Enterprise Data Catalog tool to curate the assets.

Specify the data domains that you want to move to the accepted state.

1. Export the assets to a CSV file.
2. Open the CSV file that you exported.
3. Add an AcceptedDataDomains column at the end of the worksheet.
4. For each asset that you want to curate, enter the data domain that you want to move to the accepted state in the AcceptedDataDomains column.
5. Save the file.

Perform bulk curation.

1. Access the Enterprise Data Catalog add-ons.
2. In the **My Services** page, click **CurateAssist**.
3. Click **Choose File** and select the CSV file.
4. Click **Curate** to parse the file.
5. Click the Download button to download the output file.
   The data domains that you specified in the AcceptedDataDomains column are moved to the accepted state and all other data domains that were in the Accepted or Inferred state are rejected.
6. Import the CSV file that you downloaded into the Enterprise Data Catalog tool to complete the data domain curation for the assets.

Moving Data Domains using the Data Domains Manager Utility

You can move the accepted resources in bulk from one data domain to another with the Enterprise Data Catalog Add-on.

Perform the following tasks to move accepted resources from one data domain to another data domain in bulk:

1. Access the Enterprise Data Catalog add-ons.
2. In the **My Services** page, click **Data Domains Manager**.
3. Select the source and destination data domains to use for the bulk move operation.
4. Click **Next**.
   A list of accepted resources of the source data domain appears.
5. Click **Submit**.
6. In the confirmation pop-up dialog box, click **Continue**.
   In the **Monitoring** page, view the status of job created for the bulk move operation.

Associate Business Terms from Synonym File

You can associate business terms from an Informatica Axon resource with specific columns within a tabular asset using the Enterprise Data Catalog add-on. The add-on supports tabular assets from relational database type resources. To associate business terms, use a synonym file in CSV format to define the associations. You can then specify the resource name, the Informatica Axon resource that contains the business term, upload the synonym file, and run the utility.

Define the business term associations in a synonym file.

1. Create a comma-separated values (CSV) synonym file with the following columns: TableName, ColumnName, BGMapping
2. Enter the names of columns that you want to associate with business terms, and the corresponding table names.
3. In the BGMapping column, enter the business terms that you want to associate.
4. Save the file.

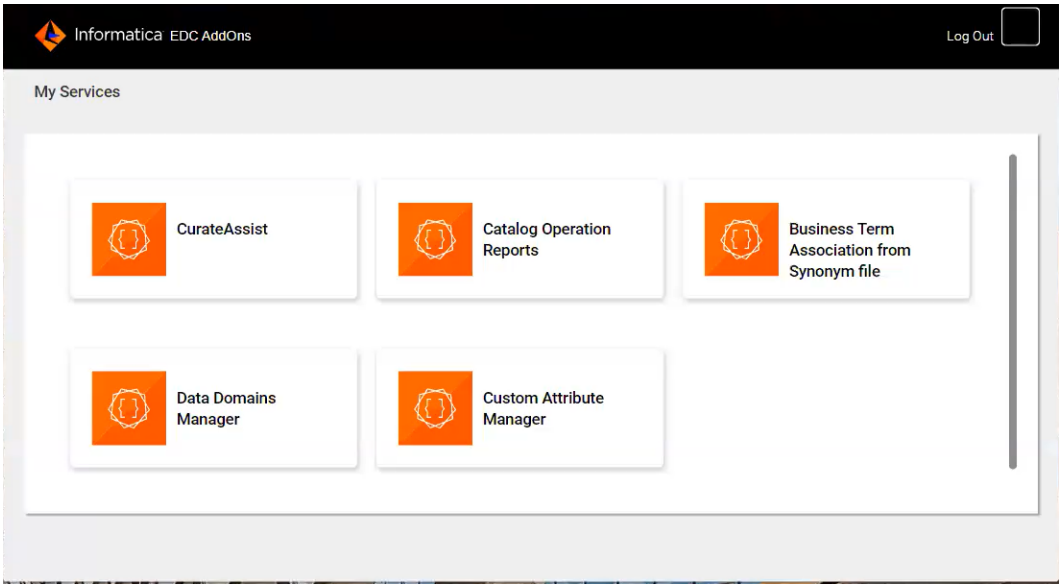
Associate business terms with specific columns.

1. Access the Enterprise Data Catalog add-ons.
2. In the **My Services** page, click **Business Term Association from Synonym file**.
3. Specify the resource that contains the columns to associate with business terms.
4. Select the Informatica Axon resource that contains the business term.
5. Choose the synonym file.
6. Click **Submit** to start the business term association.
   The job status appears in a table.
7. To download a CSV file with the status for each column that you specified, click **Download file**. You can also view error messages for failed associations.

Migrate Custom Attributes

You can migrate the value of a custom attribute to another custom attribute.

Use the Custom Attribute Manager in the Enterprise Data Catalog Marketplace Add-on portal to migrate a custom attribute.



The value of the attribute you migrate replaces or appends to the value of the destination attribute depending on the data type of the destination attribute. Migration of custom attribute is applicable for attributes with the following data types:

- String
- Boolean
- Integer
- Date
- Decimal

Follow the steps below to migrate a custom attribute:

1. From the Marketplace Add-on portal, open the Custom Attribute Manager.
2. Choose the source and destination custom attributes and click Next.
3. Click continue to initiate the migration.
4. To monitor the status, click the link that appears.

INDEX

A

accept
  data domain 182
administration
  process 15
associate
  business term 183
attribute
  custom 89
  overview 88
  system 88
attributes
  general properties 89
  search configuration properties 89

B

business glossary assets
  BG_DEFAULT_RESOURCE 46
  metadata load job 46
  synchronization 46
Business Term Association from Synonym file 183

C

catalog
  overview 18
Catalog Administrator
  Configuration workspace 56, 57
  Library workspace 57
  Monitoring workspace 56
  overview 54
  Overview workspace 56
  prerequisites 16
  tabs, viewing 54
connections
  assigning 93
  auto connections 93
  overview 92
  user-assigned connections 93
CurateAssist 182
custom attribute
  creating 90

D

Data 47
data domains
  creating a data domain 103
  overview 20
Data preview and provisioning 48

E

Enable
  provisioning 77
Enterprise Data Catalog
  accessing 15
  administration overview 14
  concepts 17
  example 19
  monitoring 98
  overview 13
Enterprise Data Catalog Add-Ons 181–183
Enterprise Unified Metadata
  architecture 14

G

global configuration
  Data Integration Service properties 97
  general properties 96

M

Manage Cloud Organization
  Overview 117
monitoring
  filters 101
Monitoring
  resources 100
  task distribution chart 99
  task status chart 99
  tasks 100

P

Prerequisites 48
Process 49, 51

R

resource
  attributes 59
  creating 63
  editing 78
  overview 18, 58
  running 79
  scanner 59
  scanner schedule 59
  viewing 80
Resource Schedule Report add-on 181
resource type
  overview 18, 64

reusable configuration
creating 97
overview 96
reusable schedule
creating 87

S

scanner
overview 18
scanner.sh
cancel 172
describeArchive 174
encryptPassword 170
generateResourceConfig 169
pause 173
publishArchive 175
resume 174
scan 171
validate 170
schedule
custom 87
overview 18
reusable 86
schedule type
overview 86
schedules
overview 86
viewing 87
Source Metadata Filter
overview 66
Standalone Utility
Best Practices 166
Components 166
Metadata archive file structure 177
Overview 163
Prerequisites 166
Process 165
Running the script 167
scanner.sh 168
synchronization
business glossary assets 46
system attribute
editing 90

V

view
resource schedules 181